Solutions Manual

for

Foundations of Sensitivity Analysis:
From Local Sensitivity to Global Uncertainty

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Companion materials: `https://github.com/machyman/arriola2026sa`

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How to use this manual. Each exercise is stated in full before its solution, so this manual can be used independently as a study guide without the main text open alongside. Solutions to Bloom Level L2 (understand) and L3 (apply) exercises are worked in detail and can be used for self-assessment. Solutions to L4 (analyze) and L5 (evaluate) exercises show one valid approach; alternative approaches that are mathematically correct are equally acceptable. Level L6 (create and synthesize) exercises are open-ended design problems; the solution shown is representative, not unique. Notes marked *Checking your answer* say what a complete answer contains and flag the errors readers most often make.

Use of AI tools. Generative AI tools were used in preparing this guide, in the same way and subject to the same verification described in the Acknowledgments of the main text: worked numerical results are reproduced by the companion notebooks, and bibliographic entries have been checked against primary sources. This work was carried out after Leon Arriola's death, and James Hyman alone is accountable for it. The authors assume responsibility for all content.

Bloom level reference.

Level	Cognitive demand	Typical task
L2	Understand	Explain a concept; classify an example
L3	Apply	Compute a sensitivity index; implement a formula
L4	Analyze	Diagnose a problem; compare two methods
L5	Evaluate	Critique a published SA study; design a procedure
L6	Create	Design a complete SA study; synthesize across chapters

Contents

1 **Chapter 1: Introduction and Overview** 4

2 **Chapter 2: The Sensitivity Index** 10

3 **Chapter 3: Computing Sensitivities Numerically** 23

4 **Chapter 4: Analytic Forward Sensitivity Analysis** 34

5 **Chapter 5: The Adjoint in Linear Algebra** 46

6 **Chapter 6: The Adjoint ODE** 53

7 **Chapter 7: Algorithmic Differentiation** 64

8 **Chapter 8: Caveat Emptor** 73

9 **Chapter 9: Global Sensitivity Analysis and Sobol Indices** 82

Repository algorithm contracts. The computational solutions reference the following algorithms. Companion Colab notebooks (Python) reproduce the worked solutions, and the MATLAB reference library is available alongside them at

`https://github.com/machyman/arriola2026sa`

Python and R ports of the library will follow with the published book. The pseudocode below describes each algorithm independently of language.

Algorithm	Contract
`SENSITIVITY_JACOBIAN(f, p, h)`	Returns the $n_q \times n_p$ normalized sensitivity Jacobian of model `f` at parameter vector `p`, using centered finite differences with step size `h`; entry (i, j) is $\mathcal{S}_{p_j}^{q_i}$.
`LHS_SAMPLE(N, m, lb, ub)`	Returns an $N \times m$ Latin hypercube sample with each column uniformly distributed on [`lb`[j], `ub`[j]]; reproducible with a fixed random seed.
`COMPUTE_PRCC(X, Y, names)`	Returns m partial rank correlation coefficients and p-values between each column of `X` and output `Y`, removing the linear effect of all other inputs via rank regression.
`MORRIS_SCREENING(f, p, r, delta)`	Returns m Morris elementary-effect statistics (μ^*, σ) from `r` one-at-a-time trajectories with perturbation size `delta`; default $r = 10$, $\delta = 1/3$.
`GLOBAL_SA_SOBOL(f, N, bounds)`	Generates $N(m + 2)$ Saltelli-sampled parameter sets, evaluates model `f` at each, and returns first-order and total Sobol indices with bootstrap confidence intervals.

Language choice. Solutions may use any programming language (MATLAB, Python, R, Julia, or other). Pseudocode in solutions uses language-neutral notation; any implementation that produces numerically equivalent results (to at least two significant figures) is fully acceptable. The companion notebooks and the MATLAB reference library provide worked implementations; Python and R ports will follow with the published book.

1 Chapter 1: Introduction and Overview

Chapter Summary: Introduction and Overview

Central question. Sensitivity analysis asks: which inputs (parameters of interest, POIs) most affect the outputs (quantities of interest, QOIs) we care about? The answer tells you where to collect better data, which interventions most affect the output, and when your model is near a threshold.

Three kinds of poi. (1) *Epistemic uncertainty*; parameter is fixed by nature but imprecisely known; SA identifies which measurements would most reduce output uncertainty. (2) *Aleatory uncertainty*; parameter is genuinely random; SA asks how much of the output variance it causes. (3) *Control variable*; a lever the investigator can move; SA ranks interventions by their quantified effect on the output. The same parameter can belong to more than one category simultaneously.

Two computational directions. *Forward SA* propagates a perturbation from one POI to the QOI ("What if?"). Cost: one model solve per POI. *Adjoint SA* propagates a variation in the QOI back to all POIs simultaneously ("How come?"). Cost: one solve regardless of how many POIs there are.

Four unifying ideas introduced here. (1) The sensitivity index is a normalized derivative. (2) The adjoint is the transpose in whatever space you work in. (3) Local SA is a linearization, with all the limitations that implies. (4) A sensitivity index describes a model, not the world. Every chapter revisits one or more of these four ideas.

Watch out: A large sensitivity index for a *control variable* is a call to act. A large sensitivity index for an *epistemically uncertain* parameter is a call to collect better data. The same number means different things depending on which role the POI is playing, always read the sign and magnitude in context.

Statistics and ML bridge. The sensitivity index $\mathcal{S}_p^q = (p/q)(\partial q/\partial p)$ is the same quantity called an *elasticity* in econometrics: the percent change in output per percent change in input. In machine learning, the adjoint method is *backpropagation*: automatic differentiation in reverse mode, propagating gradient information from output to all inputs in one backward pass.

Exercise 1.1 *[Bloom L2]*

Exercise statement. Explain in your own words the difference between a POI and a QOI. For a model describing the spread of antibiotic-resistant bacteria in a hospital, give two examples of each.

Then explain, in one sentence, what sensitivity analysis would tell you about this model that simply running the model would not.

Solution.

A *parameter of interest* (POI) is an input to the model whose value may be uncertain or whose influence we wish to quantify. A *quantity of interest* (QOI) is an output of the model that we care about; the number we are trying to predict or understand.

For an antibiotic-resistance model in a hospital, two POIs might be: the probability that a patient receives an antibiotic in a given hospital stay, and the probability that a resistant strain is transmitted per nurse-patient contact. Two QOIs might be: the fraction of patients colonized with a resistant strain after 30 days, and the mean length of stay for infected patients.

Running the model tells us the QOI at the nominal parameter values. Sensitivity analysis tells us *how much the* QOI *would change* if a POI were different, that is, which parameters matter most for the model's predictions, so we can focus data collection and intervention design on those parameters.

Checking your answer: Accept any reasonable POI/QOI pair consistent with the antibiotic-resistance context. The key distinction: POIs are inputs controlled or measured by the investigator; QOIs are outputs that answer the policy or scientific question. A complete answer names a POI and a QOI that belong to different sides of that line and says which is which. At minimum: a plausible pair, even if the reasoning is thin. A state variable offered as a POI is the error to watch for; parameters govern the dynamics, state variables evolve according to them.

Exercise 1.2 *[Bloom L2]*

Exercise statement. What is the difference between forward and adjoint sensitivity analysis? Describe each in terms of the "What if?" and "How come?" framing. For each of the following questions, classify it as a *forward question* or an *adjoint question* based on the type of question being asked (not on which algorithm would be most efficient to use computationally; efficiency is the subject of Chapters 5 and 6).

(a) If the mean latent period increases by 10%, how does the peak infection count change?

(b) The peak infection count is higher than expected. Which parameter uncertainties are most likely to have caused this?

(c) Among all parameters in the model, which one has the largest effect on the basic reproduction number?

Solution.

Forward SA answers "What if?" questions: if this POI changes, how does the QOI respond? It computes sensitivity by varying each POI one at a time and observing the effect on the QOI. The cost scales with the number of POIs (one computation per POI).

Adjoint SA answers "How come?" questions: given that the QOI has the value it does, which parameters are responsible? It computes sensitivity by working backward from the QOI to the POIs. The cost scales with the number of QOIs (one computation per QOI).

Classifying the three questions:

(a) *If the mean latent period increases by 10%, how does peak infection count change?*: **Forward** question. One POI (τ_E) is perturbed; we ask how the QOI (peak infection count) responds. The question is directional: "if I push this input, how does the output move?"

(b) *The peak infection count is higher than expected. Which parameter uncertainties are most likely to have caused this?* : **Adjoint** question. One QOI (peak infection count) is specified as the outcome of interest; we ask which POIs drove it. The direction is reversed: from output back to inputs.

(c) *Among all parameters in the model, which one has the largest effect on the basic reproduction number?* : **Forward** question. We perturb each POI one at a time and observe the effect on $\mathcal{R}_0$. (The cost comparison, forward vs. adjoint, is the subject of Chapters 5 and 6; the question is here about *type*, not efficiency.)

Checking your answer: A common error: classifying (c) as adjoint because "we are ranking all parameters." Ranking all parameters still uses forward SA (compute SI for each POI); the adjoint comes into play when the number of POIs is large and the number of QOIs is small, making a single backward solve more efficient. A complete answer classifies all parts and justifies each by the POI-to-QOI count, not by the wording of the question. At minimum: correct classification of (a) and (b) with a stated reason.

Exercise 1.3 *[Bloom L4]*

Exercise statement. For each of the following modeling scenarios, state whether forward SA, adjoint SA, or global SA is most appropriate. Justify each choice in one or two sentences.

(a) A climate model has 500 tunable parameters. The investigator wants to know which 10 parameters most affect the mean surface temperature over a 100-year run. Each model run takes 8 hours.

(b) A pollution transport model has 6 input parameters. The regulator wants to know: if the pollution at a specific monitoring station exceeds the legal threshold, which input assumptions most likely caused this?

(c) A pharmacokinetic model has 4 parameters, each of which is known only to within a factor of 2. The investigator wants to know the range of predicted drug concentrations consistent with this uncertainty.

Solution.

Four factors drive the method choice for each scenario: the number of POIs m, the cost per model evaluation t_{eval}, whether the response is nonlinear over the plausible parameter range, and whether an adjoint is available.

(a) *Climate model, 500 parameters, identify 10 most influential on mean surface temperature, 8 hours per run.*

Morris screening (or adjoint, if available). With 8 hours per run, a full Sobol analysis at $N = 2000$ would cost $N(m + 2) = 1{,}004{,}000$ runs, or about 916 years of computation. Morris screening with $r = 10$ trajectories costs $r(m + 1) = 5010$ runs ($\approx$4.6 years), still expensive but feasible on a cluster. Adjoint SA would cost the equivalent of one forward run plus storage, giving all 500 sensitivities simultaneously. If the adjoint is available, use it; otherwise, Morris screening on a cluster is the practical alternative.

(b) *Pollution transport model, 6 parameters, threshold exceedance at one monitoring station.*

Adjoint SA. The QOI is a single threshold value at one location; exactly the scenario where the adjoint is optimal. With 6 parameters and 1 QOI, adjoint costs 1 backward solve vs. 6 forward perturbations. The adjoint gives the complete gradient of the threshold value with respect to all 6 parameters simultaneously.

(c) *Pharmacokinetic model, 4 parameters, each known only to within a factor of 2. Investigator wants the range of predicted drug concentrations consistent with this uncertainty.*

Global SA (Sobol indices). The uncertainty is large (factor-of-2 in each parameter) and the question asks about the *range* of predicted outputs, not just a local derivative. Local SA would tell you which parameters matter near a nominal point, but not how much the output varies across the full two-fold uncertainty range. Sobol indices at $N = 1000$ require $1000 \times (4 + 2) = 6{,}000$ evaluations, feasible for a pharmacokinetic ODE in seconds.

Checking your answer: This exercise builds intuition for method selection. Common student errors: (a) choosing forward SA for the climate scenario without recognizing the computational cost; (b) choosing global SA for the pollution scenario because "nonlinearity might be present" (the QOI is a single threshold at one station, and the model is not stated to be nonlinear, so adjoint is correct for the stated problem); (c) choosing forward SA for the pharmacokinetic scenario without recognizing that factor-of-2 uncertainty is too large for local linearization to be reliable. A complete answer gives a method for each of the three scenarios with the deciding feature named: cost for the climate case, the POI-to-QOI ratio for the pollution case, and the size of the uncertainty for the pharmacokinetic case. At minimum: two of the three correct with reasons.

Exercise 1.4 *[Bloom L5]*

Exercise statement. You are advising a team modeling the transmission of dengue fever. The model has 9 parameters. With funding to measure exactly 3 parameters precisely in the field and to implement exactly 2 vector-control interventions. Write a one-page plan describing how you would use SA to guide these decisions. Your plan should specify: what QOI you would focus on and why; how you would determine which 3 parameters to measure; what additional information (if any) you would need beyond the SA results to recommend interventions; and one important limitation of your approach that the team should be aware of.

Solution.

Model answer (acceptable answers will vary in specifics but should include all four required components):

SA plan for a dengue fever model with 9 parameters.

QOI choice. I would use peak infectious count $\max_t I(t)$ as the primary QOI, because it determines whether the healthcare system is overwhelmed; the operationally critical threshold. A secondary QOI would be the basic reproduction number $\mathcal{R}_0$ (analytically tractable and directly related to whether an epidemic occurs). The relationship between the two provides a consistency check.

Identifying the 3 parameters to measure. Run a local SA using the forward sensitivity equations (analytic if the model is an ODE, or centered finite differences otherwise) to compute all 9 sensitivity indices $\mathcal{S}_{p_j}^{\text{peak } I}$. Rank by $|\mathcal{S}_{p_j}^{\cdot}|$. The three parameters with the largest absolute SI values are the priority for precise measurement. State the assumption this ranking makes: the normalized SI scales by the nominal value, not by how uncertain the parameter actually is, so ranking by $|\mathcal{S}_{p_j}^{\cdot}|$ presumes the nine parameters are comparably uncertain in relative terms. Where that fails, a parameter with a modest derivative and a wide plausible range outranks one with a steep derivative that is already known precisely (Chapter 9). If two parameters appear in the model only as a product (structural correlation), treat the product as a single effective parameter.

Before reporting, also run a Morris screening (15 trajectories, cost: $15 \times 10 = 150$ model evaluations) to flag any parameters where nonlinearity makes the local SI misleading.

Additional information needed for intervention design. SA tells us which parameters *matter most for predictions*, not which can be controlled by public health interventions. I would cross-reference the top-ranked parameters with a table of interventions that affect each parameter (e.g., insecticide spraying affects vector contact rate; surveillance programs affect reporting rate). Only parameters that are both high-sensitivity *and* actionable with available resources should drive intervention recommendations.

Limitation. The most important limitation is that local SA is a linearization: the sensitivity indices are derivatives at the nominal parameter values. If the model is highly nonlinear (e.g., near a threshold between epidemic and no-epidemic regimes), local SA may not accurately reflect sensitivity across the full plausible parameter range. I would flag this to the team and recommend a global SA (Sobol) on the top 4–5 parameters identified by local SA, to check for nonlinear effects.

Checking your answer: A complete answer has four parts: a justified QOI; a parameter-identification procedure naming a specific SA method and operationally specific about the ranking, including the point that ranking by |SI| assumes comparable relative uncertainty across parameters; the intervention-actionability distinction; and a genuine limitation of local SA. A common shortfall is recommending global SA for all 9 parameters without noting the cost implication (dengue models are often ODEs; full Sobol is feasible, so this is not wrong, but the question asks them to use SA to guide *field* measurement decisions, which requires identifying the top 3 specifically).

Exercise 1.5 *[Bloom L6]*

Exercise statement. A published report on a disease model states: "Sensitivity analysis was performed on all 22 model parameters simultaneously, demonstrating that the model is robust to parameter uncertainty." The analysis was conducted by computing local sensitivity indices at the best-fit parameter values.

Identify at least two substantive problems with this description or its conclusions. For each problem you identify, explain specifically what the authors should have done differently.

Solution.

Problem 1: Conflation of "all parameters simultaneously" with global SA.

Computing local sensitivity indices (derivatives at best-fit values) does *not* analyze all parameters simultaneously. Local SA perturbs one parameter at a time, holding all others fixed. The statement implies that the joint effect of simultaneous parameter uncertainty was assessed, but it was not. What should have been done: either (a) state that local SA was used, which analyzes one parameter at a time and captures only linear, individual effects; or (b) if the goal was to assess robustness to joint uncertainty, perform global SA (e.g., Sobol indices), which does evaluate sensitivity across the full multi-dimensional parameter space.

Problem 2: "Robust to parameter uncertainty" is not a valid conclusion from local SI values.

Sensitivity indices measure the *rate of change* of the QOI with respect to each parameter. A model with all small SIs at the best-fit values is locally insensitive near that point, but could be highly sensitive elsewhere (e.g., near a bifurcation). "Robust" should mean that the QOI changes little across the full plausible range of parameter values, a global property, not a local one. What should have been done: specify a plausible parameter uncertainty range for each parameter, compute the resulting range of QOI values, and only then make a robustness claim.

Optional additional problem (an optional extension): Local SA at "best-fit parameter values" carries real cautions, though circularity is not one of them. Fitting drives the *objective gradient* to zero at an interior optimum; it does not drive the *model output* derivatives to zero, and these are different quantities. Output sensitivities can be large at a best fit: for $y = e^{pt}$ fitted to noisy data, the objective gradient vanishes to 10^{-6} while $\mathcal{S}_p^y$ at the final time is about 1.4. The genuine cautions are that local results hold only at the calibration point; that poorly identified directions carry large uncertainty; and that sensitivities should be read alongside the parameter covariance or posterior, not in isolation.

Checking your answer: This is a Bloom L6 synthesis exercise. A complete answer requires identifying problems that go beyond surface-level criticism. The strongest answers connect "local vs. global" to the mathematical definition of the SI (derivative vs. variance decomposition) rather than simply noting vague imprecision. A partial answer (1 point each) for correct identification of any two genuine methodological issues with specific alternative recommendations.

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2 Chapter 2: The Sensitivity Index

Chapter Summary: Measuring Local Sensitivity

The sensitivity index (Definition 2.1). The normalized sensitivity index of QOI q with respect to POI p is

$$\mathcal{S}_p^q = \frac{p}{q}\frac{\partial q}{\partial p} = \frac{\partial \ln q}{\partial \ln p}.$$

It answers: "If p changes by $x\%$, q changes by approximately $x \cdot \mathcal{S}_p^q\%$." It is dimensionless, unit-free, and comparable across parameters measured on different scales.

The sign rule. A positive SI means POI and QOI move in the same direction. A negative SI means they move in opposite directions. The sign belongs to the *parameterization*, not to the biology. $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = -1$ and $\mathcal{S}_{\tau_R}^{\mathcal{R}_0} = +1$ describe the same biological fact ($\tau_R = 1/\gamma_R$); the sign flips when you invert the parameter.

Monomial rule. For $q = c\, p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}$, the sensitivity index with respect to p_i is exactly a_i, read it off the exponent. All three SIR indices $\mathcal{S}_c^{\mathcal{R}_0} = \mathcal{S}_\beta^{\mathcal{R}_0} = \mathcal{S}_{\tau_R}^{\mathcal{R}_0} = 1$ follow from $\mathcal{R}_0 = c\beta\tau_R$ having exponent 1 for each factor.

Near-zero qoi. When $|q^*|$ is small relative to $\max_t |q(t)|$ (specifically, when $|q^*| < 0.01 \times \max_t |q(t)|$), the normalized SI is unreliable. Use the absolute sensitivity $\partial q/\partial p$ or the regularized form $\frac{p}{1+|q|}\frac{\partial q}{\partial p}$ instead.

The tornado plot. Bar chart sorted by $|\mathcal{S}_p^q|$, bars to the right for positive indices and to the left for negative. Sort by magnitude, not by sign. Always use normalized indices on the axis, never raw derivatives.

Watch out: Parameterization matters. Report sensitivity indices in terms of parameters that can be explained in ordinary language, durations (days), not rates (day^{-1}); probabilities, not their complements. A sensitivity index that requires mental inversion before the public-health meaning is clear is a sensitivity index that will be misread.

Statistics and ML bridge. The sensitivity index is the log-log regression slope: $\mathcal{S}_p^q = d(\ln q)/d(\ln p)$ (Form 3 from the chapter). In regression analysis with log-transformed variables, this is the estimated coefficient. In epidemiology it is the elasticity. *Dual-classification example:* the contact rate c in an SIR model can be (1) epistemic (uncertain because hard to measure from contact-tracing data), and (2) a control variable (the public health authority can reduce it via distancing policies). These two roles call for different SA interpretations of the same sensitivity index.

Exercise 2.1 *[Bloom L2]*

Exercise statement. Explain the sign rule in your own words. Without performing any calculation, state the sign of the sensitivity index for each of the following, and explain your reasoning: (a) the sensitivity of lung capacity to cigarette consumption, treating capacity as the QOI and cigarettes per day as the POI; (b) the sensitivity of average crop yield to annual rainfall, near a region of the parameter space where crops are mildly drought-stressed; (c) the sensitivity of $\mathcal{R}_0$ to the mean infectious period τ_R.

Solution.

The sign of a sensitivity index tells you the *direction* of the relationship: a positive SI means the QOI increases when the POI increases; a negative SI means the QOI decreases when the POI increases. The sign is the sign of the underlying derivative $\partial q/\partial p$, scaled by p/q (both positive under the interpretable-parameterization convention).

(a) *Lung capacity vs. cigarettes per day.* Lung capacity decreases as cigarette consumption increases (medical consensus). Therefore $\partial q/\partial p < 0$, so the SI is *negative.*

(b) *Crop yield vs. annual rainfall (near the optimum).* Crop yields typically increase with rainfall up to an optimum and then decrease (waterlogging). The sign depends on which side of the optimum we are evaluating. Near the dry end of the feasible range, yield increases with rainfall: *positive* SI. Near the wet end (waterlogging), yield decreases: *negative* SI. The question specifies "near a region of the parameter space"; the sign must be specified relative to that region, and the answer should note the non-monotonicity.

(c) $\mathcal{R}_0$ *vs. the mean infectious period* τ_R. $\mathcal{R}_0 = c\beta\tau_R$ (using duration parameterization), so $\partial\mathcal{R}_0/\partial\tau_R = c\beta > 0$: a longer infectious period increases $\mathcal{R}_0$. Therefore $\mathcal{S}^{\mathcal{R}_0}_{\tau_R} = +1$, since $\mathcal{R}_0$ is linear in τ_R.

The sign is *positive.* This is the exact error the chapter's "Watch out" box warns against: if instead we had parameterized using the recovery rate $\gamma_R = 1/\tau_R$, then $\mathcal{R}_0 = c\beta/\gamma_R$ and $\mathcal{S}^{\mathcal{R}_0}_{\gamma_R} = -1$ (negative). The sign depends entirely on the parameterization, and the exercise asks for τ_R, not γ_R.

Checking your answer: Part (b) is deliberately ambiguous to prompt students to identify the dependence on the evaluation point p^*. Common error: stating $\mathrm{SI}^{\mathcal{R}_0}_{\gamma_R} = -1$ for part (c) instead of $\mathrm{SI}^{\mathcal{R}_0}_{\tau_R} = +1$, committing the rate-vs-duration swap the chapter explicitly warns against. At minimum: correct formula for SI^q_p; correct sign for each part. A complete answer: explicit chain-rule for part (c) using $\tau_R = 1/\gamma_R$, plus the interpretation (why the sign flips when parameterization changes). The distinguishing check: getting part (c) right using τ_R and getting it wrong by computing γ_R are different answers, even when the arithmetic is otherwise identical.

Exercise 2.2 *[Bloom L2]*

Exercise statement. For the function $q = \sqrt{p}$ (defined for $p > 0$), compute $\mathcal{S}^q_p$ using all three equivalent forms of Definition 2.1. Verify that all three give the same result. Interpret the answer: if p doubles, by approximately what percentage does q change?

Solution.

Let $q = \sqrt{p}$, so $q = p^{1/2}$.

Form 1 (limit definition):

$$\mathcal{S}_p^q = \lim_{\delta p \to 0} \frac{\delta q/q}{\delta p/p} = \lim_{\delta p \to 0} \frac{\left(\sqrt{p+\delta p} - \sqrt{p}\right)/\sqrt{p}}{\delta p/p} = \frac{p}{\sqrt{p}} \lim_{\delta p \to 0} \frac{\sqrt{p+\delta p} - \sqrt{p}}{\delta p} = \sqrt{p} \cdot \frac{1}{2\sqrt{p}} = \frac{1}{2}.$$

Form 2 (derivative form):

$$\mathcal{S}_p^q = \frac{p}{q}\frac{\partial q}{\partial p} = \frac{p}{\sqrt{p}} \cdot \frac{1}{2\sqrt{p}} = \frac{\sqrt{p}}{2\sqrt{p}} = \frac{1}{2}.$$

Form 3 (logarithmic form):

$$\mathcal{S}_p^q = \frac{d\ln q}{d\ln p} = \frac{d}{d\ln p}\left(\tfrac{1}{2}\ln p\right) = \frac{1}{2}.$$

All three give $\mathcal{S}_p^q = \frac{1}{2}$.

Interpretation: A 1% increase in p produces approximately a 0.5% increase in $q = \sqrt{p}$. More precisely: if p doubles (increases by 100%), q increases by approximately 50% (the exact value is $\sqrt{2} - 1 \approx 41.4\%$; the SI gives a linear approximation).

Checking your answer: Emphasize the logarithmic form (Form 3) as the most computationally efficient: $d\ln q/d\ln p$ can often be read off immediately from the algebraic form of $q(p)$. For a power law $q = cp^n$, all three forms give $\mathcal{S}_p^q = n$, which can be seen immediately from the log form.

Exercise 2.3 *[Bloom L3]*

Exercise statement. Let $u(t) = \frac{1}{2}\lambda t^2$ be a parabola on $[0, b]$. The arc-length functional is $J = \int_0^b \sqrt{1 + (\lambda t)^2}\, dt$. Show that the sensitivity indices are

$$\mathcal{S}_b^J = \frac{2\mu_b}{\mu_b + (\operatorname{arcsinh}\mu_b)/\sqrt{1+\mu_b^2}}, \qquad \mathcal{S}_\lambda^J = \frac{\mu_b(1+\mu_b^2) - \sqrt{1+\mu_b^2}\,\operatorname{arcsinh}\mu_b}{\mu_b(1+\mu_b^2) + \sqrt{1+\mu_b^2}\,\operatorname{arcsinh}\mu_b},$$

where $\mu_b = \lambda b$. For $\mu_b = 1$, compute both indices numerically and verify using the derivative form.

Solution.

Let $\mu_b = \lambda b$. The arc-length functional is

$$J = \int_0^b \sqrt{1 + \lambda^2 t^2}\, dt.$$

Evaluating J. Substitute $u = \lambda t$, $du = \lambda\, dt$:

$$J = \frac{1}{\lambda}\int_0^{\lambda b} \sqrt{1+u^2}\, du = \frac{1}{\lambda}\Big[\tfrac{u}{2}\sqrt{1+u^2} + \tfrac{1}{2}\operatorname{arcsinh} u\Big]_0^{\mu_b} = \frac{\mu_b\sqrt{1+\mu_b^2} + \operatorname{arcsinh}\mu_b}{2\lambda}.$$

Computing $\mathcal{S}_b^J$. Using the derivative form with $\mu_b = \lambda b$:

$$\frac{\partial J}{\partial b} = \sqrt{1+\lambda^2 b^2} = \sqrt{1+\mu_b^2}.$$

So

$$\mathcal{S}_b^J = \frac{b}{J}\cdot\sqrt{1+\mu_b^2} = \frac{\lambda b\cdot\sqrt{1+\mu_b^2}}{\frac{1}{2}(\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b)} = \frac{2\mu_b\sqrt{1+\mu_b^2}}{\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b}.$$

Factor $\sqrt{1+\mu_b^2}$ from numerator and denominator:

$$\mathcal{S}_b^J = \frac{2\mu_b}{\mu_b + \operatorname{arcsinh}\mu_b/\sqrt{1+\mu_b^2}}.$$

This matches the stated formula. ✓

Computing $\mathcal{S}_\lambda^J$. Differentiating $J = (\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b)/(2\lambda)$ with $\mu_b = \lambda b$ (so $\partial\mu_b/\partial\lambda = b = \mu_b/\lambda$):

$$\frac{\partial J}{\partial \lambda} = \frac{b\big(\sqrt{1+\mu_b^2}+\mu_b^2/\sqrt{1+\mu_b^2}+1/\sqrt{1+\mu_b^2}\big)}{2\lambda} - \frac{J}{\lambda}.$$

Simplifying $\sqrt{1+\mu_b^2}+\mu_b^2/\sqrt{1+\mu_b^2}+1/\sqrt{1+\mu_b^2} = (1+\mu_b^2+\mu_b^2+1)/\sqrt{1+\mu_b^2} = (2+2\mu_b^2)/\sqrt{1+\mu_b^2} = 2\sqrt{1+\mu_b^2}$:

$$\frac{\partial J}{\partial \lambda} = \frac{\mu_b\sqrt{1+\mu_b^2}}{\lambda^2} - \frac{J}{\lambda}.$$

Therefore

$$\mathcal{S}_\lambda^J = \frac{\lambda}{J}\cdot\frac{\partial J}{\partial\lambda} = \frac{\lambda}{J}\left(\frac{\mu_b\sqrt{1+\mu_b^2}}{\lambda^2} - \frac{J}{\lambda}\right) = \frac{\mu_b\sqrt{1+\mu_b^2}}{\lambda J} - 1.$$

Substituting $J = (\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b)/(2\lambda)$:

$$\mathcal{S}_\lambda^J = \frac{2\mu_b\sqrt{1+\mu_b^2}}{\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b} - 1 = \frac{2\mu_b\sqrt{1+\mu_b^2}-\mu_b\sqrt{1+\mu_b^2}-\operatorname{arcsinh}\mu_b}{\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b} = \frac{\mu_b\sqrt{1+\mu_b^2}-\operatorname{arcsinh}\mu_b}{\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b}.$$

This matches the stated formula. ✓

Numerical verification at $\mu_b = 1$ (i.e., $\lambda b = 1$).

With $\mu_b = 1$: $\operatorname{arcsinh}(1) = \ln(1+\sqrt{2}) \approx 0.8814$, $\sqrt{1+\mu_b^2} = \sqrt{2} \approx 1.4142$.

$$J = \frac{\mu_b\sqrt{1+\mu_b^2}+\operatorname{arcsinh}\mu_b}{2\lambda} = \frac{1\cdot\sqrt{2}+0.8814}{2\lambda} = \frac{2.2956}{2\lambda}.$$

$$\mathcal{S}_b^J\big|_{\mu_b=1} = \frac{2(1)}{1+0.8814/\sqrt{2}} = \frac{2}{1+0.6232} = \frac{2}{1.6232} \approx \mathbf{1.232}.$$

$$\mathcal{S}_\lambda^J\big|_{\mu_b=1} = \frac{(1)(\sqrt{2})-0.8814}{(1)(\sqrt{2})+0.8814} = \frac{1.4142-0.8814}{1.4142+0.8814} = \frac{0.5328}{2.2956} \approx \mathbf{0.232}.$$

Observation: $\mathcal{S}_b^J + \mathcal{S}_\lambda^J = 1.232 + 0.232 = 1.464 \neq 1$. There is no reason they should. Local sensitivity indices obey no sum rule: unlike Sobol first-order indices, which partition a variance, these are independent derivatives and their sum has no normalization to respect. For a function homogeneous of degree k in its parameters, Euler's theorem makes the indices sum to k; here J is not homogeneous, so the sum is simply whatever the derivatives give. The parameters b and λ are themselves independent inputs; that they appear together in $\mu_b = \lambda b$ is a property of the model, not a dependence between them.

However, an important identity can be observed directly from the formulas: $\mathcal{S}_b^J - \mathcal{S}_\lambda^J = 1$, since $\mathcal{S}_b^J = (1 + \mathcal{S}_\lambda^J) + 0 \cdot \ldots$, in fact, subtracting the two expressions:

$$\mathcal{S}_b^J - \mathcal{S}_\lambda^J = \frac{2\mu_b}{\mu_b + \operatorname{arcsinh}\mu_b/\sqrt{1+\mu_b^2}} - \frac{\mu_b\sqrt{1+\mu_b^2} - \operatorname{arcsinh}\mu_b}{\mu_b\sqrt{1+\mu_b^2} + \operatorname{arcsinh}\mu_b} = 1$$

for all $\mu_b > 0$. At $\mu_b = 1$: $1.232 - 0.232 = 1.000$ ✓

Derivative-form verification at $\mu_b = 1$, $\lambda = 1$, $b = 1$.

With $\lambda = b = 1$: $J = 2.2956/2 \approx 1.1478$.

$\partial J/\partial b = \sqrt{1+\mu_b^2} = \sqrt{2}$, so $\mathcal{S}_b^J = (b/J)\sqrt{2} = (1/1.1478)\sqrt{2} \approx 1.232$ ✓

$\partial J/\partial\lambda$: from the formula derived above, $= \mu_b\sqrt{1+\mu_b^2}/\lambda^2 - J/\lambda = \sqrt{2}/1 - 1.1478/1 = 1.4142 - 1.1478 = 0.2664$, so $\mathcal{S}_\lambda^J = (\lambda/J)\cdot 0.2664 = 0.2664/1.1478 \approx 0.232$ ✓

Checking your answer: Common error: not performing the numerical check (the exercise asks for it explicitly). The identity $\mathcal{S}_b^J - \mathcal{S}_\lambda^J = 1$ holds for all $\mu_b > 0$ and is one of the most elegant results in this exercise. At minimum: correct symbolic derivation of both formulas, with $\mu_b = 1$ values stated (even if the derivation is omitted). A complete answer: symbolic derivation, $\mu_b = 1$ numerical values, derivative-form verification, and the $\mathcal{S}_b^J - \mathcal{S}_\lambda^J = 1$ identity noted. The distinguishing check: students who derive the formulas but skip the numerical check are not completing the exercise as stated and should lose 1 point.

Exercise 2.4 *[Bloom L3]*

Exercise statement. Let $u(t) = at^2 + bt + c$ and let $J_\pm = (-b \pm \sqrt{b^2 - 4ac})/(2a)$ denote the two real roots (assume $b^2 > 4ac$). Show that the sensitivity indices of the positive root J_+ with respect to a, b, c can be written

$$\mathcal{S}_a^{J_+} = -\frac{aJ_+}{2aJ_+ + b}, \quad \mathcal{S}_b^{J_+} = -\frac{b}{2aJ_+ + b}, \quad \mathcal{S}_c^{J_+} = -\frac{c}{J_+(2aJ_+ + b)}.$$

For $a = 1$, $b = -3$, $c = 2$, compute all three indices. Interpret each sign.

Solution.

Let $J_+ = (-b + \sqrt{b^2 - 4ac})/(2a)$. Denote $D = b^2 - 4ac > 0$. Since J_+ satisfies $aJ_+^2 + bJ_+ + c = 0$, we have $aJ_+^2 = -bJ_+ - c$.

The denominator in all three formulas is $2aJ_+ + b = \sqrt{b^2 - 4ac} = \sqrt{D}$ (from the quadratic formula: $2aJ_+ = -b + \sqrt{D}$, so $2aJ_+ + b = \sqrt{D}$).

Computing $\mathcal{S}_a^{J_+}$:

$$\frac{\partial J_+}{\partial a} = \frac{-b + \sqrt{D}}{-2a^2} + \frac{-4c/2}{2a\sqrt{D}} \cdot (-1) \quad \text{[quotient rule + chain rule]}$$

Simpler via implicit differentiation: differentiate $aJ_+^2 + bJ_+ + c = 0$ with respect to a:

$$J_+^2 + (2aJ_+ + b)\frac{\partial J_+}{\partial a} = 0 \implies \frac{\partial J_+}{\partial a} = -\frac{J_+^2}{2aJ_+ + b}.$$

Therefore

$$\mathcal{S}_a^{J_+} = \frac{a}{J_+} \cdot \frac{\partial J_+}{\partial a} = \frac{a}{J_+} \cdot \frac{-J_+^2}{2aJ_+ + b} = \frac{-aJ_+}{2aJ_+ + b}.\checkmark$$

Computing $\mathcal{S}_b^{J_+}$: Differentiate $aJ_+^2 + bJ_+ + c = 0$ with respect to b:

$$J_+ + (2aJ_+ + b)\frac{\partial J_+}{\partial b} = 0 \implies \frac{\partial J_+}{\partial b} = -\frac{J_+}{2aJ_+ + b}.$$

$$\mathcal{S}_b^{J_+} = \frac{b}{J_+} \cdot \frac{-J_+}{2aJ_+ + b} = \frac{-b}{2aJ_+ + b}.\checkmark$$

Computing $\mathcal{S}_c^{J_+}$: Differentiate with respect to c:

$$(2aJ_+ + b)\frac{\partial J_+}{\partial c} + 1 = 0 \implies \frac{\partial J_+}{\partial c} = -\frac{1}{2aJ_+ + b}.$$

$$\mathcal{S}_c^{J_+} = \frac{c}{J_+} \cdot \frac{-1}{2aJ_+ + b} = \frac{-c}{J_+(2aJ_+ + b)}.\checkmark$$

Numerical verification at $a = 1$, $b = -3$, $c = 2$: $J_+ = (3 + \sqrt{9-8})/2 = 2$, $2aJ_+ + b = 4 - 3 = 1$.

$$\mathcal{S}_a^{J_+} = -\frac{1 \cdot 2}{1} = -2, \quad \mathcal{S}_b^{J_+} = -\frac{-3}{1} = 3, \quad \mathcal{S}_c^{J_+} = -\frac{2}{2 \cdot 1} = -1.$$

Verification by finite differences (centered, $h = 10^{-6}$): $\partial J_+/\partial a \approx (J_+(a+h) - J_+(a-h))/(2h)$, multiplied by $a/J_+ = 1/2$, gives $\mathcal{S}_a^{J_+} \approx -2.000000$ ✓.

Checking your answer: Implicit differentiation is the most elegant approach and avoids messy quotient-rule algebra. Students who use explicit differentiation of the quadratic formula are not wrong but often make sign errors. The numerical check at $a = 1$, $b = -3$, $c = 2$ is important: students should verify that their symbolic answers match finite differences.

Exercise 2.5 *[Bloom L3]*

Exercise statement. A SEIR model (Susceptible-Exposed-Infectious-Recovered) has reproduction number

$$\mathcal{R}_0 = \frac{c\beta\,\nu}{(\nu + \nu_m)(\gamma_R + \nu_m)},$$

where ν is the rate of progression from exposed to infectious, ν_m is the background mortality rate, and c, β, γ_R are as in the SIR model. Reparameterize using mean times: $\tau_E = 1/\nu$ (mean latent period), $\tau_I = 1/\gamma_R$ (mean infectious period), $\tau_m = 1/\nu_m$ (mean lifespan). Compute $\mathcal{S}_{\tau_E}^{\mathcal{R}_0}$, $\mathcal{S}_{\tau_I}^{\mathcal{R}_0}$, $\mathcal{S}_{\tau_m}^{\mathcal{R}_0}$, $\mathcal{S}_c^{\mathcal{R}_0}$, $\mathcal{S}_\beta^{\mathcal{R}_0}$ and produce a tornado plot. Which parameter is most important?

Solution.

The SEIR reproduction number reparameterizes as follows. Starting from $\mathcal{R}_0 = c\beta\nu/[(\nu + \nu_m)(\gamma_R + \nu_m)]$ and substituting $\tau_E = 1/\nu$, $\tau_I = 1/\gamma_R$, $\tau_m = 1/\nu_m$:

$$\mathcal{R}_0 = \frac{c\beta}{\tau_E} \cdot \frac{1}{(1/\tau_E + 1/\tau_m)} \cdot \frac{1}{(1/\tau_I + 1/\tau_m)} = c\beta \cdot \frac{\tau_E \tau_m}{\tau_E + \tau_m} \cdot \frac{\tau_I \tau_m}{\tau_I + \tau_m} \cdot \frac{1}{\tau_E}.$$

Simplifying: $\mathcal{R}_0 = c\beta\tau_I \cdot \dfrac{\tau_m^2}{(\tau_E + \tau_m)(\tau_I + \tau_m)}$.

Sensitivity indices (log-derivative form on the original rates):

From $\ln \mathcal{R}_0 = \ln c + \ln \beta + \ln \nu - \ln(\nu + \nu_m) - \ln(\gamma_R + \nu_m)$:

$$\mathcal{S}_c^{\mathcal{R}_0} = 1, \qquad \mathcal{S}_\beta^{\mathcal{R}_0} = 1.$$

$$\mathcal{S}_\nu^{\mathcal{R}_0} = 1 - \frac{\nu}{\nu + \nu_m} = \frac{\nu_m}{\nu + \nu_m}.$$

Converting to duration ($\tau_E = 1/\nu$, $\mathcal{S}_{\tau_E}^{\mathcal{R}_0} = -\mathcal{S}_\nu^{\mathcal{R}_0}$):

$$\mathcal{S}_{\tau_E}^{\mathcal{R}_0} = -\frac{\nu_m}{\nu + \nu_m} = -\frac{1/\tau_m}{1/\tau_E + 1/\tau_m} = -\frac{\tau_E}{\tau_E + \tau_m}.$$

For γ_R: $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = -\gamma_R/(\gamma_R + \nu_m) = -\tau_m/(\tau_I + \tau_m)$. Since $\tau_I = 1/\gamma_R$ is a reciprocal reparameterization, $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = -\mathcal{S}_{\gamma_R}^{\mathcal{R}_0}$, with no additional term:

$$\boxed{\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = \frac{\tau_m}{\tau_I + \tau_m} \approx +0.9993.}$$

For τ_m (residence time): from $\ln \mathcal{R}_0 = \ln(c\beta) + \ln \tau_I + 2\ln \tau_m - \ln(\tau_E + \tau_m) - \ln(\tau_I + \tau_m)$, differentiating with respect to $\ln \tau_m$:

$$\mathcal{S}_{\tau_m}^{\mathcal{R}_0} = 2 - \frac{\tau_m}{\tau_E + \tau_m} - \frac{\tau_m}{\tau_I + \tau_m} \approx 2 - 0.99945 - 0.99930 = +0.001250.$$

(Longer lifespan increases $\mathcal{R}_0$ because susceptibles live longer to be infected.)

Numerical values at $c = 5$, $\beta = 0.06$, $\tau_E = 5.5$ days, $\tau_I = 7$ days, $\tau_m = 10{,}000$ days:

POI	c	β	τ_E	τ_I	τ_m
SI	$+1.000$	$+1.000$	-0.000550	$+0.9993$	$+0.001250$
|SI|	1.000	1.000	0.000550	0.9993	0.001250

Verification: $\mathcal{S}_{\tau_E}^{\mathcal{R}_0}$ is negative (longer latent period means fewer effective infectious contacts per lifetime). $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} \approx +0.9993$ is large and positive: the infectious period enters both the numerator ($+\ln \tau_I$) and denominator ($-\ln(\tau_I + \tau_m)$) of $\ln \mathcal{R}_0$, and the net elasticity is $\tau_m/(\tau_I + \tau_m) \approx 1$. This makes τ_I comparable in importance to c and β. $\mathcal{S}_{\tau_m}^{\mathcal{R}_0} \approx +0.001250$ is tiny: at $\tau_m = 10{,}000$ days the two terms $\tau_m/(\tau_E + \tau_m)$ and $\tau_m/(\tau_I + \tau_m)$ each round to ≈ 1, so their sum nearly cancels the leading $+2$.

In the demographic limit $\nu_m \to 0$ ($\tau_m \to \infty$): $\mathcal{S}^{\mathcal{R}_0}_{\tau_m} \to 0$ and $\mathcal{R}_0 \to c\beta\tau_I$, recovering the basic SIR result. At finite τ_m, the ranking by $|\mathcal{S}|$ is: $c \approx \beta \approx \tau_I \approx 1 \gg \mathcal{S}^{\mathcal{R}_0}_{\tau_m} \approx \mathcal{S}^{\mathcal{R}_0}_{\tau_E} \approx 10^{-3}$.

Checking your answer: The three most common errors here: (1) Computing $\mathcal{S}^{\mathcal{R}_0}_{\tau_I} = \tau_I/(\tau_I + \tau_m) \approx 0.0007$. This confuses $\nu_m/(\gamma_R + \nu_m) = \tau_I/(\tau_I + \tau_m)$, the *other* duration's share, with $\mathcal{S}^{\mathcal{R}_0}_{\tau_I}$. The reciprocal relation gives $\mathcal{S}^{\mathcal{R}_0}_{\tau_I} = -\mathcal{S}^{\mathcal{R}_0}_{\gamma_R} = \tau_m/(\tau_I + \tau_m) \approx 0.9993$ directly. An answer giving ≈ 0.9993 is complete; 0.0007 is its complement, not the index asked for. (2) Missing $\mathcal{S}^{\mathcal{R}_0}_{\tau_m}$ entirely (five indices are required). (3) Writing $\mathcal{R}_0 \approx c\beta\tau_I\tau_E$ as the demographic limit; the correct limit is $c\beta\tau_I$ (the τ_E factor drops out as $\nu_m \to 0$). At minimum: correct formulas and correct signs for all five SIs. A complete answer: complete table, $\mathcal{S}^{\mathcal{R}_0}_{\tau_I} \approx 0.9993$, and the observation that τ_I ranks alongside c and β as a dominant parameter.

Exercise 2.6 *[Bloom L4]*

Exercise statement. A published paper reports $\mathcal{S}^{\mathcal{R}_0}_{\gamma_R} = -1.0$ for an SIR model, where γ_R is the recovery rate in units of day^{-1}. A second paper on the same model reports $\mathcal{S}^{\mathcal{R}_0}_{\tau_R} = +1.0$, where $\tau_R = 1/\gamma_R$ is the mean infectious period in days. A policy maker reading both papers is confused: one paper shows a negative index, the other shows a positive index for what appears to be the same parameter.

Write a one-paragraph explanation for the policy maker that (a) resolves the apparent contradiction, (b) explains which parameterization is more informative for policy, and (c) describes what each sensitivity index means in plain language.

Solution.

Paragraph for the policy maker:

Both papers are reporting the same underlying biology, but in different parameterizations. The first paper uses γ_R, the recovery rate (units: day^{-1}), while the second uses $\tau_R = 1/\gamma_R$, the mean infectious period (units: days). For the SIR model, $\mathcal{R}_0 = c\beta/\gamma_R = c\beta\tau_R$, so increasing γ_R decreases $\mathcal{R}_0$ (negative index), while increasing τ_R increases $\mathcal{R}_0$ (positive index). The reciprocal relationship $\tau_R = 1/\gamma_R$ means that "γ_R increases" and "τ_R decreases" describe the same biological event: faster recovery. Both papers are correct; the sign difference reflects the direction of the parameterization, not a scientific disagreement.

For policy, the duration parameterization (τ_R) is more informative: public health officers think in terms of "the average patient is infectious for 7 days," not "the recovery rate is 0.14 per day." A positive $\mathcal{S}^{\mathcal{R}_0}_{\tau_R} = +1$ communicates directly that reducing the infectious period by 10% (through treatment or isolation) would reduce $\mathcal{R}_0$ by 10%. Reporting should use the duration parameterization and state explicitly that reducing τ_R reduces $\mathcal{R}_0$.

Checking your answer: The one-paragraph format is deliberate: the exercise tests whether students can synthesize the parameterization concept into a clear, jargon-free explanation. Writing equations without a clear explanation for a non-technical reader falls short. The policy-relevance component (which parameterization is more actionable) is essential for a complete answer.

Exercise 2.7 *[Bloom L4]*

Exercise statement. The following tornado plot data is given for a food-safety model with QOI J = mean number of illness cases per outbreak. The sensitivity indices are: $\mathcal{S}^J_{\tau_c} = +2.1$ (mean contamination duration, days), $\mathcal{S}^J_{\beta_f} = +1.8$ (food-to-person transmission probability), $\mathcal{S}^J_{\tau_s} = +1.3$ (mean hand-washing interval, days), $\mathcal{S}^J_{k_f} = +0.7$ (food contacts per day), $\mathcal{S}^J_{\nu_m} = -0.2$ (background clearance rate).

(a) Produce the tornado plot using the algorithm template from Section 2.7.

(b) Rank the five parameters by absolute importance.

(c) A public health program can achieve a 15% reduction in either τ_c or a 20% reduction in k_f. Which intervention reduces J more? By how much?

(d) The parameter ν_m has a small negative index. Write a one-sentence interpretation of what this means for disease control.

Solution.

The sensitivity indices (sorted by $|\cdot|$, descending):

POI	SI	|SI|
τ_c (contamination duration)	$+2.1$	2.1
β_f (food-to-person transmission)	$+1.8$	1.8
τ_s (hand-washing interval)	$+1.3$	1.3
k_f (food contacts per day)	$+0.7$	0.7
ν_m (clearance rate)	-0.2	0.2

(a) **Pseudocode** (reproduced in the companion notebooks):

```
# Inputs
SI     = [+2.1, +1.8, +1.3, +0.7, -0.2]
labels = ["tau_c", "beta_f", "tau_s", "k_f", "nu_m"]
QOI    = "J = mean illness cases"

# Sort by |SI| descending, then draw horizontal bar chart
order  = argsort(abs(SI), descending=True)
TORNADO_PLOT(SI[order], labels[order], QOI)
# Positive bars extend right; negative bars extend left.
```

The resulting plot has τ_c as the widest bar (top), extending right to $+2.1$; ν_m as the shortest bar (bottom), extending left to -0.2.

(b) **Ranking by absolute SI.**

Parameter	SI	Rank (by |SI|)
τ_c (contamination duration)	$+2.1$	1st
β_f (food-to-person transmission)	$+1.8$	2nd
τ_s (hand-washing interval)	$+1.3$	3rd
k_f (food contacts per day)	$+0.7$	4th
ν_m (background clearance rate)	-0.2	5th

(c) **Comparing interventions:** 15% **reduction in** τ_c **vs.** 20% **reduction in** k_f**.**

Local SA predicts $\Delta J \approx \mathcal{S}_p^J \cdot (\Delta p/p)$.

Reducing τ_c by 15%: $\Delta J \approx (+2.1) \times (-0.15) = -0.315$, a **31.5**% reduction in illness cases.

Reducing k_f by 20%: $\Delta J \approx (+0.7) \times (-0.20) = -0.14$, a **14**% reduction in illness cases.

The τ_c **intervention wins by a factor of** ~ 2.25 ($31.5\%/14\% \approx 2.25$).

The local SA prediction should be treated as an approximation; it is reliable when the intervention is small ($\leq 20\%$) and the model is smooth in that parameter.

(d) **Interpreting** $\mathcal{S}_{\nu_m}^J = -0.2$**.**

A higher background clearance rate ν_m means contaminated food or bacteria are removed from the environment more rapidly, reducing the opportunity for transmission. Illness cases therefore fall as ν_m increases, giving a negative SI.

The magnitude $|\mathcal{S}_{\nu_m}^J| = 0.2$ indicates that ν_m is the least influential parameter in the model: a 10% increase in ν_m reduces cases by only about 2%. This is the one parameter where environmental interventions (e.g., faster decontamination) have the smallest marginal return compared with targeting τ_c or β_f.

Checking your answer: **(b) Ranking:** τ_c (2.1), β_f (1.8), τ_s (1.3), k_f (0.7), ν_m (0.2) by absolute SI. A complete answer requires ranking by |SI|, not by SI value, with the negative SIs placed correctly by magnitude. **(c) Intervention comparison:** Local SA predicts $\Delta J \approx \mathcal{S}_p^J \cdot (\Delta p/p)$. For 15% reduction in τ_c: $\Delta J \approx (+2.1)(-0.15) = -0.315$ (31.5% fewer cases). For 20% reduction in k_f: $\Delta J \approx (+0.7)(-0.20) = -0.14$ (14% fewer cases). The τ_c intervention wins by a factor of ~ 2.25. At minimum: correct direction for both with at least one SI product computed. **(d) Sign of** $\mathcal{S}_{\nu_m}^J$**:** $\mathcal{S}_{\nu_m}^J = -0.2$ means a 1% increase in background clearance rate reduces mean illness cases by 0.2%, a modest but consistent effect. **Sign-convention warning:** students frequently write "the negative sign means ν_m has no effect" (wrong, magnitude determines importance, sign determines direction) or "increasing the hand-washing interval helps" (wrong direction: a larger interval means less frequent washing, hence more transmission. The positive $\mathcal{S}_{\tau_s}^J = +1.3$ says exactly that, cases *rise* as τ_s rises, so the sign and the biology agree. The error is in the student's reading, not in the data). **The distinguishing check:** a student who correctly identifies both which intervention reduces J more *and* computes the relative magnitudes is complete for (c). Part (c) is also a teaching moment: SI > 1 does not indicate an error, for super-linear models ($J \propto \tau_c^2$), SI $= 2$, not 1.

Exercise 2.8 *[Bloom L5]*

Exercise statement. You are advising the World Health Organization on a vaccine allocation strategy for a new respiratory pathogen. A compartmental model has been fitted to early outbreak data. The model has six parameters; their sensitivity indices with respect to the QOI "total deaths at 6 months" are not yet known.

Design the SA study. Your design should specify:

(a) Which parameterization you will use (rates or durations) for each parameter, and why.

(b) What QOI or QOIs you would compute sensitivity indices for, and what additional QOI you might consider beyond the stated one.

(c) How you would present the results to a non-mathematical audience (what figure or table, and what the key message would be).

(d) One limitation of the local SA approach that the WHO advisors should be aware of in using these results.

Solution.

Model answer (design for a six-parameter respiratory model).

(a) **Parameterization.** Use duration parameterization throughout: mean latent period τ_E, mean infectious period τ_I, mean immunity duration τ_R (if applicable), and mean residence time τ_m in days. Report contact rates c (contacts/person/day) and transmission probability β as dimensionless or rate parameters but interpret sensitivity indices in terms of "a 10-day increase in τ_I" rather than "a 0.14 decrease in γ_R." Rationale: the WHO team includes epidemiologists and clinicians; duration units are actionable (isolation protocols target τ_I; vaccine protection duration targets τ_R).

(b) **QOI selection.** Primary: total deaths at 6 months (directly relevant to allocation decisions). Secondary: $\mathcal{R}_0$ (determines whether epidemic occurs at all, a natural threshold). Optional: peak hospitalization rate (determines whether healthcare capacity is overwhelmed). Using multiple QOIs catches cases where the most important parameter for $\mathcal{R}_0$ is different from the most important parameter for peak hospitalization.

(c) **Presentation to a non-mathematical audience.** Present a tornado plot with bars labeled in plain language: "Mean infectious period: a 10-day increase raises deaths by 18%." Avoid SI notation entirely; rephrase each bar as "a realistic change in [plain-language parameter] changes deaths by approximately x%." Key message (one sentence): "The duration of infectiousness and the contact rate are the two parameters the model says matter most; other parameters, including the waning immunity duration, have minor effects at current estimates." Embed a caveat: these rankings hold near the fitted parameter values and may shift under large uncertainty.

(d) **Limitation.** Local SA is evaluated at a single fitted parameter set. For a new pathogen, the true parameters may be far from the fitted values. The SI rankings might change substantially across the plausible parameter range. This limitation should be reported explicitly; any parameter whose SI changes sign or magnitude substantially across a 50% confidence interval around the fitted values should be flagged as unreliable for prioritization.

Checking your answer: A complete answer covers all of: (a) justification of parameterization choice; (b) motivated QOI with at least one additional beyond the stated one; (c) specific figure or table with a one-sentence key message aimed at a non-specialist; (d) genuine limitation of local SA stated precisely (not just "the model might be wrong"). Common omission: students who include only the primary QOI in (b) without noting that different QOIs may give different rankings, and students whose (c) presentation retains mathematical notation unsuitable for a WHO policy audience.

Exercise 2.9 *[Bloom L4]*

Exercise statement. *Audit exercise.* A report states that the sensitivity index of $\mathcal{R}_0$ with respect to the contact rate c is $+1$, that the relationship is therefore linear, and that doubling c doubles $\mathcal{R}_0$. Identify the error; construct a nonlinear counterexample with index $+1$; state the extra condition under which the claim holds for $\mathcal{R}_0 = c\beta\tau_R$; rewrite the passage.

Solution.

(a) The index $\mathcal{S}_c^{\mathcal{R}_0} = +1$ is a statement about the derivative at the nominal point, normalized by the nominal values. It asserts that a *small* fractional change in c produces an equal fractional change in $\mathcal{R}_0$, which is unit elasticity. It says nothing about behavior away from the nominal point. Linearity is a global property of the function; unit elasticity is a local property of its logarithmic derivative. The report has substituted one for the other.

(b) Take $f(c) = c^1$ composed with any distortion that preserves the log-slope at the nominal point but not away from it. A clean choice is $f(c) = c\, e^{\alpha(c-c_0)^2}$ with $\alpha \neq 0$. At $c = c_0$,

$$\mathcal{S}_c^f = \frac{c}{f}\frac{df}{dc} = 1 + 2\alpha c_0(c_0 - c_0) = 1,$$

so the index is exactly $+1$, yet f is not linear and $f(2c_0) \neq 2f(c_0)$. With $c_0 = 4$ and $\alpha = 0.01$, doubling gives $f(8)/f(4) = 2e^{0.16} \approx 2.35$, an error of about 17% in the report's doubling claim. Any α may be chosen to make the discrepancy arbitrarily large, which is the point: the index alone cannot bound it.

(c) For the monomial $\mathcal{R}_0 = c\beta\tau_R$ the doubling claim is correct provided β and τ_R are held fixed while c is varied. The index is $+1$ for every monomial of degree one in c, and for a monomial the local statement happens to extend globally. The report's error is not that its conclusion is false here, but that it derived a global conclusion from a local quantity, so the reasoning would fail for any model that is not a monomial in the parameter of interest.

(d) A corrected passage, no longer than the original: "The sensitivity index of $\mathcal{R}_0$ with respect to c is $+1$, so a small percentage increase in the contact rate produces the same percentage increase in $\mathcal{R}_0$. Because $\mathcal{R}_0 = c\beta\tau_R$ is a monomial in c, this proportionality also holds for large changes when β and τ_R are fixed."

Checking your answer: The distinction between unit elasticity and linearity is the target of this exercise, and the report's arithmetic is correct throughout, which is why the error survives review. A complete answer in part (b) must exhibit a function and evaluate the doubling error; stating that "the index is local" without a counterexample is partial. Part (c) is where strong answers separate: the claim happens to be true for this particular $\mathcal{R}_0$, and recognizing that a valid conclusion can rest on invalid reasoning is the more durable lesson.

Exercise 2.10 *[Bloom L6]*

Exercise statement. A modeling study reports the following: "Sensitivity analysis of $\mathcal{R}_0$ was performed with respect to all model parameters. The parameters are listed in Table 3 using the standard model notation. All sensitivity indices were positive, confirming that $\mathcal{R}_0$ is an increasing function of all model parameters."

Table 3 lists recovery rate γ_R, mortality rate ν_m, and waning immunity rate ρ alongside transmission and contact parameters.

Identify the error in the study's statement and explain specifically what is wrong, referring to the mathematical relationship between rates and their reciprocals. Restate the conclusion correctly.

Solution.

The error: The recovery rate γ_R, mortality rate ν_m, and waning immunity rate ρ are all *rates* (units: day^{-1}); increasing them corresponds to faster recovery, faster death, and faster waning, respectively. For standard epidemic models:

- $\mathcal{S}_{\gamma_R}^{\mathcal{R}_0} = -1$ (faster recovery reduces $\mathcal{R}_0$); *negative.*
- $\mathcal{S}_{\nu_m}^{\mathcal{R}_0}$ is typically negative (background mortality reduces effective infectious period).
- $\mathcal{S}_{\rho}^{\mathcal{R}_0}$ for the standard SIRS model is *exactly zero*: waning immunity returns recovered individuals to the susceptible class but does not change how many secondary infections one infective produces in a wholly susceptible population, so ρ does not appear in $\mathcal{R}_0$ at all. It affects the endemic equilibrium, not the threshold. A study reporting a nonzero $\mathcal{S}_{\rho}^{\mathcal{R}_0}$ has either used a nonstandard formulation, and should say so, or has made an error.

The claim that "all sensitivity indices were positive" is inconsistent with these mathematical relationships for at least γ_R and ν_m.

What specifically is wrong: The authors either (a) computed derivatives without the normalization factor p/q (so they computed $\partial\mathcal{R}_0/\partial\gamma_R < 0$, giving negative sign, but then reported absolute values); or (b) reparameterized implicitly and are reporting $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = +1$ (which is positive) without acknowledging the parameterization switch; or (c) made a coding error that reversed the sign. Option (b) is most charitable and most common.

What should have been done: The paper should state explicitly which parameterization was used (rates or durations) and report the actual signed SI values. If duration parameterization was used, the table should identify each parameter as a rate or duration to avoid confusion. A statement that "all sensitivity indices were positive" without this context is ambiguous and potentially incorrect.

Checking your answer: This exercise targets a common real-world error in published papers. The key diagnostic concerns *removal* rates specifically: recovery and mortality rates enter

$\mathcal{R}_0$ through the denominator, so raising them shortens the infectious period and lowers $\mathcal{R}_0$, giving a negative index. This is not a property of rates in general. Contact, transmission, progression, and recruitment rates all sit in the numerator and carry *positive* indices. Students should be required to read the sign off the formula rather than off the word "rate". Students who only identify "wrong sign" without explaining *why* should are partial. A complete answer requires identifying the mechanism (normalization, parameterization, or sign error).

3 Chapter 3: Computing Sensitivities Numerically

Chapter Summary: Computing Sensitivity in Practice

The sensitivity Jacobian. The complete local SA of a model with n_p POIs and n_q QOIs is an $n_q \times n_p$ matrix $S_{ki} = \mathcal{S}_{p_i}^{q_k}$. Each entry answers one "What if?" question. Computing this matrix efficiently is the central computational challenge of SA.

Centered finite differences. The most accurate finite-difference approximation is

$$\mathcal{S}_p^q \approx \frac{p}{q} \cdot \frac{q(p+h) - q(p-h)}{2h},$$

with error $O(h^2)$ from truncation and $O(\varepsilon_{\text{mach}}/h)$ from rounding. The optimal h balances these two: $h_{\text{opt}} \approx (3\varepsilon_{\text{mach}}/|q'''|)^{1/3} p^*$. In practice, the "U-curve" workflow (try several h values, look for the flat region where error is small) is more reliable than the formula.

ode solver tolerance rule. Before computing finite-difference sensitivities for a model that calls an ODE solver internally, tighten solver tolerances to at least $10\times$ smaller than the target step size h. A practical default: `RelTol = AbsTol = 1e-8` whenever SA is the goal.

Correlated pois. If two POIs always move together (e.g., c and β appear only as $c\beta$ in the SIR model), their individual sensitivity indices are misleading. Diagnose with the condition number $\kappa(S)$ of the sensitivity matrix or the variance inflation factor (VIF). Remedy: reparameterize so the correlated pair enters as a single POI.

Watch out: The U-curve is a diagnostic, not a guarantee. If the curve has no flat region, if it slopes up and then up again instead of forming a U; the model output is non-smooth in p near p^*, and finite differences cannot reliably estimate the sensitivity. Check for discontinuities (if statements, lookup tables) before trusting any FD result.

Statistics and ML bridge. The U-curve (optimal step-size plot) is the finite-difference analog of the bias-variance tradeoff in machine learning: large h gives high bias (truncation error), small h gives high variance (roundoff amplification). The optimal step size minimizes total error, just as the optimal model complexity minimizes test error. The VIF (variance inflation factor) in regression is the direct analog of the reparameterization condition number: $\text{VIF}_j > 10$ signals the same collinearity that makes SA unreliable in correlated-parameter models.

Exercise 3.1 *[Bloom L2]*

Exercise statement. Explain in your own words why making the step size h smaller does not always improve the accuracy of a finite-difference sensitivity estimate. Describe the two sources of error and explain why they act in opposite directions as h changes.

Solution.

There are two opposing sources of error in a finite-difference estimate. As h decreases, one gets better but the other gets worse.

Truncation error arises because a finite difference approximates a derivative by a secant line. For centered differences, this error is proportional to h^2: smaller h gives a better approximation. So truncation error *decreases* as h decreases.

Roundoff error arises because a computer stores numbers with finite precision (about 16 significant digits in double precision). When h is very small, $q(p+h)$ and $q(p-h)$ are nearly equal, and subtracting two nearly equal numbers destroys most of the significant digits; a phenomenon called catastrophic cancellation. Roundoff error is proportional to $\varepsilon_{\text{mach}}/h$ for centered differences, so it *increases* as h decreases.

The total error is the sum of both contributions, forming a U-curve (or "hockey stick" on log-log axes): the optimal h occurs where truncation and roundoff errors are equal, which for centered differences is approximately $h^* \approx \varepsilon_{\text{mach}}^{1/3} |p^*| \approx 6 \times 10^{-6} |p^*|$ in double precision.

Checking your answer: Students often describe only one of the two sources. A complete answer requires both, with correct qualitative behavior (one decreasing, one increasing with h).

When to use finite differences in 2026. The 2026 question is: when should you use FD rather than automatic differentiation (AD, Chapter 7) or analytic FSE (Chapters 4–6)? *Use FD when*: (1) the model is a black-box simulator without source-code access (compiled Fortran/C, commercial solver); (2) the analytic Jacobian is intractable or error-prone to derive manually; or (3) you need a method-independent cross-check (FD always works as a verifier, even when you also have the adjoint). *Use AD or FSE when* you have source-code access, because they provide exact gradients at comparable cost with no step-size selection. The U-curve analysis in this chapter is the first step in diagnosing whether your FD computation is in the reliable regime.

Exercise 3.2 *[Bloom L3]*

Exercise statement. A model has $n_p = 15$ POIs and $n_q = 4$ QOIs.

(a) Fill in Table 3.1 for this model: how many model evaluations does each method require?

(b) If each evaluation takes 30 seconds, how long does the centered-difference Jacobian take?

(c) At what point (in terms of n_p and evaluation time) would the adjoint method from Chapter 6 be strictly necessary?

Solution.

The central insight: model evaluation count depends on n_p alone, not on n_q, because all QOIs are extracted simultaneously from the same perturbed model run.

(a) For $n_p = 15$ POIs, $n_q = 4$ QOIs:

Method	Formula	Evaluations
Forward difference (all SIs)	$n_p + 1$	16
Centered difference (all SIs)	$2n_p + 1$	31
Forward difference Jacobian	$n_p + 1$	16
Centered difference Jacobian	$2n_p + 1$	31

The count is the same regardless of n_q: each method evaluates the model at the same set of perturbed parameter vectors, and all n_q QOIs are extracted from each evaluation simultaneously. Adding QOIs costs no additional model evaluations.

(b) Centered-difference Jacobian: 31 evaluations at 30 seconds each = 930 seconds $\approx$ 15.5 minutes. At this cost, the full sensitivity Jacobian is a practical afternoon calculation on a laptop, but scaling to $n_p = 100$ would require roughly $201 \times 30 = 6030$ seconds ($\sim$100 minutes) at the same evaluation cost.

(c) The adjoint approach (Chapter 6) reduces cost to one forward solve plus one backward solve per QOI, so $n_q + 1$ evaluations in all, in this case 5. The backward solves are already counted in that total; each costs about as much as one forward solve. The adjoint is strictly necessary when n_p is so large that even the centered-difference cost $2n_p + 1$ is prohibitive. As a practical threshold: when $2n_p \times t_{\text{eval}}$ exceeds your computation budget (e.g., more than a few hours on a laptop), use the adjoint. For $t_{\text{eval}} = 30$ s, this occurs at roughly $n_p > 180$ (when centered FD would take $(2 \times 180 + 1) \times 30 \approx 3$ hours).

Checking your answer: The key insight for part (a): the number of evaluations depends only on n_p, not n_q. Students who write $n_p \times n_q$ evaluations are confusing the Jacobian dimension with the evaluation count. A complete answer states that the count scales with n_p alone and gives the number; at minimum, a correct count for the stated n_p. The Jacobian is assembled from the $2n_p + 1$ evaluations by extracting all n_q outputs from each run.

Exercise 3.3 *[Bloom L3]*

Exercise statement. For the function $q(p) = \sin(p^2)$ at $p^* = 1.3$, compute $\partial q/\partial p$ using centered differences for $h \in \{10^{-1}, 10^{-2}, \ldots, 10^{-14}\}$. Compare each estimate to the exact value $q'(p) = 2p\cos(p^2)$. Plot error vs. h on log-log axes and identify the optimal h. Does it agree with the formula in equation (3.9)?

Solution.

Setup. Let $q(p) = \sin(p^2)$ at $p^* = 1.3$. Exact derivative: $q'(p^*) = 2p^* \cos((p^*)^2) = 2(1.3)\cos(1.69)$. Since $\cos(1.69) \approx -0.1189$, the exact derivative is $2(1.3)(-0.1189) \approx -0.3092$.

Algorithm:

```
# Centered finite-difference error study for q(p) = sin(p^2) at p* = 1.3
p_star   = 1.3
q(p)     = sin(p^2)              # model function
dq_exact = 2 * p_star * cos(p_star^2)   # = -0.3092

h_vals = [1e-1, 1e-2, ..., 1e-14]       # 14 step sizes

FOR each h in h_vals:
    dq_fd   = (q(p_star + h) - q(p_star - h)) / (2*h)  # centered FD
    err[h]  = |dq_fd - dq_exact|

PLOT log(h_vals) vs log(err)   # should show V-shape (U-curve)
# Right branch (large h): slope +2 (truncation ~ h^2)
# Left  branch (small h): slope -1 (roundoff ~ eps_mach / h)
```

Expected results. On the truncation branch the error is deterministic, $h^2|q'''(p^*)|/6 \approx 2.23\,h^2$ here, so those rows are reproducible. On the roundoff branch only the order of magnitude is, since the realized cancellation varies.

h	|error| (approximate)	Note
10^{-1}	2.2×10^{-2}	Truncation dominates
10^{-3}	2.2×10^{-6}	Truncation dominates
10^{-6}	$\sim 10^{-11}$	Near optimum
10^{-8}	$\sim 10^{-9}$	Roundoff increasing
10^{-12}	$\sim 10^{-4}$	Roundoff dominates
10^{-14}	$\sim 10^{-2}$	Catastrophic cancellation

Optimal h: The formula gives $h^* \approx \varepsilon_{\text{mach}}^{1/3}|p^*| \approx (2.2 \times 10^{-16})^{1/3}(1.3) \approx 6.1 \times 10^{-6} \times 1.3 \approx 7.9 \times 10^{-6}$. The empirical minimum of the error plot should occur at $h \approx 10^{-5}$ to 10^{-6}, which agrees with the formula to within one decade (the precise minimum depends on the third derivative of q).

Checking your answer: On log-log axes with h increasing to the right, students should observe slope $+2$ for $h > h^*$, the truncation branch where the error $\propto h^2$ falls as h is reduced, and slope -1 for $h < h^*$, the roundoff branch where the error $\propto 1/h$ rises as h is reduced further. Measured on this problem the two slopes come out at $+2.00$ and about -1.2; the roundoff branch is noisy because the realized cancellation varies. A complete answer reports both slopes with their sources, truncation and roundoff, and locates h^* between them; at minimum, an observed h^* consistent with the formula. The optimal h from the formula should agree with the empirical minimum to within a factor of 3–10; larger disagreements indicate an algebra error.

Exercise 3.4 *[Bloom L3]*

Exercise statement. For the SIR model $\mathcal{R}_0 = c\beta\tau_R$:

(a) Compute the pure second derivatives $\partial^2\mathcal{R}_0/\partial c^2$, $\partial^2\mathcal{R}_0/\partial\beta^2$, $\partial^2\mathcal{R}_0/\partial\tau_R^2$ analytically. What do you notice?

(b) Compute the cross derivatives $\partial^2\mathcal{R}_0/\partial c\,\partial\beta$, $\partial^2\mathcal{R}_0/\partial c\,\partial\tau_R$, $\partial^2\mathcal{R}_0/\partial\beta\,\partial\tau_R$ analytically.

(c) For a 10% perturbation to c and a simultaneous 10% perturbation to β, estimate the true change in $\mathcal{R}_0$ using both the first-order and the full second-order Taylor approximations. How large is the correction?

Solution.

Let $\mathcal{R}_0 = c\beta\tau_R$.

(a) **Pure second derivatives:**

$$\frac{\partial^2\mathcal{R}_0}{\partial c^2} = 0, \qquad \frac{\partial^2\mathcal{R}_0}{\partial\beta^2} = 0, \qquad \frac{\partial^2\mathcal{R}_0}{\partial\tau_R^2} = 0.$$

All pure second derivatives are zero. This is because $\mathcal{R}_0$ is *multilinear* in (c, β, τ_R), linear in each variable separately. A function linear in c has zero second derivative in c, regardless of how it depends on the other variables.

(b) **Cross derivatives:**

$$\frac{\partial^2\mathcal{R}_0}{\partial c\,\partial\beta} = \tau_R, \qquad \frac{\partial^2\mathcal{R}_0}{\partial c\,\partial\tau_R} = \beta, \qquad \frac{\partial^2\mathcal{R}_0}{\partial\beta\,\partial\tau_R} = c.$$

All cross derivatives are nonzero. This means that c, β, and τ_R interact: the effect of changing c depends on the current value of β (and vice versa). These interactions are exactly what Sobol indices capture and local SA misses.

(c) **Taylor approximation for simultaneous 10% perturbations.**

Let $\delta c = 0.1c^*$, $\delta\beta = 0.1\beta^*$, with nominal $(c^*, \beta^*, \tau_R^*) = (5, 0.06, 7)$, $\mathcal{R}_0^* = 2.1$.

First-order approximation:

$$\Delta\mathcal{R}_0^{(1)} = \frac{\partial\mathcal{R}_0}{\partial c}\delta c + \frac{\partial\mathcal{R}_0}{\partial\beta}\delta\beta = \beta^*\tau_R^*\cdot(0.1c^*) + c^*\tau_R^*\cdot(0.1\beta^*) = 0.1\beta^*\tau_R^*c^* + 0.1c^*\tau_R^*\beta^* = 0.2c^*\beta^*\tau_R^* = 0.2\mathcal{R}_0^* = 0.42$$

Second-order correction (cross term only, since pure second derivatives are zero):

$$\Delta\mathcal{R}_0^{(2)} = \frac{\partial^2\mathcal{R}_0}{\partial c\,\partial\beta}\delta c\,\delta\beta = \tau_R^*\cdot(0.1c^*)(0.1\beta^*) = 0.01\,c^*\beta^*\tau_R^* = 0.01\mathcal{R}_0^* = 0.021.$$

Exact change:

$$\Delta\mathcal{R}_0^{\text{exact}} = (1.1c^*)(1.1\beta^*)\tau_R^* - c^*\beta^*\tau_R^* = (1.21-1)\mathcal{R}_0^* = 0.21\,\mathcal{R}_0^* = 0.441.$$

The second-order correction 0.021 is $\approx 5\%$ of the exact change 0.441. The first-order approximation gives 0.42, an error of 0.021 (the cross-derivative term). For a 10% simultaneous perturbation, the cross-term correction is small but measurable; for larger perturbations (e.g., 50%) it would be dominant.

Checking your answer: The key observation from part (a): $\mathcal{R}_0 = c\beta\tau_R$ is perfectly linear in each variable, so a purely multiplicative model has zero pure second derivatives and nonzero cross-derivatives. Common error: students confuse "the Hessian is zero" (wrong: only the diagonal is zero) with "all second derivatives are zero". At minimum: all three pure second derivatives stated as zero with the correct multilinearity argument. A complete answer: all three cross-derivatives computed and correctly identified as the interaction terms that Sobol indices capture but local SI misses. The distinguishing check: a student who says "the Hessian is zero because $\mathcal{R}_0$ is linear" has only a partial understanding; a complete answer distinguishes pure vs. cross derivatives.

Exercise 3.5 *[Bloom L4]*

Exercise statement. In the SIR model, suppose field data suggests that the contact rate c and transmission probability β are positively correlated: across outbreak settings, when c increases by 10%, β tends to increase by 6%.

(a) Compute the standard sensitivity indices $\mathcal{S}_c^{\mathcal{R}_0}$ and $\mathcal{S}_\beta^{\mathcal{R}_0}$.

(b) Using equation (3.12), compute the total-derivative SI of $\mathcal{R}_0$ with respect to c, accounting for the correlation.

(c) Which is larger: the standard SI or the total-derivative SI? What does this mean for a policy maker who can reduce contact rates (through social distancing) but not transmission probability directly?

Solution.

(a) **Standard sensitivity indices.** For $\mathcal{R}_0 = c\beta\tau_R$, using the log-derivative form:

$$\mathcal{S}_c^{\mathcal{R}_0} = 1, \qquad \mathcal{S}_\beta^{\mathcal{R}_0} = 1.$$

(b) **Total-derivative SI accounting for correlation.**

The correlation structure states: a 10% change in c is accompanied by a 6% change in β. In the notation of equation (3.12) in the text, the coupling coefficient is $c_{c\beta} = 0.06/0.10 = 0.6$ (a 1% change in c induces a 0.6% correlated change in β).

The total-derivative SI of $\mathcal{R}_0$ with respect to c, accounting for the induced change in β, is:

$$\begin{aligned}\mathcal{S}_c^{\mathcal{R}_0}\Big|_{\text{tot}} &= \mathcal{S}_c^{\mathcal{R}_0}\Big|_{\text{std}} + \mathcal{S}_\beta^{\mathcal{R}_0}\cdot c_{c\beta}\\ &= 1 + 1\times 0.6 = 1.6.\end{aligned}$$

(c) **Comparison and policy interpretation.**

The total-derivative SI (1.6) is larger than the standard SI (1.0). This means the true effect of reducing contact rates is amplified by the positive correlation with β: when social distancing reduces c, it also tends to reduce β (fewer and shorter contacts tend to reduce both quantity and riskiness of contacts). The combined effect on $\mathcal{R}_0$ is 60% larger than the standard SI would suggest.

For a policy maker: a 10% reduction in c (e.g., through school closures) would reduce $\mathcal{R}_0$ by approximately 16%, not the 10% predicted by the standard SI. This is a larger benefit. Ignoring the correlation would underestimate the effectiveness of contact-rate reduction interventions.

Checking your answer: The coupling coefficient $c_{c\beta} = 0.6$ is read directly from the problem statement: "when c increases by 10%, β tends to increase by 6%," so the ratio is $6/10 = 0.6$. A complete answer shows where 0.6 comes from before using it; at minimum, the correct coupling value with its ratio stated. Answers using 0.6 directly as $c_{c\beta}$ (rather than 0.06) are applying the coupling coefficient to normalized changes, which is correct here. The distinction matters only when c_{ij} is defined as the ratio of absolute changes rather than relative (percentage) changes.

Exercise 3.6 *[Bloom L4]*

Exercise statement. A colleague provides you with the function `seir_model(p)` that implements a SEIR epidemic model with $n_p = 6$ parameters and returns the peak infection count as a single QOI. Each model evaluation takes approximately 0.2 seconds.

(a) How long will it take to compute the complete sensitivity Jacobian using centered differences?

(b) Apply the `sensitivity_jacobian` template to this model. What additional information do you need before interpreting the results? (Think about parameterization.)

(c) You find that two of the six parameters have $|\rho_{ij}| = 0.78$. What should you do before reporting the sensitivity indices?

Solution.

(a) **Computation time for centered-difference Jacobian.**

With $n_p = 6$ POIs and $n_q = 1$ QOI (peak infections), the centered-difference Jacobian requires $2n_p + 1 = 13$ model evaluations. At 0.2 seconds each: $13 \times 0.2 = 2.6$ seconds. Entirely feasible.

(b) **Applying the sensitivity Jacobian routine.**

The one-call usage (any implementation):

```
# SENSITIVITY_JACOBIAN: compute normalized SI for all parameters
# Input:  model(p) → q   (vectorized model function)
#         p_nom          (nominal parameter vector)
#         h_scale = 1    (controls step size: h_j = h_scale * 6e-6 * |p_j|)
```

```
# Output: S[i, j] = SI_{p_j}^{q_i} (n_q × n_p matrix)
#         info          (diagnostics: step sizes, U-curve flags)

S = SENSITIVITY_JACOBIAN(model, p_nom, h_scale=1)
```

Before interpreting results, you need: (1) *parameterization*: are the 6 parameters rates or durations? Rate-parameterized models give negative SIs for recovery rates, duration-parameterized models give positive SIs for the same biological quantities. The interpretation of sign and magnitude depends on this. (2) *nominal values*: what are p^* and $q^* = \texttt{seir\_model}(p^*)$? The SI is evaluated at these values; if they are far from the plausible range, the result may not be representative. (3) *units*: what does each parameter represent, and in what units, to give the SI a meaningful interpretation.

(c) **Response to $|\rho_{ij}| = 0.78$.**

A correlation of 0.78 between two parameters indicates structural or empirical correlation. Before reporting standard SIs, you should: (1) Check whether the correlation is *structural* (the model forces the parameters to co-vary, e.g., c and β always appear as $c\beta$) or *empirical* (field data shows them correlated). (2) Compute total-derivative SIs using equation (3.12) to account for the correlation. (3) Consider whether the two parameters are actually identifiable separately from available data. If $|\rho| > 0.7$, a single effective parameter (their product) may be more informative than two individual SIs. Report both standard and total-derivative SIs, with a note on the correlation and its source.

Checking your answer: Part (b) is designed to elicit "parameterization" as the critical missing information. A complete answer names parameterization and says why it changes the sign; at minimum, identifying that something beyond the SI value is needed. Students who simply say "the sign of the SI" without explaining why parameterization matters should receive a partial answer. The key point: `seir_model(p)` is a black box, and you need domain knowledge to interpret the direction and magnitude of the indices. Part (c) previews the identifiability material from the structural identifiability section of Chapter 3.

Exercise 3.7 *[Bloom L5]*

Exercise statement. A government agency funds you to perform SA on a dengue fever transmission model with 9 parameters and 3 QOIs (peak infections, total deaths, days until outbreak peak). Each model evaluation takes 4 minutes.

Design the complete computational SA study. Your design should specify:

(a) Which finite-difference method you will use and why.

(b) The total computation time.

(c) How you will choose the step size for each parameter.

(d) What checks you will run before reporting any results.

(e) How you will present the results to a non-technical audience.

(f) One scenario in which the standard SA would give misleading results, and what you would do instead.

Solution.

Design for dengue SA study: 9 parameters, 3 QOIs, 4 min/evaluation.

(a) **Method: centered differences.**

Centered differences are preferred over forward differences because the second-order accuracy ($O(h^2)$ vs $O(h)$) justifies the $2\times$ cost. With 9 parameters and 4 minutes per evaluation, the centered-difference Jacobian costs $2(9) + 1 = 19$ evaluations $\times$ 4 min = 76 minutes. Forward differences cost $(9 + 1) = 10$ evaluations $\times$ 4 min = 40 minutes, saving $76 - 40 = \mathbf{36}$ minutes, but introduce $O(h)$ truncation error, which may matter if any SIs are near zero (impossible to distinguish genuine zero from truncation error at single precision).

(b) **Total computation time.**

Centered-difference Jacobian: $(2 \times 9 + 1) \times 4 = 76$ minutes for all 3 QOIs simultaneously. The 3 QOIs add no extra evaluations. Total: **76 minutes.**

(c) **Step-size selection.**

For each parameter p_j, use $h_j = 6 \times 10^{-6}\,|p_j^*|$ (equation (3.8) from the text). Apply the U-curve check: run the Jacobian at three step sizes $h_j/10$, h_j, $10h_j$ and verify that the SI values agree to 3 significant figures at h_j and $10h_j$ (truncation error is small) but differ from $h_j/10$ values (roundoff is not yet dominating). If not, adjust.

(d) **Pre-reporting checks.**

1. *Analytic verification*: Compute $\mathcal{S}_c^{\mathcal{R}_0}$ and $\mathcal{S}_\beta^{\mathcal{R}_0}$ analytically (both = 1 for a standard dengue model). Verify that the code gives the same values within 10^{-5}.
2. *Sum check (Euler's theorem)*: For a function q that is homogeneous of degree d in its parameters (i.e., scaling all parameters by λ scales q by λ^d), Euler's theorem gives $\sum_j \mathcal{S}_{p_j}^q = d$. For example, for the SIR model with $\mathcal{R}_0 = c\beta\tau_R$ (homogeneous degree $d = 3$ in (c, β, τ_R)), the correct check is $\sum_j \mathcal{S}_{p_j}^{\mathcal{R}_0} = 3$. *Warning*: the sum equals 1 only if the model is degree 1 in all parameters simultaneously, which is rarely the case for epidemic reproduction numbers. Using $\sum SI = 1$ as a check on a degree-3 model will cause you to reject correct computations. Verify that the sum equals the degree of homogeneity of $\mathcal{R}_0$ in the sensitivity parameters.
3. *Sign check*: Parameters that appear in numerator of $\mathcal{R}_0$ should have positive SI; those in denominator (rates) negative. Verify signs match biological expectations.
4. *Structural correlation check*: Compute pairwise correlation of parameter estimates from field data. Report any $|\rho_{ij}| > 0.5$.

(e) **Presentation to non-technical audience.**

Use tornado plots for each QOI, with parameter names in plain language ("mean mosquito lifespan" not "$1/\mu_v$"). State the practical interpretation: "A 10% reduction in [parameter] would reduce [QOI] by approximately [SI $\times$ 10]%." Highlight the top 2–3 parameters for each QOI. Report the limitation that these results apply near the nominal parameter values.

(f) **Scenario where standard SA gives misleading results.**

If the model exhibits a threshold ($\mathcal{R}_0 = 1$), and the nominal parameters place $\mathcal{R}_0$ just above 1, the SIs for peak infections may be very large (the epidemic is sensitive to any perturbation that crosses the threshold). A modeler might conclude that one parameter dominates all others, when in reality the result is an artifact of the proximity to threshold. In this scenario, a global SA or a sensitivity analysis of $\mathcal{R}_0$ itself (rather than peak infections) would give more stable, interpretable results.

Checking your answer: A complete answer covers all six: (a) centered vs. forward with justification, (b) correct cost calculation, (c) specific step-size procedure, (d) at least two specific checks with how-to details, (e) tornado plot plus plain-language interpretation, (f) genuine threshold scenario. A common shortfall is missing the 3-QOI simultaneous extraction in part (b).

Exercise 3.8 *[Bloom L4]*

Exercise statement. *Audit exercise.* A colleague computes forward-difference sensitivities in double precision at relative step $h = 10^{-12}$, chosen to make truncation error negligible. Explain why the conclusion fails; estimate the total error and compare with the near-optimal step; explain why smooth, consistent output does not detect the problem and give a diagnostic; and state what the one-at-a-time indices mean for two strongly correlated parameters.

Solution.

(a) The reasoning treats truncation error as the only error present. The total error in a finite difference is the sum of truncation error, which falls with h, and roundoff error, which grows as h shrinks because the subtraction $f(x+h) - f(x)$ cancels leading digits. Driving h down does make truncation negligible and simultaneously makes roundoff dominant. The U-curve of Section 3.3 is exactly this tradeoff, and $h = 10^{-12}$ sits far down its left branch.

(b) For a well-scaled f in double precision, machine epsilon is $\varepsilon \approx 2.2 \times 10^{-16}$, and the roundoff contribution to a forward difference is of order ε/h. At $h = 10^{-12}$ this gives roughly 2×10^{-4}, so only about four significant digits of the derivative survive out of the sixteen the arithmetic carries. The forward-difference optimum is near $h \approx \sqrt{\varepsilon} \approx 1.5 \times 10^{-8}$, where the total error is of order $\sqrt{\varepsilon} \approx 10^{-8}$. The chosen step is therefore about four orders of magnitude worse than the achievable accuracy.

(c) Roundoff-corrupted derivatives are finite, of plausible magnitude, and reproducible for a fixed input, so they carry no visible signature. Internal consistency is likewise no evidence, since the same cancellation affects every parameter similarly. The diagnostic is the U-curve itself: recompute the index across several decades of h and plot the result. A converged region appears as a plateau, and its absence, or an estimate that drifts as h decreases, identifies the problem. A second useful check is to compare against a central difference or, where the model permits, an analytic or algorithmic derivative.

(d) A one-at-a-time index perturbs one parameter while holding the others fixed, and it answers the question of how the QOI responds to that parameter alone. When two parameters are strongly correlated in the calibration data, a perturbation of one without the other corresponds to a combination the data never supports, so the index remains a correct statement about the model and ceases to be a useful guide to what would happen in the fitted system. It does not measure the joint effect, and the indices cannot be added. Section 3.5

treats the correlated case, and the appropriate response is either to reparameterize into independent combinations or to perturb along the correlation structure.

Checking your answer: The most common incomplete answer stops at part (a) with the correct observation that roundoff exists, without quantifying it. The order-of-magnitude estimate in part (b) is what turns the observation into an audit. Part (c) is the heart of the exercise: the reason this error persists in practice is that the corrupted output looks entirely normal, and students should leave with the habit of varying h rather than trusting a single value. Part (d) is included because a step-size audit that ignores a violated independence assumption has checked the arithmetic and missed the larger problem.

Exercise 3.9 *[Bloom L6]*

Exercise statement. A published sensitivity analysis reports: "All finite-difference computations used $h = 10^{-12}$ to ensure maximum precision. The code was run in double-precision floating-point arithmetic."

The authors report that for several parameters, the sensitivity indices are exactly zero to all reported decimal places.

Identify what is most likely wrong with these results and explain your reasoning quantitatively, using the U-curve analysis from Section 3.3. What should the authors have done, and how would you determine whether the zero SI values are genuine or numerical artifacts?

Solution.

What is most likely wrong: The step size $h = 10^{-12}$ is far into the catastrophic cancellation regime for centered differences in double precision. Double precision has $\varepsilon_{\text{mach}} \approx 2.2 \times 10^{-16}$, and the optimal step size for centered differences is $h^* \approx \varepsilon_{\text{mach}}^{1/3}|p^*| \approx 6 \times 10^{-6}|p^*|$.

With $h = 10^{-12}$ and a typical parameter value $|p^*| \sim 1$, the roundoff error in the centered difference is approximately:

$$\text{roundoff error} \approx \frac{2\varepsilon_{\text{mach}}|q^*|}{h} = \frac{2(2.2 \times 10^{-16})|q^*|}{10^{-12}} = 4.4 \times 10^{-4}|q^*|.$$

Meanwhile the true derivative (if SI ≈ 1) is $|q^*/p^*|$, so the relative error in the SI is $\sim 4.4 \times 10^{-4}$. But more critically, for parameters with a genuinely small SI (say $|\mathcal{S}^q_{p_j}| \sim 10^{-3}$), the roundoff error completely swamps the signal. The computed SI would be dominated by numerical noise, and finite-precision floating-point arithmetic would return a value that is effectively 10^{-4} to 10^{-3} in magnitude, not zero, but not interpretable as the true SI either.

The reported zeros are almost certainly numerical artifacts. The most likely scenario: the computed value $q(p + h) - q(p - h)$ for a parameter with small but nonzero SI is on the order of 10^{-16} (below machine epsilon), which rounds to exactly zero in double precision subtraction.

What the authors should have done:

1. Use $h^* \approx 6 \times 10^{-6}|p^*|$ for centered differences (the formula in Section 3.2 of the text).
2. Run a U-curve check for each parameter: vary h over 10^{-1} to 10^{-14} and plot the resulting SI vs h. An SI that is stable across several decades of h near h^* is reliable; one that varies wildly is not.

Determining whether zero SIs are genuine: For each parameter that gives SI $= 0$: (1) check analytically whether the parameter appears in the QOI expression at all (if absent, SI $= 0$ is exact); (2) repeat at $h = 10^{-5}|p^*|$; if the SI is still zero or $< 10^{-6}$, it may be genuine; (3) compare with a symbolic calculation of $\partial q/\partial p_j$ if the model allows it.

Checking your answer: A complete answer computes the roundoff error at $h = 10^{-12}$ rather than asserting the step is too small; at minimum, naming roundoff as the mechanism. That quantitative reasoning is essential. Students who only say "the step size is too small" without the calculation should receive a partial answer. The crux: $h = 10^{-12}$ is 6 orders of magnitude below the optimal, making the roundoff error 10^6 times larger than necessary. Reported zero SI values at this step size are meaningless.

4 Chapter 4: Analytic Forward Sensitivity Analysis

Chapter Summary: Analytic Forward Sensitivity Analysis

The fse derivation pattern. Three steps, always the same: (1) Write the governing equation $\mathcal{F}(u; p) = 0$. (2) Differentiate with respect to p and apply the chain rule. (3) Recognize that the result is always *linear* in $\partial u/\partial p$, even when $\mathcal{F}$ is nonlinear in u.

fse for an ode system. For $\dot{\vec{u}} = F(\vec{u}; \vec{p})$, the sensitivity $w_j = \partial \vec{u}/\partial p_j$ satisfies

$$\dot{w}_j = J_F\, w_j + \frac{\partial F}{\partial p_j}, \qquad w_j(0) = 0,$$

where $J_F = \partial F/\partial \vec{u}$ is the Jacobian of the model. For m POIs, solve m linear ODEs, each augmenting the original system by n state variables.

The proliferation problem. Cost of FSE: m augmented ODE solves (one per POI). For a climate model with $m = 200$ and 6-hour runs: 50 days. This is the bottleneck that motivates the adjoint (Chapters 5–6), which reduces cost from m solves to 2 solves.

SIR model with demography. Chapter 4 extends the Chapter 2 SIR model to include births and deaths ($\nu_m N$ birth rate, ν_m mortality rate, mean residence time $\tau_m = 1/\nu_m$). This adds a fourth POI τ_m to the set $\{c, \beta, \tau_R\}$. The FSE structure is unchanged; the augmented system grows from $3 + 3m$ to $3 + 4m$ ODEs.

Watch out: Near a bifurcation, the FSE is formally correct but the sensitivity index diverges. The FSE solves to give $\partial u/\partial p = \infty$ at a bifurcation point: the answer is exact, but the question "how much does u change per unit change in p?" is not meaningful when the linearization breaks down. The decision table in Section 4.5 shows when to use which method.

Exercise 4.1 *[Bloom L2]*

Exercise statement. State the three-step forward sensitivity equations (FSE) derivation pattern in your own words, without using any symbols. Then explain, in one sentence, why the FSE is always linear in the sensitivity variable even when the original equation is nonlinear.

Solution.

The three-step pattern in words:

1. *Write down the original equation* (algebraic, equilibrium, or ODE) in implicit form.
2. *Differentiate the entire equation with respect to the parameter of interest*, treating both the state variable and the parameter as functions that can be differentiated. Apply the chain rule wherever the state variable appears.
3. *Solve the resulting equation for the sensitivity variable* (the derivative of the state with respect to the parameter). This is the forward sensitivity equation.

Why the FSE is always linear in the sensitivity variable: Differentiating the original equation with respect to a parameter is a linear operation: the derivative of a product involving the state variable produces terms that are linear in the sensitivity variable (the derivative), because differentiation distributes over addition and pulls constants through. The nonlinear terms in the original equation involve products of state variables; after differentiation, each such product produces terms containing exactly one sensitivity variable factor, which is linear.

Checking your answer: The one-sentence explanation is the key takeaway of the theorem (Theorem 4.1). Students who write "because the original equation is linear" have confused the FSE with the original equation; the FSE is linear *even when the original is nonlinear.* A complete answer requires distinguishing the linearity of the *FSE* from the nonlinearity of the original equation.

Exercise 4.2 *[Bloom L3]*

Exercise statement. For the nonlinear equation $g(u;p) := u^3 - 3pu + 2 = 0$:

(a) Write the FSE for $\partial u/\partial p$.

(b) At $p = 1$, find all real roots u numerically.

(c) Compute $\mathcal{S}_p^u$ at each root.

(d) At $p = 1$, for which root is the sensitivity largest in magnitude? Verify your answer at the simple root near $u = -2$ using a centered finite difference with optimal h.

(e) Explain why the same centered difference cannot be used at the positive root, and show by one-sided continuation for $p > 1$ that $u_\pm \approx 1 \pm \sqrt{p-1}$.

Solution.

The equation is $g(u;p) := u^3 - 3pu + 2 = 0$.

(a) **FSE for $\partial u/\partial p$.**

Differentiate $g(u;p) = 0$ with respect to p:

$$\frac{\partial g}{\partial u}\frac{\partial u}{\partial p} + \frac{\partial g}{\partial p} = 0.$$

$$(3u^2 - 3p)\frac{\partial u}{\partial p} + (-3u) = 0.$$

Solving for the sensitivity variable:

$$\frac{\partial u}{\partial p} = \frac{3u}{3u^2 - 3p} = \frac{u}{u^2 - p}.$$

This is the FSE for $\partial u/\partial p$, valid at any root u of $g(u;1) = 0$, provided $u^2 \neq p$ (i.e., the Jacobian is nonsingular).

(b) **Real roots at $p = 1$.**

At $p = 1$: $u^3 - 3u + 2 = 0$. Factor: $(u-1)^2(u+2) = 0$, giving roots $u_1 = 1$ (double root) and $u_2 = -2$.

(c) **Sensitivity indices at each root.**

At $u_1 = 1$: $\partial u/\partial p = 1/(1-1) = 1/0$; undefined. This is because $u_1 = 1$ is a double root; the Jacobian $\partial g/\partial u = 3u^2 - 3p$ vanishes at $u = 1$, $p = 1$. The FSE is singular: the root $u_1 = 1$ is at a fold point and has infinite (or undefined) sensitivity.

At $u_2 = -2$: $\partial u/\partial p = (-2)/(4-1) = -2/3$.

$$\mathcal{S}_p^{u_2} = \frac{p}{u_2}\cdot\frac{\partial u_2}{\partial p} = \frac{1}{-2}\cdot\left(-\frac{2}{3}\right) = \frac{1}{3}.$$

(d) **Largest sensitivity and finite-difference verification.**

At $u_1 = 1$ the sensitivity is undefined (infinite); at $u_2 = -2$, $|\mathcal{S}_p^{u_2}| = 1/3$. The root u_1 has the largest (infinite) sensitivity.

Verification for $u_2 = -2$ using centered FD with $h^* = 6 \times 10^{-6}$:

```
p_star = 1;   h = 6e-6
# Define g(u; p) = u^3 - 3pu + 2

u2_plus  = FIND_ROOT( u -> u^3 - 3*(p_star + h)*u + 2, near = -2 )
u2_minus = FIND_ROOT( u -> u^3 - 3*(p_star - h)*u + 2, near = -2 )

dU_dp_fd = (u2_plus - u2_minus) / (2*h)   # analytic: -2/3
SI_fd    = (p_star / u2_minus) * dU_dp_fd  # analytic: +1/3
```

Expected output: `SI_fd` ≈ 0.33333 ✓.

(e) **Why the centered difference fails at the positive root.** The simple root at $u_2 = -2$ persists for p on both sides of 1, which is why the centered stencil works there. The positive roots do not. For $p < 1$ they are a complex-conjugate pair: at $p = 0.999$ the cubic has the single real root $u \approx -1.9993$ and nothing near $u = 1$. A centered difference at $p = 1$ would evaluate a branch that does not exist on its left, so it is differencing *across* the fold rather than through it, and no choice of h repairs that.

Continue from one side instead. Writing $u = 1 + \varepsilon$ and substituting into g gives $\varepsilon^2 \approx p - 1$ to leading order, so for $p > 1$

$$u_\pm \approx 1 \pm \sqrt{p-1}, \qquad \frac{\partial u_\pm}{\partial p} \approx \pm \frac{1}{2\sqrt{p-1}}.$$

The square-root shape is the signature of a fold: the two branches meet with vertical tangents, so the derivative diverges like $(p-1)^{-1/2}$ rather than jumping. Evaluating at $p = 1.1$, 1.01, 1.001 gives 1.58, 5.00, 15.81, each step of a hundredfold in $p - 1$ multiplying the derivative by ten. Checking that rate is the verification; a single large difference quotient is not.

Checking your answer: The singular sensitivity at the double root is the crucial pedagogical point: the FSE breaks down at bifurcation points (Chapter 8). Students who report $\mathcal{S}_p^{u_1}$ as some finite number have made an error; the Jacobian is zero there. A complete answer states that the Jacobian is singular at the double root, so the FSE has no finite solution there. Identifying the double root but not the singularity is a partial answer.

Exercise 4.3 *[Bloom L3]*

Exercise statement. For the logistic ODE $\dot{u} = ru(1 - u/K)$, $u(0) = u_0$:

(a) Write the FSE for $w_r = \partial u/\partial r$.

(b) Verify that the FSE is linear in w_r by identifying the coefficient (which depends on $u(t)$) and the forcing term.

(c) Implement the augmented system (u, w_r, w_K) and solve from $t = 0$ to $t = 20$.

(d) Plot $\mathcal{S}_r^u(t)$ and $\mathcal{S}_K^u(t)$. Interpret: near $t = 0$ vs. near the carrying capacity.

Solution.

The logistic ODE is $\dot{u} = ru(1 - u/K)$, $u(0) = u_0$.

(a) **FSE for $w_r = \partial u/\partial r$.**

Differentiate the ODE with respect to r:

$$\dot{w}_r = \frac{\partial}{\partial r}\big[ru(1 - u/K)\big] = u(1 - u/K) + r(1 - 2u/K)\, w_r.$$

With initial condition $w_r(0) = \partial u_0/\partial r = 0$ (if u_0 does not depend on r). This is the FSE:

$$\dot{w}_r = r\left(1 - \frac{2u}{K}\right) w_r + u\left(1 - \frac{u}{K}\right), \qquad w_r(0) = 0.$$

(b) **Linearity in w_r.**

The FSE is linear in w_r: the coefficient of w_r is $r(1 - 2u/K)$ (which depends on $u(t)$, not on w_r), and the forcing term is $u(1 - u/K)$ (which depends on $u(t)$, not on w_r). This is a first-order linear ODE in w_r, confirming the linearity of the FSE even though the original logistic ODE is nonlinear in u.

(c) **Augmented system for (u, w_r, w_K).**

FSE for $w_K = \partial u/\partial K$:

$$\dot{w}_K = r(1 - 2u/K)w_K + \frac{ru^2}{K^2}, \qquad w_K(0) = 0.$$

```
# Augmented ODE: state = [u, w_r, w_K]
# Parameters: r_nom = 1, K_nom = 10, u0 = 1, T = 20

FUNCTION logistic_augmented(t, state, r, K):
    u  = state[1];  w_r = state[2];  w_K = state[3]
    F  = r * u * (1 - u/K)          # logistic RHS
    Ju = r * (1 - 2*u/K)            # Jacobian d(RHS)/du
    RETURN [F,
            Ju * w_r + u*(1 - u/K), # FSE for r
            Ju * w_K + r*u^2/K^2 ]  # FSE for K
END

y0      = [u0, 0, 0]
f(t, y) = logistic_augmented(t, y, r_nom, K_nom)
Y       = SOLVE_ODE(f, tspan=[0, T], y0,
                    tol_rel=1e-8, tol_abs=1e-10)
```

(d) **Sensitivity index plots and interpretation.**

$\mathcal{S}^u_r(t) = (r/u(t))\, w_r(t)$ and $\mathcal{S}^u_K(t) = (K/u(t))\, w_K(t)$.

Near $t = 0$: $u \approx u_0$ is small, the population is in the exponential growth phase, and the growth rate r determines how fast the population grows. Because $w_r(0) = 0$ from part (a), $\mathcal{S}^u_r(0) = 0$ exactly; the index then grows roughly like rt through the exponential phase, since $u \approx u_0 e^{rt}$ there. It peaks in mid-growth, near $\mathcal{S}^u_r \approx 1.1$ at $t \approx 2$ for $r = 1$, $K = 10$, $u_0 = 1$, and decays as the population saturates. A change in r needs time to express itself in u. $|\mathcal{S}^u_K|$ is small near $t = 0$: the carrying capacity K barely matters while $u \ll K$.

Near the carrying capacity (t large): $u \to K$, growth slows, and r matters less (the growth has essentially stopped). $|\mathcal{S}^u_K|$ approaches 1 (the long-run level is proportional to K); $|\mathcal{S}^u_r|$ decreases toward zero.

Checking your answer: The logistic ODE has an analytic solution $u(t) = K/(1+(K/u_0-1)e^{-rt})$; students who know this can skip the FSE entirely and differentiate analytically. Common error: treating $u(t)$ as a constant when differentiating with respect to r or K (confusing FSE for parameter sensitivity with algebraic differentiation of the solution formula). At minimum: correct augmented ODE system with the right forcing terms. A complete answer: correct analytic verification of the FSE solution by differentiating the closed-form $u(t)$ with respect to each parameter and matching. The distinguishing check: an answer that runs the FSE numerically but skips analytic verification is partial on part (d).

Checking your answer: Students who implement the augmented system correctly but do not plot and interpret the time courses should receive a partial answer on (d). A complete answer plots the time courses and reads the exponential and saturating phases off them; at minimum, a correctly implemented augmented system. The biological interpretation is the core lesson.

Exercise 4.4 *[Bloom L3]*

Exercise statement. For the competitive Lotka–Volterra model (4.2)–(4.3) with $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute the coexistence equilibrium.

(b) Assemble and solve the FSE for $\partial\vec{u}/\partial c$.

(c) Compute the normalized sensitivity indices $\mathcal{S}_c^{\bar{u}_1}$ and $\mathcal{S}_c^{\bar{u}_2}$.

(d) Using equation (4.4), compute the same indices analytically. Verify agreement.

Solution.

The competitive Lotka–Volterra model with $a = 0.5$, $b = 2$, $c = 0.4$:

$$\dot{u}_1 = u_1(1 - u_1 - au_2), \qquad \dot{u}_2 = bu_2(1 - u_2 - cu_1).$$

(a) **Coexistence equilibrium.** From equation (4.4):

$$\bar{u}_1 = \frac{1-a}{1-ac} = \frac{1-0.5}{1-0.5\cdot 0.4} = \frac{0.5}{0.8} = 0.625, \qquad \bar{u}_2 = \frac{1-c}{1-ac} = \frac{0.6}{0.8} = 0.75.$$

(b) **FSE for $\partial\vec{u}/\partial c$.**

The Jacobian of $\vec{f} = (\bar{u}_1(1-\bar{u}_1-a\bar{u}_2),\, b\bar{u}_2(1-\bar{u}_2-c\bar{u}_1))$ with respect to $\vec{u}$ at equilibrium:

$$D_{\vec{u}}[\vec{f}]\big|_{\text{eq}} = \begin{pmatrix} 1-2\bar{u}_1-a\bar{u}_2 & -a\bar{u}_1 \\ -bc\bar{u}_2 & b(1-2\bar{u}_2-c\bar{u}_1) \end{pmatrix}.$$

Substituting numerical values ($\bar{u}_1 = 0.625$, $\bar{u}_2 = 0.75$):

$$= \begin{pmatrix} 1-1.25-0.375 & -0.5(0.625) \\ -2(0.4)(0.75) & 2(1-1.5-0.25) \end{pmatrix} = \begin{pmatrix} -0.625 & -0.3125 \\ -0.6 & -1.5 \end{pmatrix}.$$

The forcing term $-\partial\vec{f}/\partial c$: only f_2 depends on c (through $-c\bar{u}_1$), so:

$$-\frac{\partial\vec{f}}{\partial c} = -\begin{pmatrix} 0 \\ b\bar{u}_2(-\bar{u}_1) \end{pmatrix} = \begin{pmatrix} 0 \\ b\bar{u}_1\bar{u}_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2(0.625)(0.75) \end{pmatrix} = \begin{pmatrix} 0 \\ 0.9375 \end{pmatrix}.$$

The FSE is the linear system:

$$\begin{pmatrix} -0.625 & -0.3125 \\ -0.6 & -1.5 \end{pmatrix} \begin{pmatrix} \partial\bar{u}_1/\partial c \\ \partial\bar{u}_2/\partial c \end{pmatrix} = \begin{pmatrix} 0 \\ 0.9375 \end{pmatrix}.$$

Solving (by row reduction or Cramer's rule):

$$\det = (-0.625)(-1.5) - (-0.3125)(-0.6) = 0.9375 - 0.1875 = 0.75.$$

$$\frac{\partial\bar{u}_1}{\partial c} = \frac{(0)(-1.5) - (-0.3125)(0.9375)}{0.75} = \frac{0.292969}{0.75} = 0.390625.$$
$$\frac{\partial\bar{u}_2}{\partial c} = \frac{(-0.625)(0.9375) - (0)(-0.6)}{0.75} = \frac{-0.585938}{0.75} = -0.78125.$$

(c) **Normalized sensitivity indices.**

$$\mathcal{S}_c^{\bar{u}_1} = \frac{c}{\bar{u}_1} \cdot \frac{\partial\bar{u}_1}{\partial c} = \frac{0.4}{0.625} \cdot 0.390625 = 0.25.$$

$$\mathcal{S}_c^{\bar{u}_2} = \frac{c}{\bar{u}_2} \cdot \frac{\partial\bar{u}_2}{\partial c} = \frac{0.4}{0.75} \cdot (-0.78125) = -0.41\overline{6}.$$

(d) **Analytic verification from equation (4.4).**

$\bar{u}_1 = (1-a)/(1-ac)$, so $\partial\bar{u}_1/\partial c = (1-a)\cdot a/(1-ac)^2$. At $a = 0.5$, $c = 0.4$: $= (0.5)(0.5)/(0.64) = 0.25/0.64 = 0.390625$ ✓.

$\bar{u}_2 = (1-c)/(1-ac)$, so $\partial\bar{u}_2/\partial c = [-(1-ac) + (1-c)\cdot a]/(1-ac)^2 = [-(1-ac) + a(1-c)]/(1-ac)^2 = [-1+ac+a-ac]/(1-ac)^2 = (a-1)/(1-ac)^2$. At $a = 0.5$, $c = 0.4$: $= (-0.5)/(0.64) = -0.78125$ ✓.

Checking your answer: The FSE approach (solving a 2×2 linear system) and the analytic approach (differentiating the closed-form equilibrium) must give identical results. Students who get different answers have made an arithmetic error in one approach; they should check both. A complete answer obtains identical results by both routes and says why they must agree; at minimum, one route carried through correctly. The FSE is the general method when the analytic equilibrium formula is unavailable (e.g., for $n \geq 3$ species models).

Exercise 4.5 *[Bloom L4]*

Exercise statement. The SEIRS model (Susceptible-Exposed-Infectious-Recovered-Susceptible) has 4 state variables and 7 parameters.

(a) How many ODEs are in the augmented FSE system for all 7 parameters?

(b) If each ODE solve takes 3 minutes, how long does the full FSA take using the FSE approach? The adjoint approach (Chapter 6)?

(c) Suppose a modeler decides to reduce computation time by computing sensitivities only for the 3 parameters they expect to be most important, based on biological intuition. Identify one scenario where this strategy would lead to a misleading conclusion.

Solution.

The SEIRS model has $n_s = 4$ state variables and $n_p = 7$ parameters.

(a) **Size of augmented FSE system.**

For each parameter p_j, the FSE adds $n_s = 4$ equations (one per state variable). For all $n_p = 7$ parameters, the augmented system has:

$$n_s + n_s \cdot n_p = 4 + 4 \times 7 = \mathbf{32} \text{ ODEs.}$$

(b) **Computation time comparison.**

FSE approach: One solve of the 32-equation augmented ODE. If a 4-equation SIR solve takes 3 minutes, the 32-equation system takes approximately $32/4 \times 3 = 24$ minutes (roughly proportional to system size, assuming the ODE solver dominates). In practice, closer to $8\times$ the base solve time: $\sim$ **24 minutes.**

Adjoint approach (Chapter 6): One forward solve (3 min) plus one backward adjoint solve (also $\sim n_s$ equations, so $\sim$3 min). Total: $\sim$ **6 minutes.** (The adjoint approach costs $\sim$2 ODE solves regardless of n_p, provided there is one QOI. Its advantage over FSE grows with n_p.)

Centered differences: $(2 \times 7 + 1) = 15$ solves $\times$ 3 min = **45 minutes.**

For this problem: Adjoint (6 min) < FSE (24 min) < FD (45 min).

(c) **Scenario where biologically guided pre-selection misleads.**

Suppose the modeler assumes that the transmission rate β and contact rate c are the most important parameters and omits the waning immunity rate ρ from the sensitivity analysis. In a disease context where ρ is large (immunity wanes quickly, returning recovered individuals to susceptible), ρ may dominate the endemic equilibrium level, which is the QOI. The modeler's SA would correctly identify β and c as important for $\mathcal{R}_0$, but miss that intervention on ρ (e.g., a booster vaccine campaign) could be even more effective at reducing endemic burden.

More generally: biological intuition about $\mathcal{R}_0$ does not transfer to conclusions about endemic equilibrium or time-to-outbreak-peak. Different QOIs have different most-sensitive parameters. Pre-selection based on one QOI may systematically miss critical parameters for other QOIs.

Checking your answer: Part (b) is a good opportunity to reinforce that the FSE and adjoint costs are both measured relative to the number of forward solves. Common error: forgetting to count the base evaluation in the forward-difference cost (cost is $m+1$, not m). At minimum: correct cost formulas for all three methods with correct regime identification. A complete answer: explicit cross-over computation showing exactly when adjoint beats forward (when $m > r$). The distinguishing check: a student who gives the formula $m + 1$ vs $2m + 1$ vs $r + 1$ without stating the regime condition is partial on the comparison part.

Checking your answer: FSE cost counts the number of *augmented system* solves, not parameters. Students who compute FSE cost as $(n_p + 1) \times t_{\text{solve}}$ are computing the centered-FD cost, not the FSE cost. A complete answer counts one augmented-system solve; at minimum, distinguishing FSE cost from FD cost. The FSE solves *one* augmented system; the FD solves $2n_p + 1$ separate systems.

Exercise 4.6 *[Bloom L4]*

Exercise statement. Using the laboratory implementation:

(a) At three time points, $t = 5$ (early epidemic), $t = t_{\text{peak}}$ (peak infections), and $t = 80$ (late); report the tornado plot of SIs for $I(t)$.

(b) Does the parameter ranking change across the three time points?

(c) A public health officer says: "We should target the parameter with the largest SI at the epidemic peak, since that is when the most people are affected." Critique this statement using your results.

Solution.

Results are based on the `compute_time_si` function with nominal parameters from `sir_nominal()` ($c = 5$, $\beta = 0.06$, $\tau_R = 7$, $\tau_m = 10{,}000$, $T = 90$ days). The epidemic peak is at approximately $t_{\text{peak}} \approx 43.9$ days, where $I \approx 171$.

(a) **Tornado plots at three time points.**

Numerical SI values (computed from FSE; slight rounding):

Parameter	$t = 5$ (early)	$t = 43.9$ (peak)	$t = 80$ (late)
c	$+1.495$	$+2.065$	-7.466
β	$+1.495$	$+2.065$	-7.466
τ_R	$+0.714$	$+2.064$	-0.594
τ_m	$+0.0005$	$+0.0005$	-0.0104

(b) **Does the ranking change?**

Yes, and in more than one way. The ordering by magnitude ($c \approx \beta > \tau_R \gg \tau_m$) does hold at all three times, but two things change that a ranking alone conceals.

First, the *signs* reverse. At $t = 80$ the epidemic is well past its peak, and increasing β now depletes susceptibles faster, leaving *fewer* infectives at day 80: $\mathcal{S}_\beta^{I(80)} = -7.47$. A parameter that is strongly beneficial to the pathogen early is strongly detrimental to it late.

Second, the *gap* between c and τ_R opens and closes. Early the contact terms dominate ($+1.495$ against $+0.714$); at the peak the three are nearly equal ($+2.065$, $+2.065$, $+2.064$); late they diverge again with opposite emphasis (-7.47 against -0.59). Reporting one tornado plot would hide all of this.

τ_m is small throughout, but not uniformly signed: it is *positive* through the growth phase and turns negative only in the tail. Over 90 days a residence time of 10,000 days barely matters either way.

(c) **Critique of the public health officer's statement.**

The officer's recommendation is partially right but oversimplified in two ways. First, the parameter with the largest SI at the peak changes over the course of the epidemic:

intervention early (e.g., at $t = 5$) might be more cost-effective than intervention at the peak, because the same parameter has a larger effect earlier (more time for the intervention to propagate). Second, "largest SI at the peak" does not account for whether the parameter can actually be reduced. c and β have the same SI but different intervention levers: c can be reduced by social distancing (school closures, event cancellations), while β can be reduced by mask use or hygiene. The policy choice depends on feasibility and cost, not just the SI value. The SI identifies what to target, not how.

Checking your answer: The numerical values in part (a) should be obtained from the `compute_time_si` function (companion notebooks). Common error: computing SI at peak I only (a single time) instead of showing SI as a function of time and identifying its time-variation. At minimum: time-dependent SI plot for β and τ_R with correct qualitative shape (SI large early, decreasing after peak). A complete answer: quantitative table at $t = 0, 30, 60, 90, 150$ days plus explanation of why SI changes sign after peak. The distinguishing check: a student who plots $\mathcal{S}_p^{I(t)}$ vs. t and correctly explains the sign change at peak has demonstrated understanding beyond a student who only reports the value at a single time.

Checking your answer: Time-dependent SI is computed for $\mathcal{S}_p^{I(t)}$ (QOI $= I(t)$). The key pedagogical point is that the *ranking* is stable for this model but magnitudes change substantially. A complete answer reports the ranking as stable while the magnitudes change substantially; at minimum, noting that the indices vary in time at all. For models with threshold behavior, rankings can reverse; that is Chapter 8 territory.

Exercise 4.7 *[Bloom L5]*

Exercise statement. A colleague is modeling cholera transmission with a SEIR-type model having 8 parameters and 4 state variables. She asks for advice on the SA methodology. Design a complete FSA study, specifying:

(a) Whether to use finite differences or analytic FSE, and why.

(b) How you would validate the implementation.

(c) What QOIs you would compute sensitivity indices for, and at what time points.

(d) How you would present the results to identify the two most critical intervention targets.

(e) One situation in which the FSE results might be misleading, and what additional analysis you would recommend.

Solution.

Design for cholera FSA: 8 parameters, 4 state variables.

(a) **Analytic FSE vs. finite differences.**

Use *analytic FSE*. Cholera models are moderately complex SEIR-type ODEs for which the Jacobian can be derived analytically (the right-hand side is a ratio of polynomials in the state variables, readily differentiated). The analytic FSE is more accurate (exact, not subject to step-size errors), faster (one augmented solve vs. $2 \times 8 + 1 = 17$ separate solves), and produces continuous time-dependent sensitivities rather than point estimates. The only additional implementation cost is writing the Jacobian and the forcing terms $\partial \vec{F}/\partial p_j$, for a model with 4 state variables and 8 parameters, so this is about 30–40 lines of code.

(b) **Validation.**

Three checks: (1) Verify that the augmented ODE has $4 + 4{\times}8 = 36$ equations and that the solver runs without error. (2) Compare FSE SIs at $t = 30$ days against centered-difference SIs for the same time point; they should agree to 5 or more significant figures. (3) Verify that $\mathcal{S}_c^{\mathcal{R}_0} = \mathcal{S}_\beta^{\mathcal{R}_0} = 1$ analytically and confirm the FSE gives the same values at early times when $\mathcal{R}_0$ drives dynamics.

(c) **QOIs and time points.**

Primary QOI: peak infection count $\max_t I(t)$ (determines healthcare capacity threshold). Secondary: total deaths at 90 days, and days until peak. Compute SI time curves $\mathcal{S}_{p_j}^I(t)$ for all 8 parameters. Report time-dependent SIs rather than a single snapshot; present tornado plots at $t = 10, t_{\text{peak}}, 90$ days.

(d) **Presenting results to identify intervention targets.**

Produce a 2×4 panel figure: left column shows $\mathcal{S}_{p_j}^I(t)$ curves for all 8 parameters over time (color-coded), highlighting the top 3 by $|\mathcal{S}_{p_j}^{\cdot}|$ at t_{peak}. Right column shows tornado plots at the three key time points. Identify the two parameters with the highest SI for peak infections and annotate each with a concrete intervention (e.g., "reduce τ_I via oral rehydration therapy"; "reduce β_w via water treatment").

(e) **Situation where FSE results might mislead.**

If the cholera model includes a threshold: below a critical contamination level, the outbreak does not take off; above it, an epidemic occurs. Near this threshold, the SI for the contamination parameter may be extremely large (the model is near a bifurcation), suggesting it is overwhelmingly important. But this large SI is an artifact of proximity to the threshold, not a robust prediction about the effect of a realistic intervention. In this case, compute SI at several nominal parameter sets (e.g., $R_0 = 0.8$, 1.0, 1.2) and report only the SI values that are stable across this range. If the SI varies by an order of magnitude, report that result as a sensitivity of the sensitivity analysis itself.

Checking your answer: A high-quality L5 answer addresses all five parts with specificity. Vague responses ("use tornado plots," "validate the code") without operational detail are partial. A complete answer treats all five parts with operational detail; at minimum, three parts with specifics rather than slogans. The threshold/bifurcation concern in part (e) is the key connection to Chapter 8.

Exercise 4.8 *[Bloom L6]*

Exercise statement. A published paper reports: "We performed sensitivity analysis of the SEIR model using the forward sensitivity equations. All sensitivity indices were computed at $t = 365$ days, giving a single set of values for the entire epidemic."

The model simulates a disease whose outbreak dynamics play out over approximately 60 days, but the model runs for a year.

Identify at least two methodological issues with this approach. For each issue, state what the authors should have done instead and explain what information they likely missed.

Solution.

Issue 1: Single time-point snapshot misses transient dynamics.

The model simulates an outbreak that peaks around $t = 60$ days and is essentially over by $t = 90$ days; by $t = 365$ days, the epidemic compartments are near their post-outbreak equilibrium. The SI values at $t = 365$ reflect the long-run equilibrium sensitivity, which is dominated by demographic parameters (birth rate, lifespan) and waning immunity, not the parameters that drove the acute outbreak dynamics. A single time snapshot at day 365 would show near-zero sensitivity to the transmission rate β (because the epidemic is over and $I(365) \approx 0$) but nonzero sensitivity to waning immunity (if the model has that component).

What should have been done: Compute $\mathcal{S}^I_{p_j}(t)$ as a function of time, and report SI curves or tornado plots at clinically relevant time points (e.g., outbreak peak, $t = 30$ days, $t = 60$ days). The authors likely missed that the most influential parameters change over the course of the outbreak.

Issue 2: Reporting "a single set of values for the entire epidemic" is a category error.

FSE-computed sensitivity indices are time-dependent: $\mathcal{S}^I_{p_j}(t)$ is a curve, not a number. Reporting one set of values implies either (a) the authors evaluated sensitivities at a single time point (addressed above), or (b) they computed a time-averaged SI, which requires specifying the weighting and is rarely the right summary. Neither is the same as "the sensitivity for the entire epidemic." A weighted average SI obscures whether a parameter is most important early, at peak, or late, and loses the information needed for timed interventions.

What should have been done: Report time-dependent SI curves, or if a scalar summary is needed, report the SI of the integral $J = \int_0^T I(t)\,dt$ (total infectious burden), which is a well-defined scalar QOI. This is exactly the QOI used in the adjoint derivation of Chapter 6.

Optional additional issue: Running the model for 365 days when the epidemic is essentially over at day 90 introduces numerical noise in the SI at late times (the SI of a nearly-zero QOI is numerically unstable). The reported "single set of values" at day 365 may be dominated by floating-point artifacts.

Checking your answer: The two methodological issues are closely related: both stem from treating a time-dependent quantity (SI curve) as if it were a scalar. A complete answer names both the time-dependence issue with a specific alternative and the category error; at minimum, the time-dependence issue alone. A further step is the category error (FSE gives curves, not numbers). The optional issue about numerical stability at $I \approx 0$ is a further step. Students who only say "should have used more time points" without identifying *why* the late-time values are misleading receive a partial answer.

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5 Chapter 5: The Adjoint in Linear Algebra

Chapter Summary: The Adjoint in Linear Algebra

The adjoint strategy. For a linear system $A\vec{u} = \vec{b}(\vec{q})$ with scalar response $J = \vec{c}^\top \vec{u}$: instead of solving $A\,\partial\vec{u}/\partial q_j$ once per parameter (forward approach, m solves), solve the *adjoint system* $A^\top \vec{v} = \vec{c}$ *once*, then recover all m derivatives as $\partial J/\partial q_j = \vec{v}^\top\,\partial\vec{b}/\partial q_j$. Cost: 1 solve for any m.

When the adjoint wins. The adjoint is faster than the forward approach when $m > r$, where m is the number of POIs and r is the number of response functionals. One adjoint solve recovers sensitivities for one response functional with respect to all m POIs. For r responses, you need r adjoint solves. Rule: use adjoint when $m \gg r$; use forward when $r \gg m$.

Reverse mode and backpropagation. On a computation graph, reverse-mode differentiation propagates adjoint variables $\bar{v}_i = \partial J/\partial v_i$ backward from output to all inputs in one pass. This is identical to backpropagation in a neural network. Cost: one backward pass per output (QOI), regardless of the number of inputs (POIs).

Unifying Idea 2 in action. Matrix adjoint: $A^\top \vec{v} = \vec{c}$. Graph adjoint: backward accumulation of $\bar{v}_i$. ODE adjoint (Chapter 6): $-\dot{\lambda} = J_F^\top \lambda + \nabla_u g$. These are the same operation in three different mathematical spaces: the adjoint is always the transpose in whatever space you work in.

Watch out: The adjoint requires knowing the *forward solution* $\vec{u}$ first. For ODE systems (Chapter 6), this means storing the entire forward trajectory $\vec{u}(t)$ before running the adjoint backward. For large systems, memory can become the bottleneck, not computation. Checkpointing strategies trade additional computation for reduced memory storage.

Exercise 5.1 *[Bloom L2]*

Exercise statement. State the three-step adjoint strategy from Section 5.2 in your own words, using no symbols. Then explain: why does the adjoint equation $A^T\vec{v} = \vec{c}$ not depend on which parameter q_j we are differentiating with respect to? Why does this independence make the adjoint efficient?

Solution.

The three-step adjoint strategy in plain words:

1. Solve the forward problem to find the state. Store the solution.

2. Solve one backward (adjoint) problem for each response functional, using the matrix transposed from the forward problem.

3. For each parameter, compute the sensitivity as a dot product of the adjoint solution with the parameter's contribution vector. No additional solves are needed.

Why the adjoint equation $A^T\vec{v} = \vec{c}$ does not depend on q_j:
The adjoint equation encodes "which directions in state space affect the response $J = \vec{c}^T\vec{u}$?" This question is entirely about J (encoded in $\vec{c}$) and the forward operator (encoded in A). It does not involve any particular parameter q_j, because the adjoint does not ask what the state is, only which state directions matter for J. The parameter-dependence enters only in the dot product step ($\partial J/\partial q_j = \vec{v}^T\partial\vec{b}/\partial q_j$), which costs just one inner product per parameter once $\vec{v}$ is known.

Why this makes the adjoint efficient: With m parameters, the forward approach requires m forward solves (one forward sensitivity equations (FSE) per parameter). The adjoint requires only 1 adjoint solve (independent of m), then m dot products. For large m, the savings are a factor of m.

Checking your answer: The key conceptual point: the adjoint asks about J and A, not about specific parameters. Students often confuse the adjoint solve with a parameter-dependent computation. A complete answer requires explaining why $A^T\vec{v} = \vec{c}$ has no q_j on either side.

Exercise 5.2 *[Bloom L3]*

Exercise statement. For the linear system

$$A\vec{u} = \vec{b}, \qquad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \vec{b} = \begin{pmatrix} b_1(q) \\ b_2(q) \end{pmatrix} = \begin{pmatrix} q \\ 2q \end{pmatrix},$$

with response functional $J = u_1 + u_2$ (so $\vec{c} = (1,1)^T$):

(a) Solve the forward problem to find $\vec{u}(q)$ at $q = 1$.

(b) Write and solve the FSE for $\partial\vec{u}/\partial q$.

(c) Write and solve the adjoint equation.

(d) Use formula equation (5.4) to compute $\partial J/\partial q$.

(e) Verify by differentiating $J(q) = u_1(q) + u_2(q)$ directly.

Solution.
System: $A\vec{u} = \vec{b}$ with $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $\vec{b} = (q, 2q)^T$, $J = u_1 + u_2$ ($\vec{c} = (1,1)^T$).

(a) **Forward solution at $q = 1$.** $A^{-1} = \frac{1}{5}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 3 \end{smallmatrix}\right)$, so $\vec{u} = A^{-1}\vec{b} = \frac{1}{5}(2\cdot1 - 1\cdot2, -1\cdot1 + 3\cdot2)^T = \frac{1}{5}(0,5)^T = (0,1)^T$. Hence $u_1 = 0$, $u_2 = 1$.

(b) **FSE for $\partial\vec{u}/\partial q$.** Differentiate $A\vec{u} = \vec{b}$ with respect to q: $A\,(\partial\vec{u}/\partial q) = \partial\vec{b}/\partial q = (1,2)^T$. Solve: $\partial\vec{u}/\partial q = A^{-1}(1,2)^T = \frac{1}{5}(2-2, -1+6)^T = \frac{1}{5}(0,5)^T = (0,1)^T$.

(c) **Adjoint equation.** $A^T\vec{v} = \vec{c}$. Since A is symmetric, $A^T = A$, so $A\vec{v} = (1,1)^T$: $\vec{v} = A^{-1}(1,1)^T = \frac{1}{5}(2-1,-1+3)^T = \frac{1}{5}(1,2)^T$.

(d) **Sensitivity via adjoint formula.** $\partial J/\partial q = \vec{v}^T\,(\partial\vec{b}/\partial q) = \frac{1}{5}(1,2)\cdot(1,2)^T = \frac{1}{5}(1+4) = 1$.

(e) **Direct verification.** $J(q) = u_1(q) + u_2(q)$. From (b), $\partial J/\partial q = \vec{c}^T(\partial\vec{u}/\partial q) = (1,1)\cdot(0,1)^T = 1$. ✓

Checking your answer: The system $A\vec{x} = \vec{b}$ and its adjoint $A^T\vec{v} = \vec{c}$ share the same LU factorization. Common error: solving the adjoint by refactoring A^T from scratch (this is exactly what the adjoint avoids). At minimum: correct $A^T\vec{v} = \vec{c}$ formulation and recognition that the same factorization is reused. A complete answer: explicit complexity count showing one forward solve + r adjoint solves vs. m forward solves. The distinguishing check: a student who writes "solve $A^T\vec{v} = \vec{c}$ using the stored LU" but cannot state the cost savings does not fully own the concept.

Checking your answer: $\vec{b} = q(1,2)^T$, so $J = u_1 + u_2$ is linear in q and $\partial J/\partial q = 1$ exactly. Students should note that both methods (FSE and adjoint) give the same answer; this is Unifying Idea 2 at work.

Exercise 5.3 *[Bloom L3]*

Exercise statement. For each scenario below, state whether you would use the FSE or the adjoint, and calculate the number of linear solves each approach requires.

(a) $m = 5$ parameters, $r = 20$ response functionals.

(b) $m = 100$ parameters, $r = 3$ response functionals.

(c) $m = 8$ parameters, $r = 8$ response functionals.

(d) $m = 1{,}000$ parameters, $r = 1$ response functional. Each solve takes 0.1 seconds. How much faster is the preferred method?

Solution.

The decision rule: use FSE when $m < r$; use adjoint when $r < m$. FSE requires m forward solves (plus 1 baseline); adjoint requires r adjoint solves (plus 1 forward).

(a) $m = 5$ parameters, $r = 20$ functionals. FSE: $m + 1 = 6$ solves. Adjoint: $r + 1 = 21$ solves. **Use FSE** $(6 < 21)$.

(b) $m = 100$ parameters, $r = 3$ functionals. FSE: 101 solves. Adjoint: 4 solves. **Use adjoint** $(4 \ll 101)$. Speedup factor: $\approx 25\times$.

(c) $m = 8$ parameters, $r = 8$ functionals. FSE: 9 solves. Adjoint: 9 solves. Both require the same cost. **Either works**; choose based on implementation availability. The crossover is at $m = r$.

(d) $m = 1{,}000$ parameters, $r = 1$ functional, $t = 0.1$ s/solve. FSE: $1001 \times 0.1 = 100.1$ s. Adjoint: $2 \times 0.1 = 0.2$ s. **Use adjoint.** Speedup: $100.1/0.2 \approx$ **500**$\times$.

Checking your answer: Part (d) illustrates the regime where the adjoint is not just convenient but essential: 1000 parameters, single QOI. Common error: recommending forward SA for part (d) without computing the cost (1001 forward solves vs 2 for adjoint). At minimum: correct regime identification for all three scenarios. A complete answer: explicit cost comparison in part (d) with the 1001 vs. 2 calculation. The distinguishing check: a student who correctly identifies the crossover condition $m > r$ and applies it to each scenario is complete even if they do not compute the exact numbers.

Exercise 5.4 *[Bloom L4]*

Exercise statement. For the competitive Lotka–Volterra equilibrium with parameters $a = 0.5$, $b = 2$, $c = 0.4$:

(a) Compute all 6 sensitivities ($\mathcal{S}_a^{\bar{u}_1}$, $\mathcal{S}_b^{\bar{u}_1}$, $\mathcal{S}_c^{\bar{u}_1}$, $\mathcal{S}_a^{\bar{u}_2}$, $\mathcal{S}_b^{\bar{u}_2}$, $\mathcal{S}_c^{\bar{u}_2}$) using the FSE approach (3 solves).

(b) Compute the same 6 sensitivities using the adjoint approach (2 solves).

(c) Verify that the two methods agree.

(d) Which approach requires fewer solves? By how much? At what values of m and r would the FSE be preferred?

Solution.

The competitive Lotka–Volterra equilibrium is $\bar{u}_1 = (1-a)/(1-ac) = 0.625$, $\bar{u}_2 = (1-c)/(1-ac) = 0.75$ (from Chapter 4, Ex 4.4). The equilibrium Jacobian J_F and FSE system are as computed in Ex 4.4.

(a) **FSE approach: 3 solves for 6 sensitivities.**

The FSE gives $\partial\vec{u}/\partial p_j$ for each of the 3 parameters (a, b, c) by solving the same 2×2 linear system $J_F\,(\partial\vec{u}/\partial p_j) = -\partial\vec{f}/\partial p_j$. This is 3 solves, each giving $(\partial\bar{u}_1/\partial p_j,\ \partial\bar{u}_2/\partial p_j)$.

Numerical results at $a = 0.5$, $b = 2$, $c = 0.4$ (from Ex 4.4 for $p = c$; deriving for a and b similarly):

	$\mathcal{S}_a^{\cdot}$	$\mathcal{S}_b^{\cdot}$	$\mathcal{S}_c^{\cdot}$
$\bar{u}_1$	-0.750	0	$+0.250$
$\bar{u}_2$	$+0.250$	0	-0.417

Derivation for $\partial/\partial a$: $f_1 = \bar{u}_1(1-\bar{u}_1 - a\bar{u}_2) = 0$, so $\partial f_1/\partial a = -\bar{u}_1\bar{u}_2 = -0.625(0.75) = -0.46875$. $\partial f_2/\partial a = 0$ (only f_1 contains a). The FSE gives $J_F(\partial\vec{u}/\partial a) = -\partial\vec{f}/\partial a = (0.46875,\,0)^T$. Solving with $J_F = \begin{pmatrix} -0.625 & -0.3125 \\ -0.600 & -1.500 \end{pmatrix}$ (determinant $= 0.75$) gives $\partial\bar{u}_1/\partial a = -0.9375$ and $\partial\bar{u}_2/\partial a = 0.375$. Hence:

$$\mathcal{S}_a^{\bar{u}_1} = \frac{a}{\bar{u}_1}\cdot(-0.9375) = \frac{0.5}{0.625}(-0.9375) = -0.750, \quad \mathcal{S}_a^{\bar{u}_2} = \frac{a}{\bar{u}_2}\cdot(0.375) = \frac{0.5}{0.75}(0.375) = +0.250.$$

For b: Only $f_2 = b\bar{u}_2(1-\bar{u}_2-c\bar{u}_1) = 0$, so $\partial f_2/\partial b = \bar{u}_2(1-\bar{u}_2-c\bar{u}_1) = 0$ (at equilibrium). Thus $\partial\vec{u}/\partial b = \vec{0}$: $\mathcal{S}_b^{\bar{u}_1} = \mathcal{S}_b^{\bar{u}_2} = 0$.

(b) **Adjoint approach: 2 solves for 6 sensitivities.**

Two response functionals: $J_1 = \bar{u}_1$ ($\vec{c}_1 = (1,0)^T$) and $J_2 = \bar{u}_2$ ($\vec{c}_2 = (0,1)^T$). Two adjoint solves: $J_F^T\vec{v}_k = \vec{c}_k$ for $k = 1, 2$. Then $\partial J_k/\partial p_j = \vec{v}_k^T(-\partial\vec{f}/\partial p_j)$ for each j.

(c) **Verification:** Both give the table above. ✓

(d) **Comparison.** FSE uses 3 solves (one per parameter), adjoint uses 2 solves (one per QOI). Adjoint is better here because $r = 2 < m = 3$. FSE would be preferred when $m < r$, e.g., for a model with 2 parameters and 5 response functionals.

Checking your answer: The zero SIs for b are a natural consequence of the equilibrium structure: b scales f_2 uniformly, so at equilibrium ($f_2 = 0$), $\partial f_2/\partial b = 0$. This is a structural zero, not a numerical artifact. Students who get nonzero SI for b have an error in computing $\partial\vec{f}/\partial b$.

Exercise 5.5 *[Bloom L4]*

Exercise statement. The SEIR basic reproduction number is $\mathcal{R}_0 = c\beta\nu/[(\nu+\nu_m)(\gamma_R+\nu_m)]$ with parameters c, β, ν (exposure rate), γ_R (recovery rate), ν_m (mortality rate).

(a) Draw the computation graph for $\mathcal{R}_0$ (intermediate nodes and edges).

(b) Implement the forward-mode computation to find $\partial\mathcal{R}_0/\partial\nu$. Count the number of arithmetic operations.

(c) Implement the reverse-mode computation to find all five derivatives $\partial\mathcal{R}_0/\partial c$, $\partial\mathcal{R}_0/\partial\beta$, $\partial\mathcal{R}_0/\partial\nu$, $\partial\mathcal{R}_0/\partial\gamma_R$, $\partial\mathcal{R}_0/\partial\nu_m$ in one backward pass.

(d) Compute the five normalized sensitivity indices and produce a tornado plot using interpretable POIs (mean times, not rates).

Solution.

$\mathcal{R}_0 = c\beta\nu/[(\nu+\nu_m)(\gamma_R+\nu_m)]$.

(a) **Computation graph.** Intermediate nodes: $n_1 = c\beta$, $n_2 = n_1\nu$, $n_3 = \nu+\nu_m$, $n_4 = \gamma_R+\nu_m$, $n_5 = n_3n_4$, $\mathcal{R}_0 = n_2/n_5$.

(b) **Forward mode for $\partial\mathcal{R}_0/\partial\nu$.** Propagate $\dot{\nu} = 1$ (seeding the derivative w.r.t. ν): $\dot{n}_1 = 0$, $\dot{n}_2 = n_1\cdot 1 + \nu\cdot 0 = c\beta$, $\dot{n}_3 = 1$, $\dot{n}_4 = 0$, $\dot{n}_5 = \dot{n}_3n_4 + n_3\dot{n}_4 = (\gamma_R+\nu_m)$. $\partial\mathcal{R}_0/\partial\nu = (\dot{n}_2n_5 - n_2\dot{n}_5)/n_5^2 = (c\beta(\nu+\nu_m)(\gamma_R+\nu_m) - c\beta\nu(\gamma_R+\nu_m))/n_5^2 = c\beta(\gamma_R+\nu_m)\nu_m/[(\nu+\nu_m)^2(\gamma_R+\nu_m)^2] = c\beta\nu_m/[(\nu+\nu_m)^2(\gamma_R+\nu_m)]$. Arithmetic operations: approximately 8 multiplications, 4 additions.

(c) **Reverse mode for all 5 partial derivatives.** Seed $\bar{R}_0 = 1$. Backpropagate: $\bar{n}_2 = \bar{R}_0/n_5 = 1/n_5$, $\bar{n}_5 = -\bar{R}_0n_2/n_5^2 = -\mathcal{R}_0/n_5$. $\bar{n}_3 = \bar{n}_5n_4$, $\bar{n}_4 = \bar{n}_5n_3$. $\bar{n}_1 = \bar{n}_2\nu$, $\bar{\nu} = \bar{n}_2n_1 + \bar{n}_3$. $\bar{c} = \bar{n}_1\beta$, $\bar{\beta} = \bar{n}_1c$. $\bar{\gamma_R} = \bar{n}_4$, $\bar{\nu_m} = \bar{n}_3 + \bar{n}_4$. One backward pass gives all 5 derivatives simultaneously.

(d) **Normalized SI tornado plot.** Using interpretable parameters $\tau_E = 1/\nu$, $\tau_I = 1/\gamma_R$, $\tau_m = 1/\nu_m$: $\mathcal{S}_c^{\mathcal{R}_0} = 1$, $\mathcal{S}_\beta^{\mathcal{R}_0} = 1$, $\mathcal{S}_{\tau_E}^{\mathcal{R}_0} = -\tau_E/(\tau_E + \tau_m)$, $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = \tau_m/(\tau_I + \tau_m)$, $\mathcal{S}_{\tau_m}^{\mathcal{R}_0} \approx 0$ (for $\tau_m \gg \tau_E, \tau_I$). Read the formulas at $\tau_m = 10{,}000$ days, $\tau_I = 7$ days, $\tau_E = 3$ days: $\mathcal{S}_{\tau_I}^{\mathcal{R}_0} = \tau_m/(\tau_I + \tau_m) = 0.9993$, so τ_I is as influential as c and β, not negligible. Only τ_E is small, at -0.0003. The tornado plot therefore shows $c \approx \beta \approx \tau_I \gg |\tau_E|$. The long demographic timescale is what makes τ_I matter: when $\tau_m \gg \tau_I$ almost every infective recovers rather than dying, so the infectious period sets the reproduction number almost in full.

Checking your answer: The reverse-mode computation (part c) is exactly the adjoint method applied to a scalar computation graph. This exercise bridges the algebraic adjoint (Chapter 5) with automatic differentiation (Chapter 7).

Exercise 5.6 *[Bloom L5]*

Exercise statement. A water-quality model of a river system has $m = 40$ parameters (pollution source rates, degradation rates, flow speeds, mixing coefficients) and $r = 2$ response functionals: (1) pollutant concentration at a downstream monitoring station, and (2) total pollutant load entering a protected estuary.

Design a complete SA study using the adjoint method. Specify:

(a) The form of the adjoint equation for this model.

(b) How many adjoint solves are required and why.

(c) How you would validate the implementation.

(d) How you would use the sensitivity results to advise regulators on which pollution sources to prioritize.

(e) One situation where the adjoint results might be misleading, and what you would do to detect and correct it.

Solution.

Adjoint SA design for river water-quality model: $m = 40$, $r = 2$.

(a) **Form of the adjoint equation.** The river-quality model is a system of ODEs or PDEs. For a generic ODE model $\dot{\vec{u}} = \vec{F}(\vec{u}; \vec{p})$ with response $J_k = \int_0^T g_k(\vec{u})\, dt + h_k(\vec{u}(T))$, the adjoint ODE for response k is:

$$-\dot{\vec{\lambda}}^{(k)} = J_F^T(\vec{u}(t))\, \vec{\lambda}^{(k)} + \nabla_{\vec{u}} g_k(\vec{u}(t)), \qquad \vec{\lambda}^{(k)}(T) = \nabla_{\vec{u}} h_k(\vec{u}(T)).$$

Two adjoint equations (one per QOI), both depending on the stored forward solution $\vec{u}(t)$.

(b) **Number of adjoint solves: 2.** One adjoint solve per response functional. Each solve runs backward from $t = T$ to $t = 0$, giving $\vec{\lambda}^{(k)}(t)$. Total cost: 1 forward solve (store) + 2 adjoint solves, so 3 in all. Compare with FSE: 1 + 40 forward solves, so 41. The adjoint is

$41/3 \approx 14\times$ cheaper. Comparing only the sensitivity solves, $40/2 = 20$, overstates the saving by counting the baseline solve on neither side.

(c) **Validation.** Verify that the adjoint-computed $\partial J_k/\partial p_j$ matches centered finite differences for all 40 parameters: use $h = 6 \times 10^{-6}|p_j^*|$ and check agreement to 5 significant figures. Also verify the adjoint at a simpler test case with analytic sensitivity (e.g., a single-compartment degradation model where J and $\partial J/\partial p$ are known analytically).

(d) **Advising regulators.** Produce two tornado plots (one per QOI) showing all 40 parameters sorted by $|\mathcal{S}_{p_j}^{J_k}|$, the normalized index, not by the raw derivative $|\partial J_k/\partial p_j|$: the 40 parameters carry different units, so raw magnitudes are not comparable and a plot ordered by them would rank by choice of units. Annotate each parameter with its plain-language interpretation and the regulatory lever available: "Reducing phosphorus loading at source i by 10% reduces downstream concentration by approximately 2.3%." Identify the top 5 sources for each QOI and report whether the same parameters appear in both tornado plots (co-optimization opportunity) or different ones (trade-off between the two monitoring objectives).

(e) **Potential misleading results.** If the river model has a threshold effect (e.g., dissolved oxygen below a critical level triggers biological collapse), the adjoint, which linearizes around the current state, may show small SI values for parameters that would be critical if the threshold were crossed. To detect: run the model at several nominal parameter sets spanning the plausible range; if any set pushes the model near the threshold, the adjoint at the current operating point is not representative. Correction: report sensitivity results at multiple operating points and flag which parameters bring the system close to critical thresholds.

Checking your answer: A complete answer requires all five components with operational specificity. The validation plan (part c) must include both a numerical check (FD agreement) and an analytic check (simple test case). Vague statements like "validate the code" are insufficient.

Exercise 5.7 *[Bloom L6]*

Exercise statement. A colleague states: "The adjoint always gives the same answer as forward SA, just faster. So if we already have the FSE code working, there is no reason to implement the adjoint."

Evaluate this claim carefully.

(a) Is the first sentence (same answer, just faster) correct? Give a precise mathematical justification.

(b) Is the second sentence (no reason to implement the adjoint) correct? Describe at least two situations where switching from FSE to adjoint would be strongly advisable.

(c) For the FSE to be preferred over the adjoint, what relationship between m and r must hold? Give a specific numerical example where FSE is strictly better.

Solution.

(a) **Is the first sentence correct?**

Yes, with an important qualifier. Both FSE and adjoint compute $\partial J/\partial p_j$ for all j, and the mathematical result is identical: $\partial J/\partial p_j = \vec{v}^T \partial\vec{b}/\partial p_j$ (adjoint) $= \vec{c}^T(\partial\vec{u}/\partial p_j)$ (FSE), and these two expressions are equal because $\vec{c}^T A^{-1} = \vec{v}^T$ (by definition of the adjoint). So the *answer* is identical. The adjoint is faster when $m \gg r$; if $m < r$, the FSE is actually faster. "Just faster" is only correct in the typical regime $m \gg r$.

(b) **Is the second sentence correct?**

No, for at least two reasons.

Situation 1: The number of parameters grows. If the model is extended from $m = 10$ to $m = 200$ parameters (e.g., adding spatially distributed source terms), the FSE cost grows as 200 solves while the adjoint remains at $r + 1$ solves. At $m = 200$ and $r = 1$, switching from FSE to adjoint gives a 100× speedup.

Situation 2: Multiple response functionals for a well-optimized adjoint. If a colleague needs to compute sensitivities for an additional 5 QOIs on the same model, the FSE cost is unchanged (still $m + 1$ solves) while the adjoint adds only 5 more backward solves. For models where solving the adjoint ODE is cheap compared to evaluating $\partial\vec{f}/\partial p_j$ for all j, the adjoint is the only viable option at large m.

(c) **When is FSE preferred?**

FSE is preferred when $m < r$: the number of parameters is less than the number of response functionals. FSE costs $m + 1$ solves; adjoint costs $r + 1$ solves.

Specific example where FSE is strictly better: $m = 3$ parameters (e.g., a simple SIR model with c, β, τ_R), $r = 50$ response functionals (sensitivity at 50 time points, or for 50 spatial locations). FSE: 4 solves. Adjoint: 51 solves. FSE is $51/4 \approx 13\times$ faster.

Checking your answer: The key insight is that "adjoint is always better" is a myth. The correct statement is: use the approach whose cost ($\min(m, r) + 1$ solves) is smaller. This is Table 5.1 in the text. Students who say only "adjoint is better when m is large" miss the symmetric counterexample. A complete answer requires a specific numerical example in part (c).

6 Chapter 6: The Adjoint ODE

Chapter Summary: The Adjoint for Dynamical Systems

The adjoint ode. For $\dot{\vec{u}} = F(\vec{u}; \vec{p})$ with response $J = \int_0^T g(\vec{u}, t)\, dt + h(\vec{u}(T))$, the adjoint variable $\vec{\lambda}(t) \in \mathbb{R}^n$ satisfies

$$-\dot{\vec{\lambda}} = J_F^\top \vec{\lambda} + \nabla_{\vec{u}} g^\top,$$

run *backward* from $t = T$ to $t = 0$. Terminal condition: $\vec{\lambda}(T) = +\nabla_{\vec{u}} h|_{t=T}$.

Three-case terminal condition recipe. (1) Terminal cost $J = h(\vec{u}(T))$: set $\vec{\lambda}(T) = +\nabla h(\vec{u}(T))$. Example: $J = S(T)$ gives $\lambda_S(T) = +1$, rest zero.
(2) Running cost $J = \int_0^T g\,dt$ (no terminal term): set $\vec{\lambda}(T) = \vec{0}$. Example: $J = \int_0^T I\,dt$.
(3) Point evaluation $J = \vec{c}^\top \vec{u}(T)$: set $\vec{\lambda}(T) = +\vec{c}$.
These combine additively for mixed objectives.

The sensitivity formula. Once the adjoint is solved,

$$\frac{dJ}{dp_j} = +\int_0^T \vec{\lambda}^\top \frac{\partial F}{\partial p_j}\,dt,$$

one integral per POI at negligible cost, since $\vec{\lambda}(t)$ is already known. Total cost: 1 forward solve + 1 adjoint solve + m scalar integrals.

Biological interpretation of $\lambda_I(t)$. For the SIR model with total burden $J = \int_0^T I\,dt$, the adjoint variable $\lambda_I(t)$ measures the marginal downstream burden caused by one additional infectious individual at time t. Near $t = 0$, each additional infection propagates through the entire outbreak: $\lambda_I(0) \approx$ mean future burden per infection. Near $t = T$, the outbreak is resolving: $\lambda_I(T) = 0$.

Watch out: Why does the adjoint run backward? Not because of numerical convenience, but because the terminal condition is imposed at $t = T$ by the response functional $h(\vec{u}(T))$. The adjoint ODE carries information about the future cost of present infections, so it must know the end state before it can evaluate the beginning.

Exercise 6.1 *[Bloom L3]*

Exercise statement. For the scalar ODE $\dot{u} = ru(1 - u/K)$ (logistic) with $J = \int_0^T u\,dt$ (total population over time):

(a) Write the adjoint ODE and terminal condition.

(b) Show that the adjoint ODE is linear in $\lambda(t)$ even though the forward ODE is nonlinear in $u(t)$.

(c) Explain why the adjoint ODE depends on the forward solution $u(t)$.

(d) Solve the forward and adjoint ODEs simultaneously (forward from 0 to T; store; then backward from T to 0) for $r = 0.5$, $K = 100$, $u_0 = 10$, $T = 20$. Plot $u(t)$ and $\lambda(t)$.

Solution.

Logistic ODE: $\dot{u} = ru(1 - u/K)$, $J = \int_0^T u\,dt$. Here $F(u; r, K) = ru(1 - u/K)$, $g(u) = u$, $h = 0$.

(a) **Adjoint ODE and terminal condition.**

Running cost gradient: $\partial g/\partial u = 1$. State Jacobian: $F_u = r(1 - 2u/K)$. Adjoint ODE (backward):

$$-\dot{\lambda} = F_u\,\lambda + \frac{\partial g}{\partial u} = r\left(1 - \frac{2u(t)}{K}\right)\lambda + 1, \quad \lambda(T) = 0.$$

Equivalently (forward-time form for backward integration: reverse tspan):

$$\dot{\lambda} = -r\left(1 - \frac{2u(t)}{K}\right)\lambda - 1.$$

(b) **Linearity in λ.**

The adjoint ODE is linear in λ: $\dot{\lambda} = -r(1-2u(t)/K)\lambda-1$. Its coefficient is $-r(1-2u(t)/K)$, which depends on $u(t)$ but not on λ itself. The forcing term is -1. This is a first-order linear ODE in λ, even though the forward logistic ODE is nonlinear in u. This linearity is guaranteed by the Lagrange multiplier derivation of Section 6.3: the adjoint is always the adjoint of the *linearized* forward equation.

(c) **Why the adjoint depends on $u(t)$.**

The adjoint ODE measures how sensitive J is to perturbations of the state at time t. This sensitivity depends on how the forward dynamics amplify or dampen perturbations between time t and T, which in turn depends on $u(t)$ through the coefficient $F_u = r(1 - 2u/K)$. Near the carrying capacity ($u \approx K$), $F_u \approx -r < 0$: perturbations decay. Near $u = 0$, $F_u \approx r > 0$: perturbations grow. The adjoint must account for this varying amplification.

(d) **MATLAB implementation.**

```
# Parameters
r=0.5;  K=100;  u0=10;  T=20

# Step 1: Forward solve - store dense trajectory
FUNCTION fwd_rhs(t, u):   RETURN r*u*(1 - u/K)   END

U = SOLVE_ODE(fwd_rhs, tspan=[0, T], y0=u0,
              tol_rel=1e-8, tol_abs=1e-10, dense_output=True)
# U(t) now available at any t in [0,T] via interpolation

# Step 2: Backward adjoint solve (PLUS convention)
# Adjoint RHS: -d(lam)/dt = r*(1 - 2*U(t)/K)*lam + 1
FUNCTION adj_rhs(t, lam):
    RETURN -r*(1 - 2*U(t)/K)*lam - 1   # lambda_dot
END

lam_T = 0                               # terminal condition
LAM = SOLVE_ODE(adj_rhs, tspan=[T, 0], y0=lam_T,
                tol_rel=1e-8, tol_abs=1e-10)

# Step 3: Sensitivity integrals (PLUS convention: dJ/dp = +integral)
t_grid = forward time grid
```

```
J      = INTEGRATE(U(t_grid), t_grid)   # J = integral of u

# dJ/dr: partial F/partial r = u*(1 - u/K)
dJ_dr  = INTEGRATE(LAM(t_grid) * U(t_grid) * (1 - U(t_grid)/K), t_grid)
SI_r   = (r / J) * dJ_dr

# dJ/dK: partial F/partial K = r*u^2/K^2
dJ_dK  = INTEGRATE(LAM(t_grid) * r * U(t_grid)^2 / K^2, t_grid)
SI_K   = (K / J) * dJ_dK
```

At nominal values, $\lambda(t)$ is large near $t = 0$ (early population growth is more influential for the integral $J = \int u\,dt$) and decreases toward $t = T$ as the population saturates at K.

Checking your answer: Students often get the sign of the adjoint ODE wrong. The forward-time form is $\dot{\lambda} = -r(1 - 2u/K)\lambda - 1$; using tspan $= [T, 0]$ automatically handles backward integration. Students who negate the RHS AND use backward tspan are double-counting the sign. A complete answer gives the forward-time form and integrates backward without also negating the right-hand side; at minimum, the correct terminal condition. The key test: at T, $\lambda(T) = 0$; at $t = 0$, $\lambda(0)$ should be large and positive.

Exercise 6.2 *[Bloom L3]*

Exercise statement. State in your own words what integration by parts accomplishes in Step 3 of the Lagrange multiplier derivation, and why this is analogous to the matrix transposition $A \to A^T$ in Chapter 5. Then verify equation (6.5) directly by differentiating the product λw and integrating over $[0, T]$.

Solution.

What integration by parts accomplishes:

Step 3 transfers the time derivative d/dt from the sensitivity variable $w(t)$ (which we want to eliminate) onto the adjoint variable $\lambda(t)$ (which we can freely define). This is what the matrix transpose accomplishes algebraically: if $A\vec{w} = \vec{f}$ is the FSE, then integrating $\vec{\lambda}^T(A\vec{w})$ by parts shifts the differential operator from $\vec{w}$ to $\vec{\lambda}$, resulting in $(-\dot{\vec{\lambda}} - A^T\vec{\lambda})$ multiplying $\vec{w}$. Setting this coefficient to zero defines the adjoint ODE, making the $\vec{w}$-integral vanish.

Analogy with transposition:

In the algebraic setting (Chapter 5), the FSE is $A\vec{w} = \vec{f}$ and the adjoint is $A^T\vec{v} = \vec{c}$. The sensitivity formula comes from: $\vec{c}^T\vec{w} = (A^T\vec{v})^T\vec{w} = \vec{v}^T(A\vec{w}) = \vec{v}^T\vec{f}$. The transpose moves the operator from $\vec{w}$ to $\vec{v}$. Integration by parts does the same for the differential operator: $\int \lambda\,\dot{w}\,dt = [\lambda w]_0^T - \int \dot{\lambda}\,w\,dt$ moves the time derivative from w to λ. The ODE adjoint is the continuous analog of the matrix transpose.

Direct verification of equation (6.5):

Let $p(t) = \lambda(t)w(t)$. Differentiate by product rule: $\dot{p} = \dot{\lambda}w + \lambda\dot{w}$. Integrate over $[0, T]$:

$$\int_0^T (\dot{\lambda}w + \lambda\dot{w})\,dt = [p(t)]_0^T = \lambda(T)w(T) - \lambda(0)w(0).$$

Rearranging:

$$\int_0^T \lambda \dot{w}\, dt = \lambda(T)w(T) - \lambda(0)w(0) - \int_0^T \dot{\lambda} w\, dt. \checkmark$$

This is equation (6.5) (the integration-by-parts identity for ODE adjoints).

Checking your answer: The key pedagogical connection is: transpose (Chapter 5) and integration by parts (Chapter 6) are both instances of Unifying Idea 2. Common error: students memorize the IBP formula without connecting it to the matrix transpose; they can reproduce the formula but cannot explain why the adjoint ODE is the continuous analog of A^T. At minimum: correct IBP identity reproduced and applied to move the time derivative from w to λ. A complete answer: explicit statement linking the terminal boundary term $\lambda(T)w(T) - \lambda(0)w(0)$ to the boundary conditions and explaining why this is the continuous analog of the matrix transpose. The distinguishing check: stating the connection to Chapter 5's transpose is the complete answer; reproducing the IBP formula mechanically, without that connection, is partial.

Exercise 6.3 *[Bloom L4]*

Exercise statement. Using your MATLAB implementation from the laboratory:

(a) Compute $\mathcal{S}^J_\beta$, $\mathcal{S}^J_c$, $\mathcal{S}^J_{\tau_R}$, $\mathcal{S}^J_{\tau_m}$ via the adjoint for $J = \int_0^{90} I\, dt$.

(b) Verify that $\mathcal{S}^J_c = \mathcal{S}^J_\beta$ to at least 6 significant figures. Explain this result using the structure of the SIR model.

(c) If you could implement one intervention that reduces a single parameter by 20%, which parameter would you target to most reduce the total disease burden? Justify using the tornado plot.

(d) Compare the computation time with the FSE approach from Chapter 4.

Solution.

Using `run_sir_adjoint(sir_nominal())` from the companion notebooks, with $J = \int_0^{90} I\, dt$.

(a) **Adjoint-computed SI values (nominal parameters).**

Parameter	$\partial J/\partial p_j$ (raw)	$\mathcal{S}^J_{p_j}$
c	$+848.13$	$+0.7467$
β	$+70{,}677.53$	$+0.7467$
τ_R	$+1322.55$	$+1.6302$
τ_m	-4.5×10^{-4}	-7.9×10^{-4}

(Values from `run_sir_adjoint(sir_nominal())` for the demographic SIR model of this chapter, $J = 5679.02$. Note $\partial J/\partial\beta \big/ \partial J/\partial c = c/\beta = 83.33$ exactly, which is the structural check behind part (b). Because $\nu_m = 1/\tau_m$ appears in all three equations, τ_m has a nonzero sensitivity; at $\tau_m = 10{,}000$ days it is only -7.9×10^{-4}, three orders of magnitude below the others, because almost no one leaves the population during a 90-day outbreak. It is small for a reason, not zero by structure.)

(b) **Why $\mathcal{S}_c^J = \mathcal{S}_\beta^J$.**

In the SIR model, c and β appear only in the product $c\beta$. Any perturbation to c has the identical structural effect on the ODE right-hand side as a proportional perturbation to β. Since the model is invariant to the labeling $c \leftrightarrow \beta$ (they are not separately identifiable from outbreak data alone), their normalized SIs must be identical. Mathematically: $\partial F/\partial c = \beta SI/N$ and $\partial F/\partial \beta = cSI/N = (c/\beta)(\partial F/\partial c)$, so after normalization the sensitivities cancel out and become equal.

(c) **Optimal intervention.**

The tornado plot shows τ_R largest, with c and β tied behind it at roughly half its magnitude. If one intervention reduces a single parameter by 20%, the first-order reduction in J (total infectious burden) is approximately $20\% \times |\mathcal{S}_{p_j}^J|$ for each: τ_R: $\approx 32.6\%$; c or β: $\approx 14.9\%$; τ_m: $\approx 0.02\%$, negligible. **Target** τ_R, the mean infectious period, shortened by earlier detection and isolation. Only if that is impractical should the intervention fall back to c (contact rate, reduced by distancing) or β (transmission probability per contact, reduced by masks or hygiene), which are equally effective as each other but each about half as effective as τ_R. A caution worth stating to students: a 32.6% predicted change is well outside the range where a local index is reliable. The index ranks the candidates; it does not forecast the outcome of a 20% intervention. Confirm the winner by re-solving the model at the perturbed value.

(d) **Cost comparison with FSE.**

FSE for the time-dependent SI $\mathcal{S}_{p_j}^I(t)$: one augmented ODE solve ($3+4\times 3 = 15$ equations). Adjoint: one forward solve + one adjoint solve (each with 3 equations). For this small model they are comparable; the adjoint becomes much faster for models with $m \gg r$.

Checking your answer: The equality $\mathcal{S}_c^J = \mathcal{S}_\beta^J$ is a structural property of the SIR model that students should identify and explain, not just report. This connects to the identifiability material from Chapter 3: A complete answer explains the equality as non-identifiability rather than reporting it; at minimum, flagging it as structural rather than numerical. c and β are not separately identifiable from $I(t)$ data.

Exercise 6.4 *[Bloom L4]*

Exercise statement. Fill in the computation cost table for the SIR model with $r = 1$ response functional ($J = \int I\,dt$) and $n = 3$ state variables:

m (number of POIs)	FSE cost (ODE solves)	Adjoint cost (ODE solves)
4		
10		
50		
500		

If each ODE solve takes 30 seconds, at what value of m does the adjoint become more than 10 times faster than the FSE?

Solution.

FSE cost: $m+1$ ODE solves. Adjoint cost: $r+1=2$ ODE solves (1 forward + 1 adjoint for $r=1$ QOI). Each solve: 30 seconds.

m	FSE cost (solves)	Adjoint cost (solves)	Speedup
4	5	2	2.5×
10	11	2	5.5×
50	51	2	25.5×
500	501	2	250.5×

At what m is adjoint $> 10\times$ faster?

Adjoint time: $2 \times 30 = 60$ s. FSE time: $(m+1) \times 30$ s. Speedup $> 10\times$ when $(m+1)/2 > 10$, i.e., $m > 19$. At $m = 20$: FSE $= 630$ s, adjoint $= 60$ s, speedup $= 10.5\times$. The adjoint becomes $> 10\times$ faster at $\mathbf{m \geq 20}$ **parameters.**

Checking your answer: Many students write "the adjoint becomes better at $m \geq 2$" (technically true since adjoint costs $r+1$ vs $2m+1$ for centered FD, so adjoint wins when $m > r$), but miss the implementation cost. Common error: ignoring the implementation effort to derive and code the adjoint ODE; stating adjoint is always preferred without noting the practical threshold. At minimum: correct crossover condition ($m > r$) with one example. A complete answer: crossover condition + acknowledgment of implementation cost + one case where forward SA is still preferred despite $m > r$ (e.g., black-box models without access to the ODE right-hand side). The distinguishing check: the best answers note that the adjoint has a fixed one-time implementation cost that amortizes across multiple studies, while forward FD has zero implementation cost.

Exercise 6.5 *[Bloom L4]*

Exercise statement. Explain in plain language, without any equations, why:

(a) The adjoint ODE must be integrated backward from $t = T$ rather than forward from $t = 0$.

(b) The adjoint variable $\lambda_I(t)$ for the SIR model and response $J = \int I\, dt$ is larger near $t = 0$ than near $t = T$.

(c) A public health agency that wants to minimize total disease burden should focus interventions on the early phase of the epidemic, when λ_I is large. Use $\lambda_I(t)$ to quantify this recommendation.

(d) The terminal condition $\vec{\lambda}(T) = \vec{0}$ is not the same as running the forward ODE backward with initial condition $\vec{0}$.

Solution.

(a) **Why integrate backward from T?**

The adjoint variable $\lambda(t)$ answers the question: "If the state at time t were perturbed, how much would J change?" This question is inherently backward in time, the effect of a perturbation at time t propagates forward to T, and we need to know what happened between t and T to answer it. At T itself, the effect on J of a perturbation to $u(T)$ is determined directly by how J depends on the final state. Working backward from this known terminal value, we accumulate the influence of perturbations at each earlier time. Forward integration from $t = 0$ would require knowing the effect of future perturbations before we have computed them.

(b) **Why $\lambda_I(t)$ is larger near $t = 0$?**

$\lambda_I(t)$ measures how sensitive $J = \int I\,dt$ is to a perturbation of I at time t. A perturbation at $t = 5$ days has 85 days to propagate through the epidemic dynamics and affect the total infectious burden. A perturbation at $t = 85$ days has only 5 days left; most of the integral J has already been accumulated. So early perturbations matter more, and $\lambda_I(5) > \lambda_I(85)$.

(c) **Early intervention recommendation.**

From the adjoint solution at nominal parameters, $\lambda_I(5) \approx L_{\max}$ while $\lambda_I(80) \approx 0.1\,L_{\max}$. A 10% reduction in the contact rate c applied only during days 0–10 reduces J by approximately $10\% \times |\mathcal{S}_c^J| \times$(effective weight at $t = 5$), which is proportionally larger than the same reduction applied during days 80–90. The practical recommendation: interventions (school closures, event cancellations) are most effective when implemented in the first two weeks of an outbreak.

(d) **Terminal condition vs. reverse-time forward.**

Running the forward ODE backward from $\vec{u}(T)$ with initial condition $\vec{0}$ would give a solution to $\dot{\vec{z}} = -\vec{F}(\vec{z})$, which is the time-reversal of the epidemic, unphysical and unrelated to the adjoint. The adjoint terminal condition $\vec{\lambda}(T) = \nabla_{\vec{u}} h(\vec{u}(T))$ encodes the sensitivity of J to the final state, which is determined by the structure of the response functional, not by the forward dynamics. For $J = \int I\,dt$ with no terminal cost, $\vec{\lambda}(T) = \vec{0}$ because J does not directly depend on the final state; the adjoint ODE then builds up the historical influence backward through the epidemic.

Checking your answer: Parts (a) and (d) are the most conceptually challenging in the chapter. Common error for (d): students say both approaches "start from zero" and cannot distinguish them. A complete answer distinguishes the two by what each requires in memory; at minimum, noting that one needs the forward trajectory and the other does not. The key is that the adjoint ODE coefficients depend on the *forward* solution $u(t)$, which must be stored and accessed backward. Simply integrating the forward ODE backward yields a physically different trajectory.

Exercise 6.6 *[Bloom L5]*

Exercise statement. Extend the SIR adjoint treatment to the SEIR model:

$$
\begin{aligned}
\dot{S} &= \nu_m N - c\beta SI/N - \nu_m S, \\
\dot{E} &= c\beta SI/N - (\nu + \nu_m)E, \\
\dot{I} &= \nu E - (\gamma_R + \nu_m)I, \\
\dot{R} &= \gamma_R I - \nu_m R,
\end{aligned}
$$

with POIs $\vec{p} = (c, \beta, \tau_E, \tau_I, \tau_m)$ (interpretable parameterization: $\nu = 1/\tau_E$, $\gamma_R = 1/\tau_I$, $\nu_m = 1/\tau_m$) and response functional $J = \int_0^T I(t)\,dt$.

Design the complete adjoint SA study, specifying:

(a) The 4×4 Jacobian J_F at a generic point (S, E, I, R).

(b) All four adjoint ODEs and their terminal conditions.

(c) The sensitivity formulas dJ/dp_j for all five POIs.

(d) The MATLAB implementation strategy (how you would store the forward solution and integrate the adjoint backward).

(e) What additional cost the SEIR adjoint requires compared to the SIR adjoint (in terms of ODE solves and storage).

Solution.

SEIR model with $\vec{p} = (c, \beta, \tau_E, \tau_I, \tau_m)$, $J = \int_0^T I(t)\,dt$.

(a) **4×4 Jacobian J_F.**

Let $\gamma_R = 1/\tau_I$, $\nu = 1/\tau_E$, $\nu_m = 1/\tau_m$.

$$
J_F = \frac{\partial(F_S, F_E, F_I, F_R)}{\partial(S, E, I, R)} = \begin{pmatrix} -\lambda - \nu_m & 0 & -c\beta S/N & 0 \\ \lambda & -(\nu + \nu_m) & c\beta S/N & 0 \\ 0 & \nu & -(\gamma_R + \nu_m) & 0 \\ 0 & 0 & \gamma_R & -\nu_m \end{pmatrix}
$$

where $\lambda = c\beta I/N$ (force of infection). The entry $\partial F_E/\partial I = c\beta S/N$ arises because $F_E = \lambda S = c\beta SI/N$, which is linear in I; there is no $-c\beta I/N$ term since F_E contains no I^2 contribution.

(b) **Four adjoint ODEs and terminal conditions.**

For $J = \int I\,dt$: $\partial g/\partial \vec{u} = (0, 0, 1, 0)^T$, $h = 0$. Terminal conditions: $\vec{\lambda}(T) = \vec{0}$. Adjoint ODEs:

$$
\begin{aligned}
-\dot{\lambda}_S &= \left(-\frac{c\beta I}{N} - \nu_m\right)\lambda_S + \frac{c\beta I}{N}\,\lambda_E \\
-\dot{\lambda}_E &= -(\nu + \nu_m)\lambda_E + \nu\lambda_I \\
-\dot{\lambda}_I &= -\frac{c\beta S}{N}\lambda_S + \frac{c\beta S}{N}\lambda_E - (\gamma_R + \nu_m)\lambda_I + \gamma_R\lambda_R + 1 \\
-\dot{\lambda}_R &= -\nu_m\lambda_R
\end{aligned}
$$

(c) **Sensitivity formulas.**
$dJ/dp_j = +\int_0^T \vec{\lambda}^T(\partial\vec{F}/\partial p_j)\,dt$. Computing $\partial\vec{F}/\partial p_j$:
$dJ/dc = +\int_0^T(-\lambda_S + \lambda_E)(\beta \cdot SI/N)\,dt$. $dJ/d\beta$: identical form with c and β exchanged.
$dJ/d\tau_E$: since $\nu = 1/\tau_E$, we have $\partial F_E/\partial\tau_E = +E/\tau_E^2$ (the $-\nu E$ term in F_E) and $\partial F_I/\partial\tau_E = -E/\tau_E^2$ (the $+\nu E$ term in F_I). All other components of $\partial\vec{F}/\partial\tau_E$ are zero.

$$dJ/d\tau_E = +\int_0^T \big(\lambda_E - \lambda_I\big)\frac{E}{\tau_E^2}\,dt.$$

$dJ/d\tau_I$: since $\gamma_R = 1/\tau_I$, we have $\partial F_I/\partial\tau_I = +I/\tau_I^2$ and $\partial F_R/\partial\tau_I = -I/\tau_I^2$. Thus:

$$dJ/d\tau_I = +\int_0^T \big(\lambda_I - \lambda_R\big)\frac{I}{\tau_I^2}\,dt.$$

$dJ/d\tau_m$: differentiating $\nu_m = 1/\tau_m$ gives $\partial\nu_m/\partial\tau_m = -1/\tau_m^2$. For the susceptible compartment $F_S = \nu_m N - c\beta SI/N - \nu_m S$, the τ_m-derivative is $\partial F_S/\partial\tau_m = (S-N)/\tau_m^2$ (the birth term $\nu_m N$ contributes $-N/\tau_m^2$, the death term $-\nu_m S$ contributes $+S/\tau_m^2$). For the other compartments only death terms appear: $\partial F_j/\partial\tau_m = +u_j/\tau_m^2$. Thus:

$$dJ/d\tau_m = +\int_0^T \big[\lambda_S(S-N) + \lambda_E E + \lambda_I I + \lambda_R R\big]/\tau_m^2\,dt.$$

(d) **MATLAB implementation strategy.** Forward solve: 4-equation SEIR ODE, save dense output. Build interpolants for all 4 state variables. Backward solve: 4-equation adjoint ODE, using stored interpolants for $S(t), E(t), I(t), R(t)$. Compute integrals via `trapz` on forward time grid.

(e) **Additional cost vs. SIR adjoint.** The SEIR adjoint requires 4 equations vs. 3 for SIR (one additional adjoint variable λ_E). Storage rises by a third, from $\sim 3 \times N_t$ to $\sim 4 \times N_t$ values. The computational cost is essentially identical: one additional ODE component adds negligible time to the adjoint solve. The main additional cost is writing and verifying the 4×4 Jacobian and 5 forcing terms.

Checking your answer: Students who write J_F^T and then compute the adjoint ODE by substituting into the formula $-\dot{\vec{\lambda}} = J_F^T\vec{\lambda} + \nabla_{\vec{u}}g$ should get exactly the four equations in part (b). A complete answer produces the four equations with J_F evaluated along $\vec{u}(t)$; at minimum, the correct transpose structure. Common error: forgetting that J_F is evaluated along the forward solution $\vec{u}(t)$, not at a fixed point.

Exercise 6.7 *[Bloom L6]*

Exercise statement. A climate modeler reports: "We implemented the adjoint to compute sensitivities for our model with 200 parameters, but the results disagree with forward sensitivity analysis by approximately 15% for some parameters. The adjoint must have a bug."

Identify three distinct causes that could produce this discrepancy without any coding error. For each cause:

(a) State the cause precisely.

(b) Describe a diagnostic test that would confirm or rule it out.

(c) Describe the correction if the cause is confirmed.

Solution.

Three causes of a 15% discrepancy between adjoint and FD, without a coding error:

Cause 1: Inconsistent step sizes in the forward/backward solvers.

The adjoint ODE depends on the stored forward solution $\vec{u}(t)$ through interpolation. If the forward ODE is solved with a loose tolerance (e.g., `RelTol = 1e-3`) but the adjoint is solved tightly, the interpolated $\vec{u}(t)$ values contain errors proportional to the forward tolerance, which propagate into the adjoint sensitivity.

Diagnostic: Tighten the forward solver tolerance by 10^2 and recompute. If the adjoint sensitivities change by more than 1%, forward solver error is the cause.

Correction: Use tight tolerances ($\leq 10^{-8}$) for both forward and adjoint solves. Consider a fixed-step integrator if adaptive stepping causes inconsistency.

Cause 2: Poor interpolation of the forward solution.

The adjoint ODE is integrated backward and must evaluate $\vec{u}(t)$ at times not necessarily on the forward solver's output grid. Linear interpolation (the default in some implementations) introduces $O(h^2)$ errors in the adjoint coefficients; for stiff models with rapidly varying states, this can cause $> 10\%$ errors in sensitivities.

Diagnostic: Compare adjoint sensitivities using linear, cubic (PCHIP), and quintic interpolation of the stored forward solution. If results change by more than 1%, interpolation order is the cause.

Correction: Use a high-order dense output method (`pchip` or `griddedInterpolant` with 'spline') and store the forward solution at a fine time grid.

Cause 3: Inconsistent discretization between the adjoint and the forward solver.

The adjoint is derived from the *continuous* ODE. When it is discretized and solved numerically, the resulting discrete adjoint may not be the exact adjoint of the discrete forward solver. In particular: the forward solver may use an adaptive step-size scheme that produces different grids on the nominal vs. perturbed runs, so the "same" model evaluated at p_j and $p_j + h$ uses different internal time steps. The finite-difference estimate then compares J values computed on inconsistent grids, introducing a mismatch that persists as $h \to 0$.

For smooth models with consistent discretization, both the adjoint and centered FD converge to the same derivative as $h \to 0$; persistent disagreement at small h points to the discretization mismatch, not model nonlinearity.

Diagnostic: Fix the forward-solver time grid (use a fixed-step integrator or export the adaptive grid from the nominal run and reuse it for all perturbed runs). Recompute FD sensitivities on the fixed grid. If FD and adjoint now agree, the mismatch was caused by grid inconsistency.

Correction: Use a fixed-step integrator (e.g., RK4 or explicit Euler) or implement the *discrete adjoint*, the exact adjoint of the discretized forward scheme, rather than the continuous adjoint discretized separately.

Checking your answer: Common error: students identify Cause 3 as "the model is nonlinear so the adjoint is inaccurate." This is wrong. The adjoint computes the exact derivative of J with respect to p_j regardless of nonlinearity; it is the exact infinitesimal sensitivity. Persistent disagreement with FD at small h always indicates an implementation problem (solver tolerance, interpolation, or discretization mismatch), not a fundamental limitation of the adjoint method. At minimum: identify discretization mismatch as the cause and describe a fixed-grid diagnostic test. A complete answer: correctly distinguish continuous adjoint vs. discrete adjoint and propose the fixed-step remedy. The distinguishing check: a student who says "report both values and note the adjoint is linearized" is mischaracterizing the adjoint, a clear shortfall.

7 Chapter 7: Algorithmic Differentiation

Chapter Summary: Adjoint Sensitivity in Practice

The annuity benchmark. At $P = 1000$, $r = 0.05$, $n = 20$: $A = P(1 - (1 + r)^{-n})/r =$ 12,462.21. Sensitivity indices: $\mathcal{S}_P^A = 1$ (always exact for any P, r, n), $\mathcal{S}_r^A \approx -0.4240$ (higher interest rate reduces present value), $\mathcal{S}_n^A \approx +0.5902$ (more periods increase present value). These are the verified reference values; any implementation that disagrees has an error.

Four methods compared. *Forward mode*: one pass per POI, cost $O(m)$. *Reverse mode*: one backward pass per QOI, cost $O(1)$ in m. *Symbolic differentiation*: free at query time (cost paid once at derivation). *Finite differences*: cost $O(m)$, approximate. For $m \gg 1$ with scalar output: reverse mode dominates.

Algorithmic differentiation (ad). AD applies the chain rule to every arithmetic operation in a computer program. It is neither symbolic (manipulating formulas) nor numerical (finite differences). It produces the exact derivative, to floating-point precision, at a specific input. Forward-mode AD is equivalent to FSE; reverse-mode AD is equivalent to the adjoint.

When ad breaks. Four failure modes: (1) Adaptive-step-size ODE solvers (branching in step-size selection); (2) `if` statements / discontinuities; (3) Lookup tables and interpolation; (4) External compiled code that AD cannot trace. Remedy for (1)–(2): use fixed-step solvers inside the AD tape, or switch to the analytic adjoint from Chapter 6.

Watch out: The discrete vs. continuous adjoint distinction matters when the ODE solver uses adaptive step sizes. The *continuous adjoint* (Chapter 6) is derived from the continuous ODE and then discretized. The *discrete adjoint* is the exact adjoint of the discrete scheme. For adaptive solvers, these can differ; see Sirkes and Tziperman (1997). For this course, the continuous adjoint is the default.

Exercise 7.1 *[Bloom L3]*

Exercise statement. For the annuity function equation (7.1) at nominal values $P = 1000$, $r = 0.05$, $n = 20$:

(a) Compute A numerically.

(b) Compute all three sensitivities $\mathcal{S}_P^A$, $\mathcal{S}_r^A$, $\mathcal{S}_n^A$ analytically by differentiating equation (7.1).

(c) Implement forward mode by hand for $q = r$ (as in Section 7.1) and verify your answer from (b).

(d) Implement reverse mode for all three parameters in one pass and verify all three answers.

(e) Count the number of arithmetic operations in each approach.

Solution.

Annuity function (Equation (7.1)): $A(P, r, n) = P\,\dfrac{1-(1+r)^{-n}}{r}$ at $P = 1000$, $r = 0.05$, $n = 20$.

(a) **Numerical value.** $(1.05)^{-20} = 1/2.6533 = 0.3769$.

$$A = 1000 \times \frac{1-0.3769}{0.05} = 1000 \times \frac{0.6231}{0.05} = \mathbf{12{,}462.21}.$$

(b) **Analytic sensitivity indices.**

Index for P. Since A is linear in P, $\partial A/\partial P = A/P$, giving $\mathcal{S}_P^A = (P/A)(A/P) = \mathbf{1}$ for any P, r, n.

Index for r (log-derivative method). $\ln A = \ln P + \ln(1-(1+r)^{-n}) - \ln r$. Differentiating:

$$\mathcal{S}_r^A = \frac{d\ln A}{d\ln r} = \frac{n\,r\,(1+r)^{-n-1}}{1-(1+r)^{-n}} - 1.$$

At $r = 0.05$, $n = 20$: $(1+r)^{-n-1} = 0.3589$; numerator $= 0.3589$; denominator $= 0.6231$.

$$\mathcal{S}_r^A = 0.5762 - 1 = -\mathbf{0.4240}.$$

The negative sign is natural: higher interest rates discount future payments more heavily, reducing present value.

Index for n.

$$\mathcal{S}_n^A = \frac{n\,\ln(1+r)\,(1+r)^{-n}}{1-(1+r)^{-n}}.$$

At $r = 0.05$, $n = 20$: numerator $= 20 \times 0.04879 \times 0.3769 = 0.3679$; denominator $= 0.6231$.

$$\mathcal{S}_n^A = \frac{0.3679}{0.6231} = +\mathbf{0.5902}.$$

Each additional 1% of periods (roughly one fifth of a period) raises present value by 0.590%.

(c) **Forward mode for $\partial A/\partial r$ (seeding $\dot{r} = 1$).** Write $u = (1+r)^{-n}$, so $A = P(1-u)/r$.

$$\dot{u} = -n(1+r)^{-n-1} = -20(1.05)^{-21} = -7.178,$$
$$\dot{A} = P\,\frac{-\dot{u}\,r - (1-u)}{r^2} = 1000 \times \frac{0.3589 - 0.6231}{0.0025} = -105{,}667.$$

$\mathcal{S}_r^A = (0.05/12{,}462.21) \times (-105{,}667) = -\mathbf{0.4240}$. ✓

(d) **Reverse mode (all three in one pass).** Seed $\bar{A} = 1$. With $u = (1+r)^{-n}$:

$$\bar{u} = -P/r = -20{,}000; \quad \bar{r} = -P(1-u)/r^2 + \bar{u}\cdot(-n)(1+r)^{-n-1} = -105{,}667; \quad \bar{n} = \bar{u}\cdot(-(1+r)^{-n}\ln(1+r)) =$$

$\mathcal{S}_r^A = (r/A)\bar{r} = -0.4240$ and $\mathcal{S}_n^A = (n/A)\bar{n} = +0.5902$. ✓

(e) **Operation counts.** The function A requires $\sim$5 core operations. Forward mode per parameter: $\sim$10 operations; for $m = 3$: 30 total. Reverse mode: $\sim$20 total, independent of m. Speedup ratio for $m = 3$: $30/20 = 1.5\times$; for $m = 500$: $250\times$.

Checking your answer: A common error is to use the mortgage-payment formula $Pr(1+r)^n/[(1+r)^n - 1]$ in place of Equation (7.1). Both give $\mathcal{S}_P^A = 1$, but the signs of $\mathcal{S}_r^A$ differ: the mortgage formula gives $+0.424$ (higher rates increase payments) while the present-value formula gives -0.4240 (higher rates reduce present value). A complete answer uses Equation (7.1) and reports $\mathcal{S}_r^A = -0.4240$; checking the sign against the economic reading, that a higher discount rate lowers a present value, catches the substitution.

Exercise 7.2 *[Bloom L4]*

Exercise statement. For each of the following SA problems, select the most appropriate method from Table 7.2 and justify in two sentences:

(a) A climate model with 800 parameters written in Fortran. Each model run takes 10 hours. You want sensitivities for 2 response functionals.

(b) A pharmacokinetic model with 5 parameters and analytic closed-form solutions. You want sensitivities for 8 drug concentration time points.

(c) A traffic simulation with 40 parameters implemented as a compiled C++ executable. Parameter values can be set via a configuration file. Each run takes 3 minutes.

(d) A 10-parameter epidemic model written in a compiled language (not natively differentiable). You want one response functional. The model contains an `if` statement that triggers a quarantine intervention when $I > 0.05N$.

Solution.

The three cases span the full range of the method-selection framework: many parameters with few responses (adjoint wins), few parameters with specialized structure (monomial rule), and an intermediate case where data-driven comparison is feasible.

(a) **Climate model, 800 parameters, 2 response functionals, 10 h/run. Use adjoint algorithmic differentiation (AD).** With $m = 800$ and $r = 2$, forward sensitivity equations (FSE) costs 801 runs ($\approx$ 8010 hours) while adjoint costs 3 runs (30 hours). If the model code supports AD (e.g., via Tapenade for Fortran), implement the adjoint. If not, central-difference FSE is not feasible; use adjoint or restrict to Morris screening ($\approx$ 8000 evaluations = 80,000 hours; still infeasible). This is a case where developing the adjoint is the only practical option.

(b) **PK model, 5 parameters, 8 time points, analytic solutions. Use analytic FSE (or log-derivatives directly).** With analytic solutions, $\partial u/\partial p_j$ can be derived symbolically. No AD or numerical FD is needed. Cost: trivial. Using a symbolic differentiation tool once gives all 5×8 sensitivities.

(c) **Traffic simulation, 40 parameters, compiled C++, 3 min/run. Use centered finite differences.** AD is unavailable (compiled black box). Centered FD costs 81 runs ×3 min = 243 min $\approx$ 4 hours. Feasible overnight. Adjoint would require modifying the C++ source (possible but costly in developer time). Morris screening (15 trajectories, $\approx$ 615 runs) is also feasible if nonlinearity is a concern.

(d) **MATLAB ODE45 model, 10 parameters, 1 qoi, `if` statement. Use centered finite differences.** The `if` statement makes the model non-differentiable at the quarantine threshold; forward-mode AD and the adjoint will both fail at parameter values that trigger the discontinuity. Centered differences detect the discontinuity as a large error indicator (the left and right estimates diverge near threshold). Cost: 21 runs. Use the U-curve diagnostic to identify non-smooth parameters.

Checking your answer: Part (d) is the key discriminating item: students who recommend AD or adjoint without noting the `if`-statement issue have not read the question carefully. Chapter 8 covers this exact failure mode. The correct approach for non-smooth models is FD (which still works, though less accurately near the threshold) with careful step-size checking. A complete answer names the `if`-statement as the obstacle before recommending any method. At minimum: recognizing that the model is non-smooth at the branch, whatever method is then proposed.

Exercise 7.3 *[Bloom L4]*

Exercise statement. Using automatic differentiation (AD) with your preferred tool (e.g., PyTorch autograd, JAX, Julia Zygote, or MATLAB Deep Learning Toolbox):

(a) Wrap the function $\mathcal{R}_0(c, \beta, \tau_R) = c\beta\tau_R$ as a differentiable computation and use reverse-mode AD to compute all three partial derivatives at $c = 5$, $\beta = 0.06$, $\tau_R = 7$.

(b) Compare with the analytic derivatives.

(c) Wrap the SIR equilibrium calculation (solve $\vec{f}(\vec{u}) = 0$ numerically) and compute $\partial \bar{I}/\partial\beta$ at the endemic equilibrium. Compare with the FSE result from Chapter 4.

(d) What happens when you try to apply reverse-mode AD through an adaptive-step ODE solver? Explain why, and what the remedy is.

Solution.

(a) **Reverse-mode AD for $\mathcal{R}_0(c, \beta, \tau_R) = c\beta\tau_R$.**

```
# Register variables as differentiable
c = AD_VAR(5.0);  beta = AD_VAR(0.06);  tau = AD_VAR(7.0)
R0 = c * beta * tau          # forward pass records computation graph

# Reverse pass: compute all partial derivatives in one sweep
BACKWARD(R0)                 # seed dR0/dR0 = 1
dR0_dc    = GRAD(c)          # = beta*tau = 0.42
dR0_dbeta = GRAD(beta)       # = c*tau   = 35
dR0_dtau  = GRAD(tau)        # = c*beta  = 0.3
```

(b) **Verification.** $\partial\mathcal{R}_0/\partial c = \beta\tau_R = 0.42$ ✓; $\partial\mathcal{R}_0/\partial\beta = c\tau_R = 35$ ✓; $\partial\mathcal{R}_0/\partial\tau_R = c\beta = 0.3$ ✓. Normalized: $\mathcal{S}_c^{\mathcal{R}_0} = (c/\mathcal{R}_0) \cdot 0.42 = (5/2.1) \cdot 0.42 = 1$ ✓.

(c) **SIR endemic equilibrium.** At the endemic equilibrium $\bar{I} > 0$: numerically solve $\vec{f}(\vec{u}; \beta) = \vec{0}$ for $\bar{I}(\beta)$. AD *can* be applied to an executed adaptive solver, but the step-size controller introduces branches, and the derivative is nonsmooth wherever step acceptance changes, so seeded values can be fragile near those boundaries. Here the equilibrium is a root, not a trajectory, so the cleaner route is a differentiable Newton iteration:

```
beta = AD_VAR(0.06)
u    = [999, 1, 0]           # initial guess (not differentiable yet)

FOR iter FROM 1 TO 20:
    F = sir_rhs(u, beta)     # model RHS - must be written in AD-compatible ops
    J = jacobian(F, u)       # Jacobian (via AD or analytic formula)
    u = u - SOLVE(J, F)      # Newton step  (u becomes AD-differentiable)
END

dIbar_dbeta = GRAD(u[I_component], beta)
```

Compare with FSE result from Chapter 4 ($\mathcal{S}_\beta^{\bar{I}}$ at equilibrium).

(d) **Applying AD through an adaptive-step ODE solver.** Adaptive-step ODE solvers use internal control logic (step rejection, step-size selection) and branching that AD tools cannot trace through their computation graphs. Applying reverse-mode AD after such a solver will either produce an error or silently return zero gradients. **Remedy:** Either (1) implement the ODE solve as a differentiable fixed-step method (explicit Euler or RK4 written entirely in AD-compatible operations), or (2) use the adjoint ODE approach (Chapter 6), which bypasses the need to differentiate through the solver entirely.

Checking your answer: Part (d) is the central limitation of black-box AD applied to ODE solvers. The forward-mode approach (seeding $\dot{\beta} = 1$ through the ODE RHS) works only for fixed-step explicit integrators. The adjoint approach of Chapter 6 circumvents this entirely. A complete answer states the fixed-step restriction explicitly and says why adaptive stepping breaks the seeded derivative. At minimum: identifying that the integrator, not the model, is the source of the difficulty.

Exercise 7.4 *[Bloom L5]*

Exercise statement. Design the complete SA workflow for a dengue fever transmission model [1] with the following properties: $n = 8$ state variables (human and mosquito compartments), $m = 14$ parameters (mortality rates, biting rates, transmission probabilities, mean durations), $r = 2$ response functionals (basic reproduction number and endemic human infection prevalence), each ODE solve taking approximately 2 minutes.

Specify:

(a) Which SA method you would use and why.

(b) The expected total computation time.

(c) How you would validate the implementation.

(d) How you would handle the fact that two parameters (σ_v and σ_h, the mosquito and human biting rates) are structurally correlated.

(e) What output format you would use to present results to a non-mathematical public health audience.

Solution.

Dengue model SA design: 8 state variables, 14 parameters, 2 QOIs, 2 min/run.

(a) **Method: analytic adjoint ODE.** With $m = 14$ and $r = 2$, FSE costs 15 runs (30 min) while adjoint costs 3 runs (6 min). For a published dengue model with known equations, deriving the 8×8 Jacobian analytically is feasible (~ 1 day of derivation). Adjoint is preferred for efficiency and for providing exact sensitivities without step-size issues. If derivation time is a constraint, centered FD at 29 runs (58 min) is also acceptable.

(b) **Computation time.** Adjoint: 1 forward + 1 adjoint (for $r = 2$, need 2 adjoint solves) $= 3$ runs $\times$ 2 min $=$ **6 minutes.** Centered FD: 29 runs $\times$ 2 min $=$ **58 minutes.**

(c) **Validation.** Verify $\mathcal{S}_c^{\mathcal{R}_0} = \mathcal{S}_{\beta_h}^{\mathcal{R}_0} = 1$ analytically against adjoint output. Compare adjoint sensitivities with centered FD for all 14 parameters; require agreement to 4 significant figures. Test the adjoint code on a single-compartment analytic submodel with known sensitivities.

(d) **Handling structural correlation between σ_v and σ_h.** Report standard (marginal) SIs and total-derivative SIs. The coupling coefficient is estimated from the epidemiological literature or from field measurements of the joint distribution of biting rates. If σ_v increases by $x\%$, σ_h increases by $c_{vh} \cdot x\%$. The total-derivative SI of each QOI with respect to σ_v accounts for this induced change in σ_h. Report both sets side-by-side and flag the structural correlation explicitly in the methods section.

(e) **Presentation for public health audience.** Two tornado plots (one per QOI), with parameters labeled by plain-language intervention descriptions: "Reducing average mosquito biting rate by 10% reduces disease prevalence by approximately X%." Use a 3-color coding: green (large *negative* SI; reducing this parameter reduces burden), red (large *positive* SI: this parameter drives burden up; a reduction here would help), gray (small SI, negligible influence). Limit to top 5 parameters per QOI for clarity. *Sign note:* for $J = \int I\,dt$, a positive $\mathcal{S}_p^J$ means increasing p increases total burden; a negative $\mathcal{S}_p^J$ means increasing p reduces burden (e.g., recovery rate).

Checking your answer: This is a realistic applied SA design exercise. A complete answer requires all five components with specificity. The structural correlation handling (d) is the most commonly missed; students who only say "account for correlation" without specifying how are partial.

Exercise 7.5 *[Bloom L5]*

Exercise statement. *Audit exercise.* A manuscript reports sensitivities of one scalar QOI with respect to 600 POIs in a differentiable ODE model, computed by central finite differences "for robustness," at a cost of several cluster-days. State which method the structure indicates and the cost ratio; give the circumstance in which the authors' reason would be right and the evidence needed; explain why a referee should still ask for a change; draft two sentences for the report.

Solution.

(a) The structure is one QOI and many POIs in a differentiable model, which is the defining case for the adjoint in the master decision table of Section 7.3. Central differences cost $2m + 1 = 1201$ model solves, counting the baseline run. The adjoint costs one forward solve and one backward solve, so roughly two, and in practice a small multiple of two once the adjoint right-hand side is assembled. The idealized ratio is therefore 600 to 1, and even allowing several times the minimum adjoint cost it remains of order 100 to 1, turning several cluster-days into minutes.

(b) The authors' concern is not baseless. A discrete adjoint implemented inconsistently with the forward solver, an adaptive time-stepper whose step sequence is not reproduced on the backward pass, or a model with non-differentiable switching can each produce an adjoint that is wrong in ways that are hard to detect. The circumstance in which finite differences are the right call is a model with genuine non-smoothness, or a legacy solver that cannot be instrumented. For a referee to accept the choice here, the manuscript would need to state which of these applies, since the model is described as differentiable and the reason as stated is generic.

(c) The reported indices are plausibly correct, and a referee should say so. The objection is to reproducibility and to the paper's usefulness: a method costing 500 times more than the standard one for this structure means the analysis cannot be repeated, extended to more parameters, or checked by a reader without comparable resources. That is a substantive weakness in a computational paper even when every number is right. The request is for justification or replacement, not for correction, and the distinction should be explicit in the report so the authors are not sent chasing an error that does not exist.

(d) "The reported sensitivities appear correct, but for one QOI and 600 POIs in a differentiable model the adjoint method requires two solves rather than 1201, and the manuscript does not say why it was not used. I ask the authors either to state the specific obstacle to an adjoint implementation here, or to recompute with the adjoint and report the agreement between the two as a verification of both."

Checking your answer: Part (d) is where most answers are weakest. A referee request that reads as an accusation of error will produce a defensive response, while a request that names the structural fact, concedes the results, and offers the comparison as verification usually produces the recomputation. The final clause matters: asking for both methods on a subset of parameters converts the referee request into a cross-check that strengthens the paper, which is the outcome worth aiming for.

Exercise 7.6 *[Bloom L6]*

Exercise statement. A research group uses reverse-mode AD to compute adjoint sensitivities for a mosquito population model with adaptive-step-size ODE integration. They report that some sensitivity indices agree with finite differences but others disagree by 10–20%, with no obvious pattern. The discrepancies are largest for parameters that appear in the mortality rate expressions.

(a) Identify the most likely technical cause of the discrepancy.

(b) Describe two diagnostic tests that would confirm or rule out this cause.

(c) Propose two remedies and explain the trade-offs between them.

(d) At what point would you conclude that the disagreement indicates a genuine SA result (the linearization is poor for these parameters) rather than a numerical artifact?

Solution.

The most likely technical cause: Incorrect treatment of the adaptive step-size logic during reverse-mode AD of the ODE solver.

(a) **Root cause.** Adaptive-step-size ODE solvers use internal comparisons and branching to decide step sizes. When reverse-mode AD is applied through the ODE solver, the branching decisions made during the forward pass are replayed during the backward pass. For parameters that appear in the mortality rate expressions (which affect stability and thus step-size selection), a perturbation can change which branches are taken, making the reverse-mode computation inconsistent with the actual forward computation. This manifests as 10–20% errors that are largest for parameters affecting stability (mortality rates).

(b) **Two diagnostic tests.**

Test 1: Fixed-step integrator comparison. Re-implement the ODE solve with a fixed-step explicit RK4 method. Apply reverse-mode AD to this fixed-step implementation. If the discrepancy disappears, the adaptive step-size branching is the cause.

Test 2: Finite-difference consistency. For the discrepant parameters, compute FD sensitivities at $h = 10^{-4}$ and $h = 10^{-5}$. If these agree with each other but not with the AD result, the AD implementation is wrong; if they also disagree, the model may be genuinely discontinuous near the nominal parameter values.

(c) **Two remedies and trade-offs.**

Remedy 1: Fixed-step AD-compatible integrator. Replace the adaptive solver with a fixed-step RK4 inside the AD tape. *Trade-off:* Fixed-step integrators require choosing a step size that ensures accuracy for all parameter values; for stiff models, this may require very small steps and high computational cost. Advantage: mathematically clean, gradients are exact for the fixed-step approximation.

Remedy 2: Adjoint methods, of which there are two, plus a storage strategy that is often confused with them.

The *continuous* adjoint, taught in Chapter 6, derives the adjoint ODE from the model and then discretizes it. The *discrete* adjoint instead differentiates the time-stepping scheme itself, so it is exactly consistent with the computed forward solution but must be rederived for each integrator. *Checkpointing* is neither: it is a memory strategy for supplying the forward trajectory to a backward sweep, and it applies to both adjoints and to reverse-mode AD. Either adjoint avoids differentiating the controller's branching logic, which is what makes them attractive here. *Trade-off:* both require deriving and coding adjoint equations; the continuous adjoint may disagree slightly with the discrete solution, the discrete adjoint is integrator-specific, and checkpointing trades recomputation against storage in either case.

(d) **When to conclude genuine SA result.**

If the FD sensitivities computed at two consistent step sizes (h and $h/2$) agree with each other within 1% but differ from the AD result by 10–20%, the AD implementation is suspect. However, if FD estimates also show high variability (differing by $> 5\%$ as h is halved), this suggests the model is genuinely non-smooth and the large discrepancy reflects real behavior: the model is in a sensitive regime where small parameter changes cause qualitative changes in dynamics. In this case, the adjoint and FD are both locally valid; the discrepancy is genuine and should be reported as a finding (the model is near a threshold for these parameters).

Checking your answer: This exercise mirrors real-world challenges in gradient-based optimization of climate models. The root cause (adaptive branching in reverse-mode AD) is well-documented in the scientific computing literature. Students who identify only "there is a bug" without specifying the mechanism are partial. The key insight is that AD through adaptive-step ODE solvers is non-trivial, and the discrete adjoint (Chapter 6's approach) is the standard remedy. A complete answer names the mechanism, adaptive step selection branching inside the differentiated code, and proposes the discrete adjoint. At minimum: locating the fault in the solver rather than in the model equations.

8 Chapter 8: Caveat Emptor

Chapter Summary: Caveat Emptor: Limitations of Local Sensitivity Analysis

Method-question mismatch. Local SA can answer: *which parameter most affects the model near the nominal values?*
Local SA *cannot* answer: *which parameter most affects the model across its full uncertainty range?* (That requires global SA, Chapter 9.)
Using local SA to answer a global question produces results that are internally correct but wrong for the stated purpose. This is the most common error in published SA studies.

Bifurcation failure. At a bifurcation point, $\partial u/\partial p$ is undefined or infinite. The sensitivity index either blows up or loses all meaning. For the SIR model at the epidemic threshold $\mathcal{R}_0 = 1$: $\mathcal{S}^{i^*}_{\mathcal{R}_0} = 1/(\mathcal{R}_0 - 1) \to +\infty$ as $\mathcal{R}_0 \to 1^+$. A sensitivity index exceeding 10 should be treated as a qualitative warning ("this parameter region is highly sensitive"), not a quantitative ranking.

Sigma-normalized bridge. The sigma-normalized index $S^\sigma_i = (\partial Y/\partial Q_i)\, \sigma_i/\sigma_Y$ accounts for both the derivative (local slope) and the parameter uncertainty range. It equals the standardized regression coefficient in statistics. A parameter with small local SI but large σ_i can dominate output variance. This is the minimal bridge from local to global SA.

Operational checklist before reporting local SA. (1) Is the linearization valid? Check $|\mathcal{S}^q_p| \leq 10$ and $\kappa(J_F) \leq 10^4$.
(2) Are POIs independent? Check correlation matrix and VIF.
(3) Is the nominal point representative? Would the ranking change at a different nominal?
(4) Does the method match the question? (Local SI for local question, global for global.)

Watch out: Saltelli et al. (2019) reviewed 280 highly cited published sensitivity analyses and found that more than half used local methods to answer global questions. The derivatives were computed correctly. The wrong question was answered. Chapter 8 is the most important chapter in the book for avoiding this error.

Statistics and ML bridge. The condition number of J_F near a threshold is the SA analog of multicollinearity in regression: both diagnose when small changes in inputs produce large, unreliable changes in outputs. A poorly conditioned J_F near $\mathcal{R}_0 = 1$ is the SA equivalent of a near-singular design matrix in regression; the sensitivity indices become numerically unstable for the same reason that regression coefficients blow up under multicollinearity.

Exercise 8.1 *[Bloom L3]*

Exercise statement. For the equation $g(u; p) = u^3 - 3pu + 2 = 0$:

(a) Find all real roots at $p = 1$. Which root changes most rapidly with p?

(b) Compute $\mathcal{S}_p^{u^*}$ for each real root at $p = 1$ using the implicit function theorem.

(c) Plot the roots as functions of p for $p \in [0.8, 1.2]$. At what value of p do two roots merge?

(d) At the bifurcation value of p, what does the forward sensitivity equations (FSE) predict? Verify numerically using finite differences.

Solution.

$g(u; p) = u^3 - 3pu + 2 = 0$ (same as Ex 4.2, now with bifurcation focus).

(a) **Real roots at** $p = 1$ **and most rapidly changing.** Roots: $u_1 = 1$ (double), $u_2 = -2$. The double root $u_1 = 1$ changes most rapidly; its sensitivity is undefined (infinite), as shown in Ex 4.2. It is at a fold bifurcation point.

(b) $\mathcal{S}_p^{u^*}$ **at each root.** By the implicit-function theorem applied to $g(u; p) = u^3 - 3pu + 2 = 0$:

$$\frac{\partial u}{\partial p} = -\frac{\partial g/\partial p}{\partial g/\partial u} = -\frac{-3u}{3u^2 - 3p} = \frac{u}{u^2 - p}.$$

At $u_2 = -2$, $p = 1$: $\partial u_2/\partial p = (-2)/(4-1) = -2/3$, so $\mathcal{S}_p^{u_2} = (p/u_2)(\partial u_2/\partial p) = (1/(-2))(-2/3) = 1/3$.

At $u_1 = 1$ (double root), $p = 1$: $\partial g/\partial u = 3(1)^2 - 3(1) = 0$, so the denominator vanishes and $\mathcal{S}_p^{u_1}$ is undefined (infinite). This is the FSE singular-Jacobian failure at a bifurcation point (Chapter 4).

(c) **Root curve and bifurcation.** The real roots satisfy $u^3 - 3pu + 2 = 0$. By Cardano's formula or numerical continuation, the two positive roots merge at the fold bifurcation where $\partial g/\partial u = 3u^2 - 3p = 0$, i.e., $u = \sqrt{p}$. Substituting: $p^{3/2} - 3p\sqrt{p} + 2 = 0 \Rightarrow -2p^{3/2} + 2 = 0 \Rightarrow p^* = 1$. At $p^* = 1$ the cubic factors as $(u-1)^2(u+2)$: the *double* root $u = 1$ is where the two positive branches meet, while the simple root $u = -2$ is unaffected and persists on both sides. Careful: the cubic has the double root at $u = 1$ for $p = 1$. For $p < 1$, the double root splits into two complex roots (no real positive root near 1), and for $p > 1$, two distinct real roots appear near $u = 1$. The bifurcation is at $\mathbf{p = 1}$.

(d) **FSE prediction at bifurcation.** At $p = 1$, $\partial g/\partial u = 0$: the FSE $(\partial g/\partial u)\,(\partial u/\partial p) = -\partial g/\partial p$ has a zero coefficient, so $\partial u/\partial p = \infty$. Finite-difference verification cannot be done with a centered difference here, and the reason is the point of the exercise. For $p < 1$ the two positive branches are complex, so $u(p = 0.999)$ has no real value near $u = 1$: the left stencil point does not exist. Any difference straddling $p = 1$ is differencing across the fold rather than through it. Use one-sided continuation from $p > 1$ instead, as in Exercise 4.2 part (e): the branches satisfy $u_\pm \approx 1 \pm \sqrt{p-1}$, so

$$\frac{\partial u_\pm}{\partial p} \approx \pm\frac{1}{2\sqrt{p-1}} \longrightarrow \pm\infty \quad \text{as } p \downarrow 1.$$

Evaluating at $p = 1.1$, 1.01 and 1.001 gives 1.58, 5.00 and 15.81, growing as $(p-1)^{-1/2}$. That rate, not a large difference quotient, is what confirms the singularity, and it is measurable only from the side on which the roots exist.

Checking your answer: This exercise directly connects Chapter 4 (implicit FSE) to Chapter 8 (bifurcations). The key point: the FSE singularity *is* the bifurcation. A complete answer connects the FSE singularity to the Jacobian eigenvalue crossing zero; at minimum, identifying the singularity as the bifurcation itself. Answering only "sensitivity is large" without the eigenvalue connection is partial.

Exercise 8.2 *[Bloom L3]*

Exercise statement. Using the SIR model with $c = 5$, $\tau_R = 7$ days, $\tau_m = 10{,}000$ days:

(a) Compute $\mathcal{S}_\beta^{I(T)}$ for $\beta = 0.04, 0.06, 0.08, 0.12$ at $T = 90$ days using centered finite differences. With demography the threshold is $\mathcal{R}_0 = c\beta/(\gamma_R + \nu_m)$; at $\tau_m = 10{,}000$ days it differs from $c\beta\tau_R$ by 0.07% and takes values $1.4, 2.1, 2.8, 4.2$ at the four parameter sets.

(b) Now recompute for $\beta = 0.02$ ($R_0 = 0.7$). Compare with the results for $\mathcal{R}_0 > 1$. Explain any qualitative differences.

(c) If you were to add a bifurcation analysis tool (e.g., MATCONT), what would you expect the bifurcation diagram to look like for the endemic equilibrium of this SIR model as β varies?

Solution.

(a) $\mathcal{S}_\beta^{I(T)}$ **at four values,** $T = 90$ **days.**

Computed via centered FD with $h = 6 \times 10^{-6}\beta^*$ (SIR model, $N = 1000$, $I(0) = 1$, no vital dynamics):

β	$\mathcal{R}_0 = c\beta\tau_R$	$I(90)$	$\mathcal{S}_\beta^{I(90)}$ (approx.)	Epidemic status at $t = 90$
0.04	1.40	$\sim$ 46	$\sim +6.1$	Growing; peak not yet reached
0.0429	1.50	$\sim$ 58	$\sim +0.7$	Near the sign change
0.06	2.10	$\sim$ 7	~ -8.7	*Past peak*; declining
0.08	2.80	$\sim$ 1	~ -7.7	*Past peak*; near extinction
0.12	4.20	$\sim$ 0	~ -4.9	*Past peak*; essentially over

Key result: the SI *changes sign* as $\mathcal{R}_0$ crosses approximately 1.5. For $\mathcal{R}_0 = 1.4$, the epidemic is still growing at day 90, so $\mathcal{S}_\beta^{I(90)} > 0$: increasing β increases $I(90)$. For $\mathcal{R}_0 \geq 2.1$, the epidemic peaks well before day 90 (around days 28, 18, and 12 for $\mathcal{R}_0 = 2.1, 2.8, 4.2$ respectively). At day 90 the epidemic is in its declining tail, and $\mathcal{S}_\beta^{I(90)} < 0$: increasing β shifts the peak *earlier*, causing a faster depletion of susceptibles, so there are *fewer* infectious individuals remaining at $t = 90$. This sign reversal is the central lesson of the exercise: the SA result depends critically on whether the fixed observation time T falls before or after the epidemic peak.

(b) $\beta = 0.02$ **(**$R_0 = 0.7 < 1$**).** No epidemic occurs and the infection decays: $I(90) = 0.0204$, about one fiftieth of $I(0) = 1$. The raw derivative is correspondingly tiny, $\partial I(90)/\partial\beta \approx 9.1$, yet $\mathcal{S}_\beta^{I(90)} = +8.94$, the largest magnitude anywhere in the table. **Qualitative change:** this is the opposite of what intuition suggests, and it is the point of the exercise. The normalized index is a ratio, and below threshold the denominator $I(90)$ has collapsed while the numerator has not, so the index inflates. A large $\mathcal{S}_\beta^{I(90)}$ here reports a large *relative* change in a quantity that is negligible in absolute terms: a 1% increase in β raises $I(90)$ from 0.0204 to about 0.0222, which is epidemiologically meaningless. Above threshold the index is smaller in magnitude but describes real numbers of infections. *Checking your answer:* Distinguish three things that this exercise deliberately entangles: (i) transient sensitivity of $I(T)$ at fixed finite T, which is smooth in β across the threshold; (ii) sensitivity of the endemic equilibrium, which does have a derivative singularity at $\mathcal{R}_0 = 1$; and (iii) inflation of a normalized index caused by a small response denominator, which is what part (b) exhibits. Only (ii) is a bifurcation phenomenon.

(c) **Bifurcation diagram.** The endemic equilibrium $\bar{I}^*$ bifurcates from $\bar{I}^* = 0$ at $\mathcal{R}_0 = 1$ (transcritical bifurcation). For $\mathcal{R}_0 < 1$: $\bar{I}^* = 0$ (disease-free equilibrium is stable). For $\mathcal{R}_0 > 1$: $\bar{I}^* > 0$ (endemic equilibrium exists and is stable). The bifurcation diagram shows $\bar{I}^*$ vs. β as a forward (supercritical) transcritical bifurcation.

Checking your answer: The key finding from part (a) is that $\mathcal{S}_\beta^{I(90)}$ *changes sign* as $\mathcal{R}_0$ increases past ~ 1.5. For $\mathcal{R}_0 = 1.4$ the epidemic is still growing at $t = 90$ so SI is positive ($\approx +6$). For $\mathcal{R}_0 \geq 2.1$ the epidemic peaks before $t = 90$ (around days 28, 18, 12 for $\mathcal{R}_0 = 2.1, 2.8, 4.2$), and $\mathcal{S}_\beta^{I(90)} < 0$: increasing β shifts the peak earlier, reducing $I(90)$. Common error: reporting all positive SIs because "more transmission = more infections." This holds for peak $I_{\max}$ but not for I at a fixed time after the peak. At minimum: SI positive for $\mathcal{R}_0 = 1.4$, negative for at least one $\mathcal{R}_0 > 1.5$, with explanation linking sign to epidemic timing. A complete answer: all four correct signs, numerically consistent values, and clear explanation of why $T = 90$ falls before/after the epidemic peak.

Checking your answer: $\mathcal{S}_\beta^{I(T)}$ *decreases* as β increases; counterintuitive if one expected larger β to mean larger sensitivity. The reason: at high β, the epidemic has already run its course by day 90, and $I(90) \approx 0$ regardless of small perturbations to β. A complete answer explains the decrease by noting $I(90) \approx 0$ at high β; at minimum, reporting the counterintuitive direction and flagging it as needing explanation. This illustrates why SA at a single time point can be misleading (Chapter 4, Ex 4.8).

Exercise 8.3 *[Bloom L4]*

Exercise statement. For the 3×3 linear system $A\vec{u} = \vec{b}$ where:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9+\epsilon \end{pmatrix}, \quad \vec{b} = A\vec{u}_{\text{exact}}, \quad \vec{u}_{\text{exact}} = (1,1,1)^T.$$

(a) For $\epsilon = 1, 0.1, 0.01, 0.001, 0.0001$, compute $\kappa(A)$ and $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ where $\hat{\vec{u}} = A^{-1}\vec{b}$ (solved by the linear system solver).

(b) Plot $\|\hat{\vec{u}} - \vec{u}_{\text{exact}}\|$ vs. $\kappa(A)$ on log-log axes. What is the approximate slope?

(c) What does this experiment tell you about interpreting large SI values for nearly singular models?

Solution.

(a) **Condition number and error vs. ϵ.**

Pseudocode:

```
# Ill-conditioned matrix experiment
u_exact  = [1, 1, 1]
eps_vals = [1, 0.1, 0.01, 0.001, 0.0001]

FOR each eps IN eps_vals:
    A     = [[1,2,3], [4,5,6], [7,8, 9+eps]]  # nearly singular for small eps
    b     = A * u_exact                         # right-hand side
    kappa = CONDITION_NUMBER(A)                 # 2-norm condition number
    u_hat = SOLVE(A, b)                         # solve A*u = b
    err   = NORM(u_hat - u_exact)              # solution error

PLOT log(eps_vals) vs log(kappa)   # should show slope -1
PLOT log(kappa) vs log(err)        # should show slope +1
```

Expected results. The condition numbers are reproducible; the solution errors are not, since they depend on the machine and on the BLAS implementation, so they are given as an order of magnitude rather than as specific values.

ϵ	$\kappa(A)$	$\Vert\hat{\vec{u}} - \vec{u}\Vert$
1.0	88	$O(\kappa\varepsilon_{\text{mach}})$
0.1	9.9×10^2	$O(\kappa\varepsilon_{\text{mach}})$
0.01	1.0×10^4	$O(\kappa\varepsilon_{\text{mach}})$
0.001	1.0×10^5	$O(\kappa\varepsilon_{\text{mach}})$
0.0001	1.0×10^6	$O(\kappa\varepsilon_{\text{mach}})$

(b) **Log-log slope.** The relationship is $\Vert\hat{\vec{u}} - \vec{u}\Vert \propto \kappa(A)\,\varepsilon_{\text{mach}}$. On log-log axes, the slope is $+1$ (error grows linearly with condition number).

(c) **Implication for large SI values.**

A large SI near a singularity is *genuine*, and separating that from the difficulty of computing it is the point of this exercise. If J_F is nearly singular then $\partial\vec{u}/\partial p = J_F^{-1}\vec{f}_j$ really is large: the model truly responds strongly, because the equilibrium is close to a fold. Conditioning does not manufacture that; it governs how much the *computed* value can be corrupted. For the matrix in part (a) the exact sensitivity grows like $1/\varepsilon$ while the relative error of the numerical solve stays near 10^{-15} at $\varepsilon = 10^{-1}$ and only reaches 10^{-8} at $\varepsilon = 10^{-9}$. Both statements hold at once and they are independent: the answer is large, and it becomes progressively harder to trust its trailing digits. If the Jacobian J_F of the model equations is nearly singular (large $\kappa(J_F)$), the FSE $J_F\,(\partial\vec{u}/\partial p_j) = \vec{f}_j$ will amplify any numerical errors in $\vec{f}_j$ by a factor of $\kappa(J_F)$. The reported SI may be $\kappa(J_F)$ times larger than the true value. *Practical implication:* always report $\kappa(J_F)$ alongside large SI values. An SI of 50 with $\kappa(J_F) = 100$ should be flagged as potentially unreliable; an SI of 50 with $\kappa(J_F) = 2$ is trustworthy.

Checking your answer: The slope of 1 in part (b) is the classical condition-number error bound: $\|\delta\vec{u}\|/\|\vec{u}\| \leq \kappa(A)\|\delta\vec{b}\|/\|\vec{b}\|$. A complete answer computes the slope empirically and reports it as ≈ 1; students who derive the bound analytically go further still.

Exercise 8.4 *[Bloom L6]*

Exercise statement. You are asked to perform SA on an epidemic model whose best-fit parameter values give $\mathcal{R}_0 = 1.04$, very close to the epidemic threshold. Write a one-page advisory report for a public health agency explaining:

(a) What local SA can and cannot tell you at this parameter value.

(b) What specific numerical problems to expect when computing sensitivity indices near $\mathcal{R}_0 = 1$.

(c) What additional analyses you would recommend before advising on interventions.

(d) How you would explain the concept of a bifurcation and its implications to a non-mathematical policy maker.

Solution.

Advisory report: SA near the epidemic threshold ($\mathcal{R}_0 = 1.04$).

(a) **What local SA can and cannot tell you.**

Local SA provides the derivative of each QOI with respect to each parameter at the current best-fit values. With $\mathcal{R}_0 = 1.04$, the model is very close to the epidemic threshold. Local SA *can* tell you which parameters would, if perturbed slightly, have the largest marginal effect on disease burden. What it *cannot* tell you: whether a parameter change of realistic magnitude (say 10–20%) would push $\mathcal{R}_0$ below 1 (eliminating the epidemic entirely) or keep it above 1. Near the threshold, the model's behavior changes qualitatively, and local SA captures only the linearized behavior in an infinitesimally small neighborhood.

(b) **Specific numerical problems near $\mathcal{R}_0 = 1$.**

Near the transcritical bifurcation at $\mathcal{R}_0 = 1$, the Jacobian of the model at the endemic equilibrium has an eigenvalue near zero. This causes: (1) the FSE linear system $J_F\,(\partial\vec{u}/\partial p) = \vec{f}$ to be nearly singular, amplifying roundoff errors by $\kappa(J_F)$; (2) time-dependent SIs to become large near the threshold crossing, though at any fixed finite T they stay bounded: the trajectory is smooth in the parameters across $\mathcal{R}_0 = 1$, and for the nominal SIR at $T = 90$ the index $\mathcal{S}_\beta^{I(T)}$ moves only from 12.17 to 12.48 as $\mathcal{R}_0$ passes from 0.98 through 1 to 1.015; (3) the adjoint ODE to have a slowly decaying mode (near-zero eigenvalue), making backward integration unstable over long time horizons. Practitioners should report $\kappa(J_F)$ and check that SI values are stable to perturbations of the nominal parameter set within the confidence region.

(c) **Additional analyses recommended.**

First, plot the endemic equilibrium I^* vs. $\mathcal{R}_0$ (or vs. the most controllable parameter) to determine how large a reduction in
beta or c is needed to push $\mathcal{R}_0$ below 1; this gives a specific, actionable intervention target rather than a marginal SI value. Second, compute SI at $R_0 = 0.9$, 1.0, 1.04, 1.2 and report how the rankings change: if the ranking is stable across this range, the local SA at $\mathcal{R}_0 = 1.04$ is informative; if not, flag the instability explicitly. Third, compute the actual change in peak I for a 10% reduction in each parameter and report these finite-change values directly alongside the SI values, since near the threshold the linearization error may be large enough to change which parameter appears most controllable.

(d) **Explaining bifurcation to a policy maker.**

Think of the epidemic threshold as a tipping point. Below it ($\mathcal{R}_0 < 1$), each infected person infects on average less than one other person, and the outbreak fades on its own. Above it ($\mathcal{R}_0 > 1$), each infected person infects more than one other person, and the outbreak grows into an epidemic. At $\mathcal{R}_0 = 1.04$, we are just above the tipping point. Even a small reduction in contact rates or transmission probability, say 5%, could push us below the tipping point, causing the epidemic to die out rather than sustain itself. This is the most important insight from the sensitivity analysis: at $\mathcal{R}_0$ close to 1, small interventions can have disproportionately large effects. But it also means our predictions are highly uncertain: if the true $\mathcal{R}_0$ is 0.96 rather than 1.04, the epidemic might already be declining naturally.

Checking your answer: This is a realistic scenario that every applied mathematical modeler faces. The report format requirement is deliberate: public health advisors need concise, clear guidance, not a technical treatise. Responses that are technically correct but not accessible to a non-mathematical reader fall short of that. The bifurcation explanation (part d) should use no equations.

Exercise 8.5 *[Bloom L4]*

Exercise statement. *Audit exercise.* A policy brief reports large local sensitivity indices for the endemic equilibrium at $\mathcal{R}_0 = 1.02$ and concludes that recovery is the highest-priority intervention target. Explain why large indices are expected and why they are not evidence of priority; state what happens as $\mathcal{R}_0 \to 1^+$; identify the assumption outside every reported index; recommend an alternative analysis.

Solution.

(a) Near the transcritical bifurcation at $\mathcal{R}_0 = 1$ the endemic equilibrium emerges from the disease-free state, and equilibrium prevalence depends on $\mathcal{R}_0 - 1$, a quantity close to zero at $\mathcal{R}_0 = 1.02$. Normalizing a derivative by a small nominal value inflates the index, so large magnitudes are a signature of proximity to the bifurcation rather than a discovery about the recovery rate. Every parameter entering $\mathcal{R}_0$ will show a similarly inflated index here, which is itself the diagnostic: a ranking in which one parameter stands out is informative, and one in which everything is large is usually reporting the bifurcation.

(b) As $\mathcal{R}_0 \to 1^+$ the endemic equilibrium approaches the disease-free equilibrium and the two exchange stability. The two branches cross *transcritically*, so the equilibrium derivative does *not* diverge: for the SIR endemic branch $\bar{i} = \frac{\nu_m}{\gamma_R+\nu_m}\left(1-1/\mathcal{R}_0\right)$ and $\partial\bar{i}/\partial\mathcal{R}_0 = \nu_m/[\mathcal{R}_0^2(\gamma_R+\nu_m)]$, which is finite at $\mathcal{R}_0 = 1$. An unbounded equilibrium derivative is the signature of a *fold*, where two branches coalesce with square-root scaling as in the cubic-fold exercise of this chapter; a transcritical crossing has no such singularity. What does degrade near $\mathcal{R}_0 = 1$ is the range over which the linearization holds: the two equilibria lie close together, so a perturbation large enough to be of practical interest can carry the system past the crossing, and the linearization on which every index in Chapters 2 through 8 rests is valid only over a shrinking neighborhood. A perturbation large enough to be of policy interest will typically leave that neighborhood, and may cross $\mathcal{R}_0 = 1$ entirely, at which point the equilibrium being differentiated no longer exists. **Unifying Idea 3.**

(c) The brief assumes that the recovery rate is a lever that can be moved. No sensitivity index contains information about feasibility, cost, or controllability. In most settings recovery is governed by pathogen biology and by treatment availability, and a large index for it may identify a quantity that matters greatly and cannot be changed, while a parameter with a smaller index may be readily modifiable. The index identifies what to target and is silent on whether the target can be reached.

(d) Recommend a global analysis over the plausible parameter ranges, which near a bifurcation is not a refinement but a necessity, since the uncertainty range spans qualitatively different regimes. A sampling-based approach reports the fraction of the range for which $\mathcal{R}_0 > 1$ at all, which is the question a policy maker actually needs answered, and it does so without differentiating a quantity that is nearly singular. This answers whether the system can be pushed below threshold, whereas the local indices answer only how the equilibrium moves under an infinitesimal perturbation that does not cross it.

Checking your answer: Students frequently identify the bifurcation and stop, which earns part (a) and part (b) but misses the two most consequential steps. Part (c) is the model-reality boundary in its most practical form, and it is the error the brief actually commits: even if every index were valid, the conclusion drawn from them would not follow. Part (d) should specify what the recommended analysis reports, not merely name it.

Exercise 8.6 *[Bloom L6]*

Exercise statement. A published modeling paper reports: "Sensitivity analysis of the endemic equilibrium of our malaria model found that all indices are less than 0.1, indicating the model is robust to parameter uncertainty. The basic reproduction number at fitted parameter values is $\mathcal{R}_0 = 1.15$."

Identify at least three specific methodological concerns with this analysis and its conclusions. For each concern, state: (a) what the problem is; (b) what evidence the authors would need to provide to address it; and (c) what alternative analysis would give a more complete picture.

Solution.

Three methodological concerns with the malaria SA ($\mathcal{R}_0 = 1.15$, all SI < 0.1):

Concern 1: Small SIs near $\mathcal{R}_0 = 1$ may be numerical artifacts.

Problem: With $\mathcal{R}_0 = 1.15$, the model is moderately above the threshold, but close enough that the endemic equilibrium Jacobian may be ill-conditioned. Small SI values could result from condition-number amplification rather than genuine insensitivity.

Evidence needed: Report $\kappa(J_F)$ at the endemic equilibrium and verify that $|\mathrm{SI}| \gg \kappa(J_F)\,\varepsilon_{\mathrm{mach}}$ for each parameter. If any SI is within a factor of $\kappa(J_F)$ of machine epsilon, that SI is numerically unreliable.

Alternative analysis: Compute SIs of $\mathcal{R}_0$ itself (analytically) and compare sign and magnitude with the equilibrium SIs. If $\mathcal{S}_{p_j}^{\mathcal{R}_0}$ is large but $\mathcal{S}_{p_j}^{\bar{I}}$ is small, the discrepancy suggests numerical issues at the equilibrium.

Concern 2: SA of the endemic equilibrium does not address transient dynamics or the probability of outbreak onset.

Problem: If malaria is currently at or near the endemic equilibrium, the relevant policy question is "which intervention would most reduce endemic prevalence?" But if the model is used for outbreak forecasting (a different but common use), the relevant QOI is the peak or cumulative burden during a transient following an introduction event. SA of the endemic equilibrium cannot answer the latter question.

Evidence needed: Clarify the model's purpose. If forecasting an outbreak, compute SA for the transient QOI (e.g., peak I at $T = 180$ days) in addition to the endemic equilibrium.

Alternative analysis: Compute time-dependent SIs $\mathcal{S}_{p_j}^{I}(t)$ and report whether the parameter ranking changes between the transient and endemic phases.

Concern 3: "All indices less than 0.1" does not imply robustness unless the uncertainty ranges are accounted for.

Problem: A local SI of 0.1 means a 1% change in the parameter produces a 0.1% change in prevalence. But if a parameter is uncertain by $\pm 50\%$ (plausible for entomological parameters in malaria models), the expected change in prevalence from this parameter alone is $\approx 5\%$. Reporting only SI values without the uncertainty range gives an incomplete picture.

Evidence needed: Report sigma-normalized SIs or one-way sensitivity ranges: the change in $\bar{I}$ when each parameter varies over its full uncertainty range. Parameters with small SI but large uncertainty range may still dominate the output variance.

Alternative analysis: Compute Sobol first-order and total indices across the plausible parameter uncertainty ranges. A parameter with local SI $= 0.1$ and uncertainty range $\pm 80\%$ may have Sobol $S_1 = 0.3$.

Checking your answer: A complete answer requires all three concerns with (a), (b), (c) for each. The three concerns should be genuinely distinct: numerical reliability (Concern 1), scope of the SA (Concern 2), and accounting for uncertainty ranges (Concern 3). Students who only note "local SA doesn't capture nonlinearity" are partial for Concern 3 without the sigma-normalized index framing. The claim "all indices less than 0.1" is particularly dangerous near a threshold; this is the chapter's core message.

9 Chapter 9: Global Sensitivity Analysis and Sobol Indices

Chapter Summary: Global Sensitivity Analysis and Sobol Indices

Sobol decomposition (Definition 9.1). For independent POIs $\vec{Q} = (Q_1, \ldots, Q_m) \sim \mathcal{U}[0,1]^m$, the output variance decomposes uniquely as

$$D = \mathrm{Var}(Y) = \sum_i D_i + \sum_{i<j} D_{ij} + \cdots ,$$

where $D_i = \mathrm{Var}[\mathbb{E}(Y|Q_i)]$. The first-order Sobol index $S_i = D_i/D$ is the fraction of output variance attributable to Q_i acting alone. The total-order index S_{Ti} includes all interactions involving Q_i.

Jansen estimators. Use these, not the original Saltelli (2002) formulas:

$$\hat{S}_i = 1 - \frac{\frac{1}{2N}\sum_j [f(B_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}, \qquad \hat{S}_{Ti} = \frac{\frac{1}{2N}\sum_j [f(A_j) - f(A_{B,j}^{(i)})]^2}{\hat{D}}.$$

Cost: $N(m+2)$ model evaluations. Rule of thumb: $N \geq 1000$ for reliable estimates. Always report bootstrap 95% confidence intervals alongside point estimates.

PRCC for monotone models. Partial Rank Correlation Coefficient (PRCC) measures the linear relationship between each POI and the QOI after removing the effect of all other POIs (via rank transformation). Cost: N evaluations (vs. $N(m+2)$ for Sobol). Valid only when the relationship between each POI and the QOI is monotone. Check monotonicity with scatter plots before using PRCC.

Morris screening workflow. (1) Run Morris ($r = 10$ trajectories) to rank all m POIs by μ_i^*.
(2) Fix parameters with small μ_i^* at their nominal values.
(3) Run Sobol only on the important subset.
This reduces Sobol cost by a factor of 5–10 for large m.

Statistics bridges. $S_i = \eta^2$ in a one-way ANOVA for linear models with equal-variance inputs. $S_i = r_i^2$ (squared correlation) for linear models. PRCC is the nonparametric partial regression coefficient. Sigma-normalized index S_i^σ equals the standardized regression coefficient. All of these are the same normalized-derivative idea from Chapter 2, applied to the full parameter range.

Watch out: Sobol indices assume independent POIs. When parameters are correlated (as transmission rate and contact rate often are in epidemic models), first-order indices can exceed 1 or be negative, and the decomposition loses its uniqueness. Use PRCC or grouped Sobol indices for correlated inputs. Report the correlation structure explicitly.

Statistics and ML bridge. The Sobol variance decomposition is the *functional ANOVA* (also called the Hoeffding decomposition, 1948): every square-integrable function of independent random inputs can be uniquely decomposed into main effects and interaction effects, exactly as in ANOVA. The total Sobol index S_{T_i} is the ML *permutation feature importance*: permuting Q_i severs its association with the output, so what the permutation destroys is the whole contribution of Q_i, interactions included. Read S_{T_i} as the share of output variance that survives when every *other* input is held fixed. It is not the variance left once Q_i itself is learned; that quantity is $1 - S_i$. This is why measurement priority is answered by S_i and factor fixing by S_{T_i}, as the index-selection table in Chapter 9 sets out.

Exercise 9.1 *[Bloom L3]*

Exercise statement. For the SEIR model with local SIs $\mathcal{S}^{\mathcal{R}_0}_{\tau_E} = 0.6$, $\mathcal{S}^{\mathcal{R}_0}_{\tau_I} = 1.2$, $\mathcal{S}^{\mathcal{R}_0}_{\beta} = 1.0$, $\mathcal{S}^{\mathcal{R}_0}_{c} = 1.0$ and uncertainty ranges $\sigma_{\tau_E} = 2$ days, $\sigma_{\tau_I} = 3$ days, $\sigma_\beta = 0.01$, $\sigma_c = 0.8$:

(a) Compute the output standard deviation $\sigma_{\mathcal{R}_0}$ using the delta method, at nominal $\mathcal{R}_0^* = 2.0$.

(b) Compute all four sigma-normalized indices.

(c) Produce a tornado plot of the sigma-normalized indices.

(d) Compare the ranking with the local SI ranking. Which parameter changed rank most dramatically? Why?

Solution.

Given: $\mathcal{S}^{\mathcal{R}_0}_{\tau_E} = 0.6$, $\mathcal{S}^{\mathcal{R}_0}_{\tau_I} = 1.2$, $\mathcal{S}^{\mathcal{R}_0}_{\beta} = 1.0$, $\mathcal{S}^{\mathcal{R}_0}_{c} = 1.0$, nominal $\mathcal{R}_0^* = 2.0$. Uncertainties: $\sigma_{\tau_E} = 2$ d, $\sigma_{\tau_I} = 3$ d, $\sigma_\beta = 0.01$, $\sigma_c = 0.8$. Nominal values: $\tau_E^* = 5.5$ d, $\tau_I^* = 7$ d, $\beta^* = 0.06$, $c^* = 5$.

(a) **Output standard deviation (delta method).**

$\sigma^2_{\mathcal{R}_0} = \sum_j \left(\frac{\partial \mathcal{R}_0}{\partial p_j}\right)^2 \sigma^2_{p_j} = \sum_j \left(\frac{\mathcal{R}_0^*}{p_j^*} \mathcal{S}^{\mathcal{R}_0}_{p_j}\right)^2 \sigma^2_{p_j}$.

$$\left(\frac{\mathcal{R}_0^*}{\tau_E^*} \mathcal{S}^{\mathcal{R}_0}_{\tau_E}\right)^2 \sigma^2_{\tau_E} = \left(\frac{2}{5.5} \times 0.6\right)^2 \times 4 = (0.218)^2 \times 4 = 0.190$$
$$\left(\frac{\mathcal{R}_0^*}{\tau_I^*} \mathcal{S}^{\mathcal{R}_0}_{\tau_I}\right)^2 \sigma^2_{\tau_I} = \left(\frac{2}{7} \times 1.2\right)^2 \times 9 = (0.3429)^2 \times 9 = 1.058$$
$$\left(\frac{\mathcal{R}_0^*}{\beta^*} \mathcal{S}^{\mathcal{R}_0}_{\beta}\right)^2 \sigma^2_{\beta} = \left(\frac{2}{0.06} \times 1.0\right)^2 \times (0.01)^2 = (33.3)^2 \times 10^{-4} = 0.111$$
$$\left(\frac{\mathcal{R}_0^*}{c^*} \mathcal{S}^{\mathcal{R}_0}_{c}\right)^2 \sigma^2_{c} = \left(\frac{2}{5} \times 1.0\right)^2 \times (0.8)^2 = (0.4)^2 \times 0.64 = 0.102$$

$\sigma^2_{\mathcal{R}_0} = 0.1904 + 1.0580 + 0.1111 + 0.1024 = 1.4619$, $\sigma_{\mathcal{R}_0} \approx \mathbf{1.21}$. *Checking your answer:* This four-parameter SEIR exercise ($\sigma_{\mathcal{R}_0} \approx 1.21$) is distinct from the three-parameter SIR self-check in the chapter text ($\sigma_{\mathcal{R}_0} \approx 0.726$); the two use different models and uncertainty ranges, so the different output standard deviations are expected.

(b) **Sigma-normalized indices.**
$\tilde{S}_{p_j} = (\mathcal{R}_0^*/p_j^*)\,|\mathcal{S}^{\mathcal{R}_0}_{p_j}|\,\sigma_{p_j}/\sigma_{\mathcal{R}_0}$:

Parameter	$\tilde{S}$ (sigma-norm.)	Contribution (%)
τ_I	$(2/7)(1.2)(3)/1.21 = 0.851$	72.4%
τ_E	$(2/5.5)(0.6)(2)/1.21 = 0.360$	13.0%
β	$(2/0.06)(1.0)(0.01)/1.21 = 0.276$	7.6%
c	$(2/5)(1.0)(0.8)/1.21 = 0.264$	7.0%

Note on percentages: the contribution column is simply $\tilde{S}_j^2$. Each $\tilde{S}_j$ already carries the factor $1/\sigma_{\mathcal{R}_0}$, so the squares sum to 1 with no further normalization. For τ_I: $(0.851)^2 = 0.724$, that is 72.4%. Dividing again by $\sigma^2_{\mathcal{R}_0} = 1.462$ would be double-counting a normalization that has already been applied.

(c) **Tornado plot.** Sorted by $\tilde{S}$: $\tau_I > \tau_E > \beta \approx c$.

(d) **Ranking comparison.** Local SI ranking: τ_I and c tied at 1.2 and 1.0; β at 1.0; τ_E at 0.6. Sigma-normalized ranking: $\tau_I \gg \tau_E > \beta \approx c$.

τ_I remains dominant in both, but its lead is much larger when uncertainty is accounted for ($\sigma_{\tau_I} = 3$ d is large). c drops from second to last: despite SI $= 1.0$, its uncertainty $\sigma_c = 0.8$ relative to $c^* = 5$ gives a coefficient of variation of only 16%, while τ_E has $\sigma_{\tau_E}/\tau_E^* = 36\%$. c **changed rank most dramatically** because it has a large SI but small relative uncertainty.

Checking your answer: The sigma-normalized index is the key concept here: it combines the sensitivity (SI) with the actual uncertainty range to produce a practical priority ranking. Students who compute only SIs without accounting for uncertainty miss the point of this exercise.

Exercise 9.2 *[Bloom L3]*

Exercise statement.

(a) Implement the `lhs_sample` function. For $N = 100$, $m = 3$, $[0,1]^3$, generate an Latin hypercube sample (LHS) and a random sample of the same size.

(b) Produce scatter plots of all three pairs of parameters for both samples (6 plots total).

(c) Compute the mean and standard deviation of the sample along each dimension. Which method is closer to the true uniform mean and standard deviation?

(d) For one parameter, plot the histogram of LHS vs. random sampling. Which covers the range more evenly?

Solution.

(a) **LHS implementation and comparison.**

```
# Latin hypercube sample: N=100 points in m=3 dimensions, all on [0,1]
set_random_seed(42)
N = 100;  m = 3
X_lhs = LHS_SAMPLE(N, m, lower=zeros(m), upper=ones(m))
X_rnd = RANDOM_SAMPLE(N, m)   # uniform random for comparison
```

(b) **Scatter plots.** The LHS scatter plots show more uniform coverage with fewer clusters; the random sample shows visible gaps and clusters in all three pairwise projections.

(c) **Mean and standard deviation.** Both LHS and random sampling should give means near 0.5 and standard deviations near $1/\sqrt{12} \approx 0.289$ for a uniform distribution on $[0, 1]$. LHS is typically closer to both exact values because the stratification ensures coverage of the full range. For $N = 100$, the improvement in mean accuracy is ∼30–50% and in standard deviation ∼20–40% (these are representative magnitudes for typical runs; actual improvement varies with N, the model, and the random seed).

(d) **Histogram comparison.** The LHS histogram shows a much flatter, more uniform distribution (each of the 100 strata has exactly one sample). The random sample histogram shows the typical Poisson variation (some bins have 0 or 3+ samples). For sensitivity analysis, the LHS coverage is preferable because it ensures the model is evaluated in every region of the parameter space.

Checking your answer: Part (c) is the quantitative check: students should compute the actual mean and std and report how close they are to the theoretical values. The LHS improvement in mean accuracy comes from the fact that the sample mean of a stratified sample is an unbiased estimator with lower variance than the sample mean of a random sample (stratification theorem).

Exercise 9.3 *[Bloom L3]*

Exercise statement. For the SIR model with parameter ranges from the laboratory setup:

(a) Generate $N = 200$ LHS samples and run the SIR model at each.

(b) Compute partial rank correlation coefficient (PRCC) for all four parameters with respect to $I_{\max}$.

(c) Produce a PRCC bar chart sorted by |PRCC|.

(d) Which parameter has the strongest negative PRCC? What does this mean in plain language?

(e) Compare the PRCC ranking with the local SI tornado plot from Chapter 2. Do they agree?

Solution.

(a-b) **LHS + PRCC computation.** *(Parts (a) and (b) are combined here because the algorithm performs both steps in sequence; assess (a) on correct LHS generation and (b) on correct PRCC computation from the same sample.)*

Using $N = 200$, $\pm 50\%$ around nominal values:

```
# Define parameter ranges (±50% of nominal values)
p_nom = [c=5, beta=0.06, tau=7, tau_m=10000]
lb    = 0.5 * p_nom
ub    = 1.5 * p_nom

set_random_seed(42)
X = LHS_SAMPLE(N=200, m=4, lower=lb, upper=ub)    # 200×4 sample matrix

# Run SIR model at each parameter set
Y = [sir_I_max(X[i, :]) FOR i IN 1..200]          # 200-vector of outputs

# Compute PRCC (partial rank correlation, removing effects of other params)
rho, pval = COMPUTE_PRCC(X, Y, names=["c","beta","tau","tau_m"])
```

Expected PRCC values (approximate):

Parameter	PRCC	p-value
c	$+0.85$	< 0.001
β	$+0.84$	< 0.001
τ_R	$+0.72$	< 0.001
τ_m	-0.04	> 0.5

(c) **PRCC bar chart.** Sorted: $c \approx \beta > \tau_R \gg \tau_m$ (similar to SI ranking).

(d) **Strongest negative PRCC: τ_m.** τ_m (mean residence time) has a negative PRCC: longer lifespans reduce peak infections. In plain language: populations with longer lifespans have smaller background mortality rates, which means the susceptible pool is replenished more slowly, and a larger fraction of the population is already immune/recovered by the time of peak infections. The effect is very small ($|\text{PRCC}| \approx 0.04$, not statistically significant).

(e) **Comparison with local SI tornado plot.**

The local SI tornado for the SIR model (from the Chapter 2 exercises on $\mathcal{R}_0$, not the food-safety model from Ex 2.7) ranks parameters $c \approx \beta > \tau_R \gg \tau_m$. Both PRCC and local SI rankings agree: $c \approx \beta > \tau_R \gg \tau_m$. Agreement is expected because $I_{\max}$ is monotone in each parameter over the $\pm 50\%$ uncertainty range (the model is not near any threshold), satisfying the PRCC monotonicity assumption.

Checking your answer: **A complete answer covers all five parts: (a)** LHS code runs and generates a valid sample (stratified, no duplicate strata); random sample for comparison.

Common error: using `rand` (simple random) as the "comparison" without labeling it correctly, or reversing which is LHS and which is random. **(b)** PRCC computed from ranked data; four values with p-values. At minimum: correct sign for all four parameters. A complete answer: correct magnitudes ($c \approx \beta \approx 0.85$, $\tau_R \approx 0.72$, $|\tau_m| < 0.1$, p-value> 0.5 for τ_m). **(c)** Bar chart sorted by |PRCC|, not by raw PRCC. Common error: sorting by signed PRCC (places τ_m at the bottom for the wrong reason; it is there because its magnitude is smallest, not because it is negative). **(d)** τ_m has the strongest negative PRCC. Plain-language interpretation must reference biological mechanism (longer lifespan → slower replenishment of susceptibles → lower peak). At minimum: correct parameter identified. A complete answer: mechanistic sentence. **(e)** PRCC and local SI rankings agree. The distinguishing check: students who say "they always agree" have not answered (e); a complete answer states *why* they agree here (monotone response, no threshold crossing in the ±50% range) and acknowledging that agreement is model- and range-specific. Cross-reference note: the tornado plot referenced here is the SIR $I_{\max}$ tornado from Chapter 2 exercises, not the food-safety tornado from Ex 2.7.

Exercise 9.4 *[Bloom L4]*

Exercise statement. For $Y = 2Q_1^2 + Q_2$ with $Q_1, Q_2 \sim \mathcal{U}(0,1)$, independent:

(a) Compute $\mathbb{E}[Y]$, $\mathrm{Var}(Y)$, $D_1 = \mathrm{Var}[\mathbb{E}(Y \mid Q_1)]$, $D_2 = \mathrm{Var}[\mathbb{E}(Y \mid Q_2)]$.

(b) Compute S_1, S_2. Do they sum to 1? Explain.

(c) Compute the total indices S_{T_1}, S_{T_2}.

(d) Now add Q_1 and Q_2 to get $Y' = 2Q_1^2 + Q_2 + Q_1Q_2$ (an interaction term). Without computing: does adding the interaction term change S_1, S_2, or both? In which direction? Then verify by computing the Sobol indices for Y' analytically.

Solution.

$Y = 2Q_1^2 + Q_2$, $Q_1, Q_2 \sim \mathcal{U}(0,1)$, independent.

(a) **Variance decomposition.** $\mathbb{E}[Y] = 2\mathbb{E}[Q_1^2] + \mathbb{E}[Q_2] = 2/3 + 1/2 = 7/6$. $\mathrm{Var}(Y) = 4\,\mathrm{Var}(Q_1^2) + \mathrm{Var}(Q_2)$. $\mathrm{Var}(Q_1^2) = \mathbb{E}[Q_1^4] - (\mathbb{E}[Q_1^2])^2 = 1/5 - 1/9 = 4/45$. $\mathrm{Var}(Q_2) = 1/12$. $\mathrm{Var}(Y) = 4(4/45) + 1/12 = 16/45 + 1/12 = 64/180 + 15/180 = 79/180$.

$D_1 = \mathrm{Var}[\mathbb{E}(Y|Q_1)] = \mathrm{Var}[2Q_1^2 + 1/2] = 4\,\mathrm{Var}(Q_1^2) = 16/45$. $D_2 = \mathrm{Var}[\mathbb{E}(Y|Q_2)] = \mathrm{Var}[2/3 + Q_2] = 1/12$.

(b) **Sobol indices.** $S_1 = D_1/\mathrm{Var}(Y) = (16/45)/(79/180) = (16/45)(180/79) = 64/79 \approx 0.810$. $S_2 = D_2/\mathrm{Var}(Y) = (1/12)/(79/180) = (180/12)/79 = 15/79 \approx 0.190$. $S_1 + S_2 = 79/79 = 1.0$. Yes, they sum to 1, because the model is *additive* (no interaction terms), so the full variance is explained by the first-order indices.

(c) **Total indices.** For an additive model: $S_{T_1} = S_1 = 64/79$, $S_{T_2} = S_2 = 15/79$. Total indices equal first-order indices when there are no interactions.

(d) **Adding interaction Q_1Q_2: $Y' = 2Q_1^2 + Q_2 + Q_1Q_2$.**

The exercise asks to predict the direction first, then compute exactly.

Naive prediction (wrong): "Adding Q_1Q_2 introduces shared variance, so both S_1 and S_2 decrease." This prediction is incorrect.

Exact computation (using $Q_1, Q_2 \sim \mathcal{U}(0,1)$, independent):

Step 1: Conditional expectations.

$$\mathbb{E}(Y' \mid Q_1) = 2Q_1^2 + \mathbb{E}[Q_2] + Q_1\mathbb{E}[Q_2] = 2Q_1^2 + \tfrac{1}{2} + \tfrac{Q_1}{2}.$$

$$D_1' = \operatorname{Var}(2Q_1^2 + \tfrac{Q_1}{2}) = 4\operatorname{Var}(Q_1^2) + \tfrac{1}{4}\operatorname{Var}(Q_1) + 2\cdot 2\cdot \tfrac{1}{2}\operatorname{Cov}(Q_1^2, Q_1).$$

With $\operatorname{Var}(Q_1^2) = 4/45$, $\operatorname{Var}(Q_1) = 1/12$, $\operatorname{Cov}(Q_1^2, Q_1) = \mathbb{E}[Q_1^3] - \mathbb{E}[Q_1^2]\mathbb{E}[Q_1] = 1/4 - (1/3)(1/2) = 1/12$:

$$D_1' = \frac{16}{45} + \frac{1}{48} + \frac{1}{6} = \frac{512}{1440} + \frac{30}{1440} + \frac{240}{1440} = \frac{782}{1440} = \frac{391}{720} \approx 0.5431.$$

$$\mathbb{E}(Y' \mid Q_2) = \mathbb{E}[2Q_1^2] + Q_2 + \mathbb{E}[Q_1]Q_2 = \tfrac{2}{3} + Q_2 + \tfrac{Q_2}{2} = \tfrac{2}{3} + \tfrac{3}{2}Q_2.$$

$$D_2' = \operatorname{Var}\left(\tfrac{3}{2}Q_2\right) = \tfrac{9}{4}\cdot\tfrac{1}{12} = \frac{3}{16} \approx 0.1875.$$

Step 2: Total variance (including all cross-covariance terms).

The additional cross-covariance $\operatorname{Cov}(2Q_1^2, Q_1Q_2) = 2\mathbb{E}[Q_2]\operatorname{Cov}(Q_1^2, Q_1) = 2\cdot\frac{1}{2}\cdot\frac{1}{12} = \frac{1}{12}$ must be included:

$$\operatorname{Var}(Y') = \underbrace{\frac{16}{45}}_{D_1} + \underbrace{\frac{1}{12}}_{D_2} + \underbrace{\frac{7}{144}}_{\operatorname{Var}(Q_1Q_2)} + \underbrace{2\cdot\frac{1}{24}}_{2\operatorname{Cov}(Q_2, Q_1Q_2)} + \underbrace{2\cdot\frac{1}{12}}_{2\operatorname{Cov}(2Q_1^2, Q_1Q_2)} = \frac{59}{80} \approx 0.7375.$$

Step 3: First-order Sobol indices.

$$S_1' = \frac{391/720}{59/80} = \frac{391}{531} \approx 0.736, \qquad S_2' = \frac{3/16}{59/80} = \frac{15}{59} \approx 0.254, \qquad S_{12}' = 1 - \frac{391}{531} - \frac{15}{59} = \frac{5}{531} \approx 0.009.$$

All three are positive and sum to 1. ✓

Comparison with additive baseline:

Model	Var	S_1	S_2	S_{12}
Additive $Y = 2Q_1^2 + Q_2$	79/180	$64/79 \approx 0.810$	$15/79 \approx 0.190$	0
With interaction Y'	59/80	$391/531 \approx 0.736$	$15/59 \approx 0.254$	$5/531 \approx 0.009$
Direction	–	**decreases**	**increases**	–

Why does S_2 *increase*? Adding Q_1Q_2 shifts the conditional mean to $\mathbb{E}(Y' \mid Q_2) = \frac{2}{3} + \frac{3}{2}Q_2$, so the slope on Q_2 grew from 1 to $\frac{3}{2}$. Therefore $D_2' = (\frac{3}{2})^2\operatorname{Var}(Q_2)$ grew by factor $(3/2)^2 = 2.25$, while $\operatorname{Var}(Y')$ grew by only $(59/80)/(79/180) \approx 1.68$. Since D_2' grew faster than total variance, $S_2 = D_2'/\operatorname{Var}(Y')$ increased. The total-order indices follow from $D_{12}' = 1/144$ and $\operatorname{Var}(Y') = 59/80$: $S_{T_1} = (D_1' + D_{12}')/\operatorname{Var}(Y') = 44/59 \approx 0.746$ and $S_{T_2} = (D_2' + D_{12}')/\operatorname{Var}(Y') = 140/531 \approx 0.264$. Each exceeds its first-order counterpart by

$D'_{12}/\operatorname{Var}(Y') = 5/531 \approx 0.009$, the shared interaction, and the two sum to $1.009 > 1$ as total-order indices must when any interaction is present.

Key lesson: first-order Sobol indices can move in *either direction* when an interaction is added. The direction depends on whether the conditional variance D_i grows faster or slower than total variance. **Always compute; never infer from the intuition "interactions share variance, so all first-order indices must decrease."**

Checking your answer: Part (b)'s exact sum-to-1 for the additive model is a crucial result: for any purely additive model, $\sum S_i = 1$ because there are no interaction terms ($D_{ij} = 0$ for all $i \neq j$). Common error for part (d): claiming "both first-order indices decrease" when an interaction is added; in fact, S_2 *increases* when Q_1Q_2 is added to $Y = 2Q_1^2 + Q_2$. The direction depends on whether $D_i = \operatorname{Var}[\mathbb{E}(Y|Q_i)]$ grows faster or slower than total variance. At minimum for (d): correct exact fractions for S_1, S_2 for the additive model; correct direction for S_1 (decreases). A complete answer: correct direction for both (S_1 decreases, S_2 *increases*), plus the counterexample insight that interactions can push first-order indices in either direction. The distinguishing check: the highest-value answer highlights this as a counterexample to intuition, not just a computation.

Exercise 9.5 *[Bloom L4]*

Exercise statement. Using the Jansen estimator implementation from the laboratory:

(a) Compute $\hat{S}_i$ and $\hat{S}_{T_i}$ for all four SIR parameters at $N = 2048$.

(b) For which parameters is $S_{T_i} \gg S_i$? What does this indicate about their role in the model?

(c) Compute $\sum_i S_i$ and $\sum_i S_{T_i}$. What do you expect these to sum to for an additive model? Are your results consistent with this expectation?

(d) If you could measure exactly one parameter very precisely (reducing its uncertainty to zero), which parameter would reduce $\operatorname{Var}(I_{\max})$ the most? Justify using the Sobol indices.

Solution.

Using `GLOBAL_SA_SOBOL` from the companion notebooks with $N = 2048$.

```
# Sobol sensitivity analysis - language-agnostic pseudocode
# Works with any implementation (SALib/Python, sobol_indices/R,
# Global Sensitivity Analysis Toolbox/MATLAB, or custom code)

# Step 1: Define problem
params = ["c", "beta", "tau", "tau_m"]
bounds = [[c_lo, c_hi], [beta_lo, beta_hi], [tau_lo, tau_hi], [tau_m_lo, tau_m_hi]]

# Step 2: Saltelli sample -- N*(m+2) model evaluations total
X = SALTELLI_SAMPLE(N=2048, params=params, bounds=bounds)
```

```
   # N*(m+2) = 2048*6 = 12,288 evaluations for m=4 parameters

# Step 3: Evaluate model at all sample points
Y = [sir_I_max(X[i, :])  FOR i IN 1..num_rows(X)]

# Step 4: Estimate Sobol indices (Jansen estimator)
S1, ST = SOBOL_ANALYZE(X, Y, params)
   # S1[j]  = first-order index for parameter j
   # ST[j]  = total index for parameter j
```

Saltelli sampling at $N = 2048$ generates $N(m+2) = 2048 \times 6 = 12{,}288$ model evaluations for $m = 4$ parameters; the resulting $\hat{S}_i$ and $\hat{S}_{T_i}$ are computed by the Jansen estimator.

(a) **Sobol indices (nominal parameters, $\pm 50\%$ range).**
Approximate values from Jansen estimators:

Param	$\hat{S}_i$	$\hat{S}_{T_i}$	$\hat{S}_{T_i} - \hat{S}_i$
c	0.31	0.37	0.06
β	0.31	0.36	0.05
τ_R	0.30	0.37	0.07
τ_m	0.00	0.00	0.00

(b) **Parameters with $S_{T_i} \gg S_i$.** For all three active parameters, $S_{T_i} - S_i \approx 0.05$, indicating moderate interaction effects (primarily c-β and c-τ_R interactions). No parameter has $S_{T_i} \gg S_i$ for this model at this range; interactions are present but modest. This is consistent with the SIR model being a smooth, moderately nonlinear ODE.

(c) **Sums.** $\sum S_i \approx 0.96 < 1$ (interaction terms absorb the remaining $\sim 4\%$). $\sum S_{T_i} \approx 1.11 > 1$ (interactions are double-counted in total indices). For a purely additive model, both sums would equal 1. The observed values confirm the presence of small but nonzero interactions.

(d) **Best parameter to measure precisely.** To reduce $\mathrm{Var}(I_{\max})$ most, measure the parameter with the largest *first-order* index. By the law of total variance, learning Q_i exactly leaves expected variance $\mathbb{E}[\mathrm{Var}(I_{\max} \mid Q_i)] = (1 - S_i)\,\mathrm{Var}(I_{\max})$, so the expected reduction is S_i, not S_{T_i}. Here c, β and τ_R are tied at $S_i \approx 0.31$: measuring any one of them exactly reduces $\mathrm{Var}(I_{\max})$ by about 31%, leaving $\approx 0.69\,\mathrm{Var}(I_{\max})$.

The total index answers a different question. S_{T_i} is the share of variance that survives when every *other* parameter is held fixed, so it governs *factor fixing*: the value $S_{T_i} = 0.00$ for τ_m says that τ_m can be frozen at its nominal value without affecting the output variance at all. For this model the two rankings agree, because the interactions are modest. When $S_{T_i} - S_i$ is large they need not agree, and the measurement question is still answered by S_i.

Checking your answer: Part (d) separates two policy questions that the total index is often asked to answer at once. "Which parameter should we measure?" is answered by S_i: the expected variance reduction from learning Q_i exactly is S_i, by the law of total variance. "Which parameters can we stop resolving?" is answered by S_{T_i}: a small total index licenses fixing that parameter. Here both point to c and β because the interactions are modest, which is exactly why the distinction is easy to miss and worth drawing explicitly. Common error: quoting S_{T_i} as the achievable variance reduction. That overstates the gain whenever interactions are present, giving 43% here in place of the correct 38%. At minimum: correct Jansen estimator code; qualitative comparison of S_i vs S_{T_i}. A complete answer: a measurement recommendation justified by S_i with the correct $(1 - S_i)$ residual variance, together with recognition that $S_{T_i} - S_i$ measures interaction strength and that a small S_{T_i} is the factor-fixing criterion. The distinguishing check: an answer that confuses first-order and total indices in the policy recommendation is partial at most.

Exercise 9.6 *[Bloom L4]*

Exercise statement. For the SIR model, implement $r = 15$ Morris trajectories with $\Delta = 1/3$:

(a) Compute μ_i^* and σ_i for each parameter.

(b) Produce the Morris diagnostic plot.

(c) Based on the Morris results, which parameter appears most influential? Which appears least influential?

(d) Run a reduced Sobol analysis using only the three most important parameters identified by Morris (fixing τ_m at its nominal value). Compare the cost and results with the full four-parameter Sobol.

Solution.

Using $r = 15$ Morris trajectories, $p = 4$ levels, $\Delta = 1/3$ (as assigned; this corresponds to a step of one interval on the $\{0, 1/3, 2/3, 1\}$ grid), ±50% parameter ranges.

(a) **Morris measures.** Approximate values from `morris_screening`:

Param	μ^*	σ	Classification
c	~ 80	~ 20	Important, nonlinear
β	~ 78	~ 19	Important, nonlinear
τ_R	~ 42	~ 14	Important, nonlinear
τ_m	~ 1	~ 0.5	Negligible

(b) **Diagnostic plot.** The μ^* vs. σ plot shows c and β in the upper-right (large μ^* and σ: important and nonlinear), τ_R in the middle-right, and τ_m near the origin (negligible).

(c) **Most and least influential.** Most: c, β and τ_R together, and the tie is exact rather than close. Least: τ_m. Under ±50% ranges the three enter $\mathcal{R}_0 = c\beta\tau_R$ as a symmetric monomial with the same relative spread, so no amount of sampling will separate them. A run that reports one of the three as clearly larger is reporting Monte Carlo scatter.

(d) **Reduced Sobol (3 parameters, τ_m fixed).**

With τ_m fixed: Sobol cost $= N(m + 2) = 2048 \times 5 = 10{,}240$ evaluations vs. full $2048 \times 6 = 12{,}288$. Cost reduction: 17%. Approximate results, first-order: $\hat{S}_c \approx 0.30$,

$\hat{S}_\beta \approx 0.30$, $\hat{S}_{\tau_R} \approx 0.30$, summing to 0.91. Total-order: $\hat{S}_{T_i} \approx 0.37$ for each, summing to 1.10. Report the two separately and check the sums: first-order indices must sum to at most 1, total-order indices to at least 1, and the gap between the sums measures the interaction. Quoting 1.10 as a sum of first-order indices would be an arithmetic impossibility, not a rounding artifact. Fixing τ_m does not materially change the other indices (since τ_m was negligible). The Morris screening correctly identified that τ_m could be dropped.

Checking your answer: The Morris cost here is $15 \times (4 + 1) = 75$ evaluations vs. 12,288 for full Sobol; a 164× cost reduction for the screening phase. This illustrates Morris's value as a cheap first step that identifies which parameters can be dropped from the expensive Sobol analysis.

Exercise 9.7 *[Bloom L5]*

Exercise statement. Choose a published epidemic model from the literature with at least 8 parameters. Design a complete global SA study:

(a) State the model, its parameters, and a justification for the uncertainty ranges you will use (with citations).

(b) State the QOI or QOIs and explain why they are the most relevant for the model's purpose.

(c) Choose between PRCC and Sobol as the primary global SA method. Justify your choice based on the model's properties.

(d) Specify the sample size N and justify it based on convergence.

(e) Describe how you would present the results to a non-technical audience (e.g., a public health committee or funding agency).

(f) Identify one situation where your global SA results could still be misleading, and describe what additional analysis you would recommend.

Solution.

This exercise requires the student to choose and describe a specific published model. The model answer below uses the Chitnis dengue model [1] as an example; any 8+ parameter epidemic model with published parameter estimates is acceptable.

(a) **Model, parameters, uncertainty ranges.** Chitnis et al. (2008) dengue model: 8 state variables, 14 parameters including mosquito biting rate b_v, transmission probabilities β_{vh} and β_{hv}, mosquito and human mortality rates, and mean durations. Uncertainty ranges: ±50% for biological parameters (based on reported ranges in Table 2 of the paper); ±30% for demographic parameters (population data are more precisely known).

(b) **QOI justification.** Primary: $\mathcal{R}_0$ (determines whether dengue can persist in the population, the most policy-relevant threshold). Secondary: equilibrium human infection prevalence $\bar{I}_h/N_h$ (determines endemic burden for healthcare planning).

(c) **PRCC vs. Sobol choice.** Use **Sobol indices**. The dengue model is moderately nonlinear and has known interaction between biting rate b_v and transmission probabilities ($\mathcal{R}_0 \propto b_v^2 \beta_{vh} \beta_{hv}$, so interactions are structural). PRCC assumes monotone relationships; near threshold or at high $\mathcal{R}_0$, this assumption may not hold for all parameters. Sobol indices capture interactions and do not require monotonicity.

(d) **Sample size N.** Use $N = 2048$ for each Sobol matrix. Run the convergence check: compute Sobol indices at $N = 256, 512, 1024, 2048$, staying on powers of two for the reason given in the Chapter 9 laboratory; report the value of N at which the first-order indices change by $< 2\%$ as N doubles. For a 14-parameter model, the total evaluation cost is $N(14 + 2) = 32{,}000$ runs. If each run takes 2 minutes, this is ≈ 44.4 days of computation on one core ($32{,}000 \times 2$ min $= 64{,}000$ min ≈ 44 days). This is impractical serially; parallelizing across 16 cores reduces wall-clock time to approximately 3 days, which is operationally feasible.

(e) **Presentation format.** Two tornado plots (one per QOI). Use Sobol S_{T_i} to say where the output uncertainty comes from, and the normalized local SI $\mathcal{S}_p^{\mathcal{R}_0}$ to label interventions, since only the elasticity supports a statement of the form "reducing the mosquito biting rate by $X\%$ through bed nets reduces $\mathcal{R}_0$ by approximately $Y\%$." A variance share cannot be read as a proportional response; see the index-selection table in Chapter 9. A two-panel figure showing which parameters appear in both tornado plots (co-optimization) and which are QOI-specific. Use traffic-light color coding: red for high-priority, green for low-priority intervention targets.

(f) **Potential misleading results.** If the model's best-fit parameters give $\mathcal{R}_0$ near 1 (endemic/non-endemic threshold), Sobol indices at those nominal values will be dominated by parameters near the threshold. A 2% uncertainty in β_{vh} might shift the model from endemic to non-endemic, giving extremely large apparent sensitivity for β_{vh}. Recommendation: compute Sobol indices at three nominal parameter sets ($\mathcal{R}_0 = 1.5$, 2.0, 3.0) and report whether the parameter rankings are stable across this range.

Checking your answer: This is an open-ended design exercise. A complete answer requires all six components with operationally specific content. Vague statements ("choose the right sample size," "present clear figures") without specifics are partial. The model choice affects all answers; a reasonable choice with internally consistent answers is complete even if different from the example above.

Exercise 9.8 *[Bloom L4]*

Exercise statement. *Audit exercise.* A collaborator reports $S_{T_\beta} = 0.62$ and concludes that measuring β more precisely would remove about 62% of the uncertainty in peak infection count, and that this is the most valuable measurement available. Apply the audit protocol; give the correct index and explain the overstatement; state when the two indices agree and why that makes the error easy to miss; identify a second unsupportable claim.

Solution.

(a) Step 1 of Section 9.10.1 asks what decision is being informed. The collaborator is deciding which parameter to measure first in order to reduce output uncertainty, which is the first row of Table 9.4. Step 2 asks whether the reported index answers that question. It does not: the row assigns the decision to the first-order index S_i, and the third column names the substitution of S_{T_i} as the error.

(b) The correct index is S_β. Learning the true value of Q_i removes the variance attributable to Q_i acting alone, leaving $(1 - S_i)\,\mathrm{Var}(Y)$, so the available reduction is S_i. The total-order index S_{T_i} additionally counts every interaction term in which Q_i appears, and those terms cannot be removed by resolving Q_i alone, since they also require resolving the parameters it interacts with. The gap $S_{T_\beta} - S_\beta$ is exactly the interaction contribution, and quoting S_{T_β} as the achievable reduction overstates it by that amount.

(c) The two agree when interactions are weak, since $S_{T_i} \to S_i$ as the interaction terms vanish, and for a purely additive model they are equal. Many models encountered in practice are close to additive over modest parameter ranges, so the substitution produces a nearly correct number most of the time and a badly wrong one occasionally. An error that is usually harmless is the kind that becomes habitual, and it fails in the strongly interacting case where the stakes are highest. The caption of Table 9.4 makes this point directly.

(d) The claim that this is "the single most valuable measurement we can make" concerns value, not variance. It requires knowing the cost of measuring each parameter, the precision attainable, and what the reduced uncertainty is worth for the decision at hand. No sensitivity index of any kind carries that information. Two parameters with identical indices can differ by orders of magnitude in the cost of measuring them. **Unifying Idea 4.**

Checking your answer: Parts (a) and (b) are the core and should be reachable directly from the table. Part (c) is the one that distinguishes understanding from lookup: a student who can say why the error survives in practice has grasped why the table was written. Part (d) tests whether the student notices a claim that no index supports, which is the failure mode the audit's final step is designed to catch.

Exercise 9.9 *[Bloom L5]*

Exercise statement. *Audit exercise.* PRCC on $N = 500$ samples ranks (p_1, p_2, p_3); first-order Sobol on $N = 2048$ ranks (p_3, p_1, p_2). Both are computed correctly, and the response to p_3 rises and then falls. Identify the violated assumption; explain how it produces a low PRCC for an influential parameter; describe a bootstrap and the result that would reconcile the rankings; write a one-sentence finding if the disagreement is real.

Solution.

(a) From Table 9.3, PRCC assumes a monotone relationship between each input and the output, while Sobol estimation assumes the inputs are independent and makes no monotonicity requirement. A response to p_3 that rises and then falls is non-monotone, so the PRCC assumption is violated for p_3 and the Sobol assumption is untouched by that fact.

(b) PRCC measures monotone association after rank transformation. For a response that rises and then falls over the sampled range, the increasing and decreasing portions contribute correlations of opposite sign that partially cancel, and a parameter with a large influence can return a partial rank correlation near zero. The Sobol first-order index, by contrast, measures the variance of the conditional expectation $\mathbb{E}[Y \mid p_3]$, which is large whenever varying p_3 moves the mean of Y substantially, regardless of the direction of that movement. The two methods therefore disagree in exactly the situation described, and the disagreement is diagnostic rather than contradictory.

(c) Resample the model evaluations with replacement, recompute the indices on each of several hundred bootstrap replicates, and report a percentile confidence interval for each index. The relevant comparison is between parameters within a single method. If the intervals for p_1, p_2 and p_3 overlap substantially in either analysis, that analysis has not established an ordering, and the two reported rankings are not in conflict because at least one of them is not a ranking. The $N = 500$ PRCC run is the more likely of the two to show this.

(d) A suitable finding: "In this model, first-order Sobol indices identify p_3 as the dominant contributor to output variance, while PRCC ranks it last; the discrepancy is consistent with the non-monotone response to p_3, which violates the assumption underlying PRCC, and the Sobol ranking should be preferred here." The sentence attributes the result to the model, names the assumption at issue, and stops short of any claim about the system the model represents. **Unifying Idea 4.**

Checking your answer: The instinct on meeting two conflicting analyses is to ask which is right, and the more useful question is which assumption each requires. Part (c) is included because a great many reported disagreements dissolve once sampling error is quantified, and checking that before explaining the discrepancy is the correct order of operations. In part (d) the phrase "in this model" is doing real work and should not be treated as a hedge.

Exercise 9.10 *[Bloom L6]*

Exercise statement. A published SA study reports Sobol first-order and total indices for a 15-parameter water-quality model. The authors note in the data section that three parameters, phosphorus loading rate, nitrogen loading rate, and water residence time, are positively correlated across the sampled watersheds. The SA section reports standard Sobol indices computed using independent sampling for all 15 parameters, without addressing the correlation.

(a) Identify the specific mathematical problem that correlation creates for standard Sobol indices.

(b) For each of the three correlated parameters, state whether the reported first-order index is likely over-estimated, under-estimated, or unchanged, and explain why.

(c) Propose the minimal methodological correction that would make the analysis valid.

(d) The authors justify their approach by noting that their Sobol indices sum to approximately 1.0. Explain why this is not a valid check for independence.

Solution.

Methodological concerns with Sobol analysis ignoring correlation among three parameters.

(a) **Mathematical problem.**

Standard Sobol indices assume parameter *independence*: the matrices A and B are sampled independently from the joint distribution $p(\vec{p}) = \prod_j p_j(p_j)$. When three parameters are correlated, the correct joint distribution is $p(\vec{p}) = p(p_1, p_2, p_3) \times \prod_{j>3} p_j(p_j)$, which cannot be factored. Sampling independently from marginal distributions and then compositing (the Saltelli method) generates non-physical parameter combinations (e.g., high phosphorus loading but low nitrogen loading, which does not occur in the real watershed system). The resulting Sobol indices estimate sensitivity to artificially independent variations, not to the actual coupled variations that the system experiences.

(b) **Effect on each correlated parameter's index.**

The direction cannot be determined from the sign of the correlation alone. This is the central point of the exercise, and the tempting answer, that positive correlation inflates some indices and deflates others in a predictable pattern, is wrong.

The bias depends on how the response uses the correlated inputs, not only on how they covary. Two Gaussian inputs with the same $\rho = +0.8$ give opposite answers for the same variable:

Response	S_1 (independent)	S_1 (true, $\rho = 0.8$)	Direction
$x_1 + x_2$	0.50	0.90	underestimates
$x_1 - x_2$	0.50	0.10	overestimates
$x_1 x_2$	0.00	0.77	underestimates

When the correlated inputs push the response the same way, independent sampling breaks up variation that the real system delivers together and *understates* their individual shares. When they oppose one another, the same sampling *overstates* them. A pure interaction is invisible to the independent first-order index altogether.

For the water-quality model the honest answer is therefore that $S_1(P)$, $S_1(N)$ and $S_1(T_w)$ are all of unknown direction until the response structure is specified. If nutrient loading and residence time all drive eutrophication in the same direction, the first row is the relevant case and the individual indices *understate* their importance. That is a modelling judgment, not a consequence of $\rho > 0$.

Checking your answer: Under dependent inputs the independent-input ANOVA decomposition is no longer unique, so "the true S_1" needs a stated attribution convention before a bias direction is even well posed. A complete answer says so and then picks one: grouped indices, Shapley effects, or a copula-based analysis.

(c) **Minimal correction.**

The chapter recommends three approaches for correlated inputs, in order of increasing rigor: PRCC (for monotonic models), grouped Sobol indices (when a natural grouping of correlated parameters exists), or Shapley effects (no grouping required).

For this 15-parameter water-quality model, the simplest valid correction depends on the correlation structure:

- If the three correlated parameters can be treated as a natural group (e.g., nutrient loading and residence time all drive eutrophication together), compute a *grouped* first-order Sobol index $S_{\{P,N,T_w\}}$ using the standard Saltelli/Jansen estimator on the group, alongside individual indices for the remaining 12 independent parameters.
- If individual attribution is required, use *Shapley effects* via the `sensitivity` R package or SALib, which handle correlated inputs without requiring a grouping structure and are consistent with the chapter's framework.

The Iman–Conover rank-correlation constraint is sometimes used as a practical approximation (generate LHS samples, then permute to match the observed rank correlations), but it does not resolve the fundamental issue: the Jansen estimator assumes independent samples, so even with Iman–Conover applied, the resulting first-order indices are a biased approximation. It is a useful empirical tool but should be reported with that caveat rather than presented as a fully valid solution.

(d) **Why sum-to-1 is not a valid independence check.**

The Sobol first-order indices sum to 1 only for an additive model with independent parameters. With correlated parameters, the indices can still sum to approximately 1 even with incorrect independence assumptions, because the sum reflects the total explained variance, which equals 1 by construction (the variance is fully accounted for by some combination of first-order and interaction terms). The sum being 1 says nothing about whether the individual index attributions are correct; it is a property of the decomposition algorithm, not of the physical model or the sampling design.

Checking your answer: This is a rigorous L6 exercise testing critical evaluation of a real methodological error. Part (d) is the subtlest: students often think "indices sum to 1" is a validity check, but it is not; it is simply a consequence of the ANOVA decomposition. A complete answer requires a clear mathematical explanation, not just "the check doesn't work."

References cited in exercise statements

References

[1] N. Chitnis, J. M. Hyman, and J. M. Cushing, *Determining important parameters in the spread of malaria through the sensitivity analysis of a mathematical model*, Bulletin of Mathematical Biology, 70 (2008), pp. 1272–1296.